38. USB 2.0 High-Speed Module (USBHS)

38.1 Overview

The MCU provides a USB 2.0 High-Speed Module (USBHS) that operates as a host or a device controller compliant with the Universal Serial Bus (USB) Specification revision 2.0. The host controller supports USB 2.0 high-speed, full-speed, and low-speed transfers, and the device controller supports USB 2.0 high-speed and full-speed transfers. The USBHS has an internal USB transceiver and supports all of transfer types defined in the USB 2.0 specification.

The USBHS has FIFO buffer for data transfers, providing a maximum of 10 pipes. Any endpoint number can be assigned to pipes 1 to 9, based on the peripheral devices or the communication requirements for your system.

Table 38.1 lists the USBHS specifications, Figure 38.1 shows a block diagram, and Table 38.2 lists the I/O pins.

Table 38.1 USBHS specifications

Parameter	Specifications
Features	● USB Device Controller (UDC) and USB 2.0 transceiver supporting host controller, device controller, and On-The-Go (OTG) functions ● Software can switch between host and device controller modes
	Host controller features: ● High-speed transfer (480 Mbps), full-speed transfer (12 Mbps), and low-speed transfer (1.5 Mbps) ● Automatic scheduling for SOF and packet transmissions ● Programmable intervals for isochronous and interrupt transfers ● Communications with multiple peripheral devices connected through a single hub
	Device controller features: ● High-speed transfer (480 Mbps) and full-speed transfer (12 Mbps) ● Control transfer stage control function ● Device state control function ● Auto response function for SET_ADDRESS request ● SOF complementation
Supported transfer types	● Control transfer ● Bulk transfer ● Interrupt transfer ● Isochronous transfer
Pipe configuration	● FIFO buffer of up to 8.5 KB for USB communications ● Up to 10 pipes selectable, including the default control pipe ● Programmable pipe configurations ● Pipes 1 to 9 assignable to any endpoint number
	Transfer conditions specifiable for each pipe: ● Pipe 0: Control transfer with 64-byte single buffer ● Pipes 1 and 2: Bulk isochronous transfer continuous transfer mode with programmable buffer size up to 2 KB and optional double buffer ● Pipes 3 to 5: Bulk transfer continuous transfer mode with programmable buffer size up to 2 KB and optional double buffer ● Pipes 6 to 9: Interrupt transfer with 64-byte single buffer
Other features	● Force-end transfer function using transaction count ● Function that changes the BRDY interrupt event notification timing ● Automatic clearing of the FIFO buffer after data for the pipe specified in the DnFIFO port (n = 0, 1) is read ● NAK setting function for response PID generated on transfer end ● On-chip pull-up and pull-down resistors for D+ and D- ● Support for Link Power Management (LPM) ECN, including a new Sleep state (the L1 state) ● Compliance with Battery Charging Class Specification Revision 1.2 ● For power reduction, selectable classic-only mode (CL-only mode) in which operation is only USB 1.1-compliant
Module-stop function	Module-stop state can be set to reduce power consumption
TrustZone Filter	Security and Privilege attribution can be set

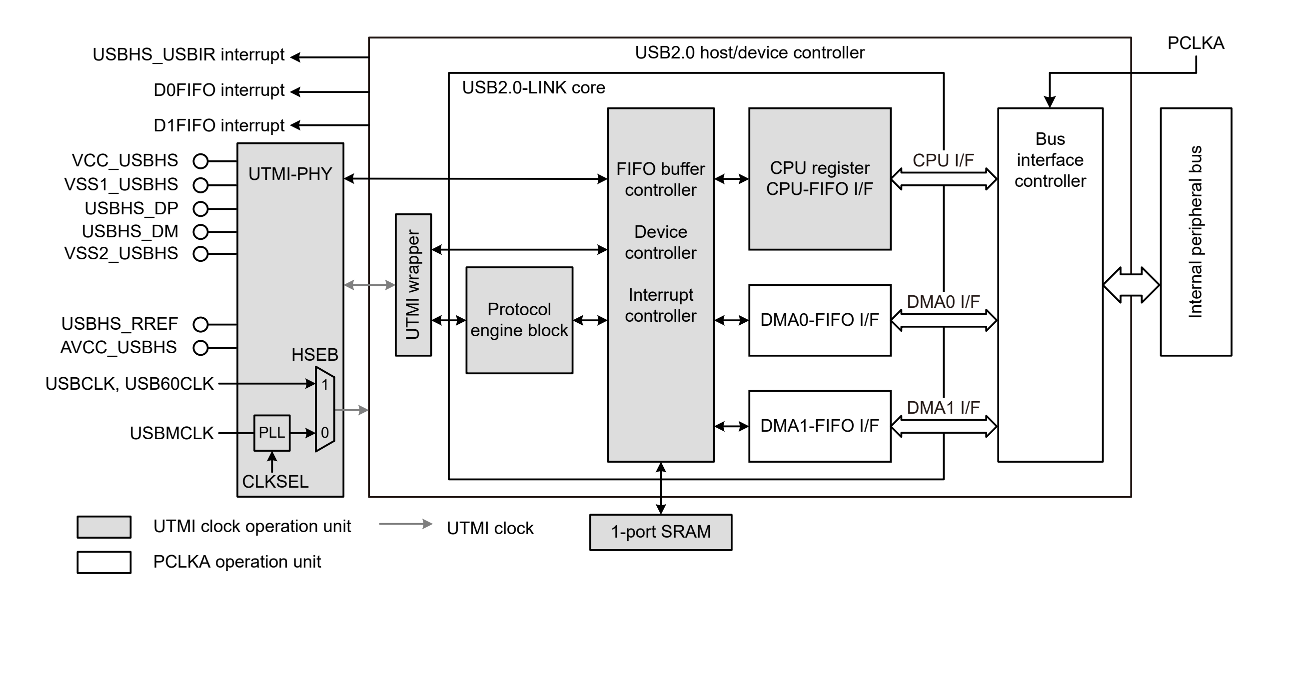


Figure 38.1 USBHS block diagram

Table 38.2 USBHS I/O pins

Pin name	I/O	Function
VCC_USBHS	Input	Power supply pin for the USBHS
VSS1_USBHS VSS2_USBHS	Input	Ground pin for the USBHS
AVCC_USBHS	Input	Analog power supply pin for the USBHS
USBHS_RREF	I/O	Reference current source pin for the USBHS Must be connected to the VSS2_USBHS pin through a 2.2-kΩ (±1%) resistor.
USBHS_DP	I/O	Input/output pin for the D+ data line of the USB bus
USBHS_DM	I/O	Input/output pin for the D- data line of the USB bus
USBHS_EXICEN	Output	Must be connected to the OTG power supply IC
USBHS_ID	Input	Must be connected to the OTG power supply IC
USBHS_VBUSEN	Output	VBUS power supply enable pin for the USBHS
USBHS_OVRCURA, USBHS_OVRCURB, USBHS_OVRCURA-DS, USBHS_OVRCURB-DS	Input	Overcurrent pin for the USBHS USBHS_OVRCURA or USBHS_OVRCURB pins can be used in Software standby mode or in normal mode. USBHS_OVRCURA-DS, USBHS_OVRCURB-DS are dedicated pins that can generate interrupt in Software standby mode or in normal mode as well as for cancel Deep Software Standby mode 1.
USBHS_VBUS	Input	USB cable connection monitor input pin

38.2 Register Descriptions

38.2.1 SYSCFG : System Configuration Control Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x000

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CNEN	HSE	DCFM	DRPD	DPRPU	—	—	—	USBE
Value after reset:	x	x	x	x	x	x	x	0	0	0	1	0	x	x	x	0

Bit	Symbol	Function	R/W
0	USBE	USBHS Operation Enable 0: Disable 1: Enable	R/W
3:1	—	The read values are undefined. The write value should be 0.	R/W
4	DPRPU	D+ Line Resistor Control 0: Disable line pull-up 1: Enable line pull-up	R/W
5	DRPD	D+/D- Line Resistor Control 0: Disable line pull-down 1: Enable line pull-down	R/W
6	DCFM	Controller Operation Select 0: Select device controller mode 1: Select host controller mode	R/W
7	HSE	High-Speed Operation Enable 0: Disable Device controller mode: full-speed Host controller mode: full-speed or low-speed 1: Enable The controller detects the communication speed	R/W
8	CNEN	Single-ended Receiver Enable 0: Disable 1: Enable	R/W
15:9	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Writing to the SYSCFG register can proceed while the PHY clock is stopped. However, written values are only reflected in the SYSCFG register after the PHY clock is oscillating again.

USBE bit (USBHS Operation Enable)

The USBE bit enables or disables operation of USBHS.

Changing the USBE bit from 1 to 0 initializes the bits listed in Table 38.3. Only change this bit after specifying the input clock in the PHYSET.CLKSEL[1:0] bits and confirming that the PLLSTA.PLLLOCK flag is 1. In CL-only mode, change the USBE bit after setting the PHYSET.HSEB bit to 1. At that time, the USBCLK must be set to 48 MHz and USB60CLK must be set to 60 MHz. For the clock settings, see section 38.3.3. Supplying the Clock.

In host controller mode, always set this bit to 1 after setting the DRPD bit to 1, eliminating SYSSTS0.LNST[1:0] bit chattering, and confirming that the USB bus state is stable.

Table 38.3 Bits initialized by writing SYSCFG.USBE = 0

Selected function	Register	Bit	Remarks
Device controller (DCFM = 0)	SYSSTS0	LNST[1:0]	Value is saved in host controller mode
	DVSTCTR0	RHST[2:0]	—
	PL1CTRL1	DVSQ[3:0]	Value is saved in host controller mode
	USBADDR	USBADDR[6:0]	Value is saved in host controller mode
	USBREQ	• BREQUEST[7:0] • BMREQUESTTYPE [7:0]	Value is saved in host controller mode
	USBVAL	WVALUE[15:0]	Value is saved in host controller mode
	USBINDX	WINDEX[15:0]	Value is saved in host controller mode
	USBLENG	WLENTUH[15:0]	Value is saved in host controller mode
Host controller (DCFM = 1)	DVSTCTR0	RHST[2:0]	—
	FRMNUM	FRNM[10:0]	Value is saved in device controller mode
	UFRMNUM	UFRNM[2:0]	Value is saved in device controller mode

DPRPU bit (D+ Line Resistor Control)

The DPRPU bit enables or disables pulling up the D+ line in device controller mode.

When the DPRPU bit is set to 1 in device controller mode, the USBHS pulls up the D+ line to notify the USB host that it attached. Changing the DPRPU bit from 1 to 0 releases the pull-up, thereby notifying the USB host that it detached.

Set this bit to 1 in device controller mode and to 0 in host controller mode.

DRPD bit (D+/D- Line Resistor Control)

The DRPD bit enables or disables pulling down D+ and D- lines in host controller mode.

Set this bit to 1 in host controller mode. Set it to 0 when OTG is not used in device controller mode.

DCFM bit (Controller Operation Select)

The DCFM bit selects the host or device function of the USBHS.

Only change this bit when the DPRPU and DRPD bits are both 0.

HSE bit (High-Speed Operation Enable)

The HSE bit enables or disables high-speed operation.

When this bit is 1, the USBHS operates in high-speed or full-speed based on the results of the reset handshake.

In host controller mode, setting this bit to 0 allows the USBHS to operate in full-speed or low-speed. If the DVSTCTR0.RHST[2:0] flags indicate that a low-speed device has attached, set the HSE bit to 0.

In host controller mode, setting this bit to 1 allows the USBHS to operate in high-speed or full-speed based on the results of the reset handshake. Change the HSE bit after detection of an attach event (ATTCH interrupt) and before the USB bus reset (when DVSTCTR0.USBRST = 1), or after detection of a detach event.

In device controller mode, setting this bit to 0 allows the USBHS to operate in full-speed. Setting the bit to 1 allows the USBHS to perform the reset handshake and then operate in high-speed or full-speed, based on the results.

In device controller mode, only change this bit when the DPRPU bit is 0.

CNEN bit (Single-ended Receiver Enable)

Setting the CNEN bit to 1 enables single-ended receiver operation and selects monitoring of the D+ and D- line states in the SYSSTS0.LNST[1:0] flags. Use this bit to prevent through-current damage that might otherwise be caused during single-ended receiver operation, where the terminals are floating while the USBHS is detached.

In host controller mode, set this bit to 1 after confirming that the PHY clock is being supplied. In device controller mode, set this bit to 1 when the VBUS is detected because of a VBUS interrupt, and set it to 0 when the VBUS line is removed.

38.2.2 BUSWAIT : CPU Bus Wait Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x002

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	BWAIT[3:0]			
Value after reset:	x	x	0	0	0	0	0	0	x	x	0	0	1	1	1	1

Bit	Symbol	Function	R/W
3:0	BWAIT[3:0]	CPU Bus Access Wait Specification 0x0: 0 waits (2 access cycles) : 0x2: 2 waits (4 access cycles) : 0x4: 4 waits (6 access cycles) : 0xF: 15 waits (17 access cycles) (initial value)	R/W
5:4	—	These bits are read as 0. The write value should be 0.	R/W
7:6	—	The read values are undefined. The write value should be 0.	R/W
13:8	—	These bits are read as 0. The write value should be 0.	R/W
15:14	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

BWAIT[3:0] bits (CPU Bus Access Wait Specification)

The BWAIT[3:0] bits specify the wait time for access to the USBHS registers.

When accessing the registers at offset addresses in range 0x004 to 0x15F for writing and range 0x004 to 0x167 for reading, set the cycle time for consecutive access to at least 41 ns. The initial value is 0xF (17 cycles), but Renesas recommends that you satisfy this condition by setting the best wait time for the frequency of PCLKA in your application.

This setting is the same as the wait time for accesses to the FIFO port register. The maximum speed of access to the FIFO port is as follows:

- MBW[1:0] = 10b (32-bit width): Maximum 60 MB/s
- MBW[1:0] = 01b (16-bit width): Maximum 30 MB/s
- MBW[1:0] = 00b (8-bit width): Maximum 15 MB/s

38.2.3 SYSSTS0 : System Configuration Status Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x004

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVCMON[1:0]		—	—	—	—	—	—	—	HTACT	SOFEA	—	—	IDMON	LNST[1:0]	
Value after reset:	x	x	x	x	x	x	x	x	x	0	0	x	x	x	x	x

Bit	Symbol	Function	R/W
1:0	LNST[1:0]	USB Data Line Status Monitor Flag Indicates the status of the USB data lines. See Table 38.4.	R
2	IDMON	USBHS_ID Pin Monitor Flag 0: USBHS_ID pin is low 1: USBHS_ID pin is high	R

Bit	Symbol	Function	R/W
4:3	—	The read values are undefined.	R
5	SOFEA	SOF Active Monitor Flag While Host Controller Operation Is Selected 0: SOF output stopped 1: SOF output operating	R
6	HTACT	Host Sequencer Status Monitor Flag 0: Host sequencer stopped 1: Host sequencer operating	R
13:7	—	The read values are undefined.	R
15:14	OVCMON[1:0]	External USBHS_OVRCURA or USBHS_OVRCURA-DS, and USBHS_OVRCURB or USBHS_OVRCURBB-DS Input Pin Monitor Flag OVCMON[1] indicates the USBHS_OVRCURA or USBHS_OVRCURA-DS pin status. OVCMON[0] indicates the USBHS_OVRCURB or USBHS_OVRCURB-DS pin status.	R

Note: S-TYPE-3, P-TYPE-3

LNST[1:0] flags (USB Data Line Status Monitor Flag)

The LNST[1:0] flags indicate the state of the USB data lines, D+ and D-. For details, see Table 38.4.

In device controller mode, read the LNST[1:0] flags after setting the SYSCFG.CNEN and SYSCFG.USBE bits to 1. In host controller mode, read them after setting the SYSCFG.DRPD bit to 1.

When you are checking hardware contacts for the battery charging function in device controller mode, read the LNST[1:0] flags after setting the SYSCFG.DRPD, SYSCFG.CNEN, and BCCTRL.IDPSRCE bits to 1. For details, see section 38.3.15. Battery Charging Detection Processing.

Table 38.4 Status of USB data bus lines (D+ and D-)

LNST[1]	LNST[0]	Low-speed operation (host controller mode only)	Full-speed operation	High-speed operation	Chirp operation
0	0	SE0	SE0	Squelch	Squelch
0	1	K-State	J-State	Unsquelch	Chirp J
1	0	J-State	K-State	Invalid	Chirp K
1	1	SE1	SE1	Invalid	Invalid

Chirp: The reset handshake protocol is being executed when high-speed operation is enabled (HSE bit is 1).

Squelch: SE0 or idle state

Unsquelch: High-speed J-state or high-speed K-state

Chirp J: Chirp J-State

Chirp K: Chirp K-State

SOFEA flag (SOF Active Monitor Flag While Host Controller Operation Is Selected)

The SOFEA flag is used in host controller mode to check whether the output of the last SOF is complete when the USBHS is suspended because of a 0 setting to the DVSTCTR0.UACT bit.

In host controller mode, check that both the HTACT and SOFEA flags are 0 before setting the SYSCFG.USBE bit to 0 to stop the USBHS or setting the LPSTS.SUSPENDM bit to 0 to stop the clock signal supply during communication.

HTACT flag (Host Sequencer Status Monitor Flag)

The HTACT flag clears to 0 when the host sequencer of the USBHS is completely stopped.

In host controller mode, check that the HTACT flag is 0 before setting the DVSTCTR0.UACT bit to 0 to place the USBHS in the Suspend state or setting the LPSTS.SUSPENDM bit to 0 to stop the clock signal supply during communication.

OVCMON[1:0] flags (External USBHS_OVRCURA or USBHS_OVRCURA-DS, and USBHS_OVRCURB or USBHS_OVRCURBB-DS Input Pin Monitor Flag)

The OVCMON[1:0] flags indicate the status of the overcurrent signals from an external power supply IC.

38.2.4 PLLSTA : PLL Status Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x006

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PLLLOCK
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

Bit	Symbol	Function	R/W
0	PLLLOCK	PLL Lock Flag 0: PLL not locked 1: PLL locked	R
15:1	—	The read values are undefined.	R

Note: S-TYPE-3, P-TYPE-3

PLLLOCK flag (PLL Lock Flag)

The PLLLOCK flag indicates whether the USB-PHY internal PLL is locked. When not using CL-only mode, make sure that the PLL is locked before starting USB communication.

38.2.5 DVSTCTR0 : Device State Control Register 0

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x008

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	HNPBTOA	EXICEN	VBUSEN	WKUP	RWUPE	USBRST	RESUME	UACT	—	RHST[2:0]		
Value after reset:	x	x	x	x	0	0	0	0	0	0	0	0	x	0	0	0

Bit	Symbol	Function	R/W
2:0	RHST[2:0]	USB Bus Reset Status Flag 0 0 0: Communication speed indeterminate (powered state or no connection) 0 0 1: Host controller mode Low-speed connection 0 1 0: Host controller mode Full-speed connection Device controller mode USB bus reset in progress or full-speed connection 0 1 1: Host controller mode High-speed connection Device controller mode USB bus reset in progress or high-speed connection Others: Host controller mode USB bus reset in progress Device controller mode Setting prohibited	R
3	—	The read value is undefined. The write value should be 0.	R/W
4	UACT	USB Bus Operation Enable for the Host Controller Operation 0: Disable downstream port (disable SOF or micro-SOF transmission) 1: Enable downstream port (enable SOF or micro-SOF transmission)	R/W
5	RESUME	Resume Signal Output for the Host Controller Operation 0: Do not output resume signal 1: Output resume signal	R/W

Bit	Symbol	Function	R/W
6	USBRST	USB Bus Reset Output for the Host Controller Operation 0: Do not output USB bus reset signal 1: Output USB bus reset signal	R/W
7	RWUPE	Remote Wakeup Detection Enable for the Host Controller Operation 0: Disable downstream port remote wakeup 1: Enable downstream port remote wakeup	R/W
8	WKUP	Remote Wakeup Output for the Device Controller Operation 0: Do not output remote wakeup signal 1: Output remote wakeup signal	R/W
9	VBUSEN	USBHS_VBUSEN Output Pin Control 0: Output low on external USBHS_VBUSEN pin 1: Output high on external USBHS_VBUSEN pin	R/W
10	EXICEN	USBHS_EXICEN Output Pin Control 0: Output low on external USBHS_EXICEN pin 1: Output high on external USBHS_EXICEN pin	R/W
11	HNPBTOA	Host Negotiation Protocol (HNP) Control Use this bit when switching from device B to device A in OTG mode. If the HNPBTOA bit is 1, the internal function control remains in the Suspend state until the HNP processing ends even if SYSCFG.DPRPU = 0 or SYSCFG.DCFM = 1 is set.	R/W
15:12	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

RHST[2:0] flags (USB Bus Reset Status Flag)

The RHST[2:0] flags indicate the USB bus reset status.

In host controller mode, writing 1 to the USBRST bit causes the RHST[2:0] flags to set to 100b. When 0 is written to the USBRST bit and the USBHS ends the SE0 state, the RHST[2:0] flags update to a new value.

In device controller mode, if the USBHS detects a USB bus reset, the RHST[2:0] flags set to 010b if an attach event occurs while the DPRPU bit is 1, and a DVST interrupt is generated.

UACT bit (USB Bus Operation Enable for the Host Controller Operation)

When set to 1 in host controller mode, the UACT bit enables USB bus operation by controlling SOF packet transmission to the USB bus in addition to data and reception. The USBHS starts SOF packet output within one frame period after this bit is set to 1. If UACT is set to 0, the USBHS enters the idle state after the SOF packet output.

The USBHS sets the bit to 0 on any of the following conditions:

- A DTCH interrupt is detected during communication (while UACT = 1)
- An EOFERR interrupt is detected during communication (while UACT = 1)

Always write 1 to the UACT bit at the end of the USB bus reset processing (on a 0 write to the USBRST bit) or at the end of resume processing from the Suspend state (on a 0 write to the RESUME bit).

The USBHS clears the UACT bit to 0 if it receives an ACK response to an LPM token while the HL1CTRL1.L1REQ bit is set to 1. The USBHS sets the UACT bit to 1 when it finishes resume processing from the L1 state.

In device controller mode, always set this bit to 0.

RESUME bit (Resume Signal Output for the Host Controller Operation)

The RESUME bit controls the resume signal output in host controller mode. When this bit is set to 1, the USBHS drives the USB port to the K-state and outputs the resume signal. The USBHS sets the bit to 1 on detection of a remote wakeup signal while the RWUPE bit is 1 and in the USB suspend state. The USBHS continues outputting the K-state while the RESUME bit is 1, until the bit is cleared to 0 by software. The RESUME bit must be 1 (= resume period) for the time defined in the USB 2.0 specification. Only set this bit to 1 while the interface is in the Suspend state. Write 1 to the UACT bit simultaneously with the end of the resume processing (0 write to the RESUME bit).

Setting the RESUME bit to 1 during transition to the L1 state allows the USBHS to drive the USB port to the K-state and output the resume signal. The USBHS clears the RESUME bit to 0 at the end of the resume period, the value set in the HL1CTRL2.HIRD[3:0] bits.

Always set this bit to 0 in device controller mode.

USBRST bit (USB Bus Reset Output for the Host Controller Operation)

The USBRST bit controls the output of the USB bus signal in host controller mode. When this bit set to 1, the USBHS drives the USB port to the SE0 state to reset the USB bus. The USBHS continues outputting SE0 while the USBRST bit is 1, until the bit is cleared to 0 by software. The USBRST bit must be 1 (= USB bus reset period) for the time defined in the USB 2.0 specification. Writing 1 to the USBRST bit during communication (UACT bit = 1) or during resume processing (RESUME bit = 1) prevents the USBHS from starting USB bus reset processing until both the UACT and RESUME bits clear to 0. Write 1 to the UACT bit simultaneously with the end of the USB bus reset processing (0 write to the USBRST bit).

Always set this bit to 0 in device controller mode.

RWUPE bit (Remote Wakeup Detection Enable for the Host Controller Operation)

The RWUPE bit enables or disables remote wakeup signals (resume signals) from downstream peripheral devices in host controller mode. When this bit is set to 1, the USBHS detects a remote wakeup signal (K-state for 2.5 µs) from a downstream peripheral device, and it performs resume processing, driving the K-state. When the RWUPE bit is set to 0, the USBHS ignores remote wakeup signals (K-states) from peripheral devices connected to the USB port.

Do not stop the PHY clock while the RWUPE bit is 1, even in the Suspend state (the LPSTS.SUSPENDM bit must be set to 1). Also, do not reset the USB bus (setting USBRST to 1) from the Suspend state. This is prohibited in the USB 2.0 specification.

The RWUPE bit is also used to enable or disable detection of a remote wakeup signal during transition to the L1 state.

Always set this bit to 0 in device controller mode.

WKUP bit (Remote Wakeup Output for the Device Controller Operation)

The WKUP bit enables or disables remote wakeup signals (resume signals) to the USB bus in device controller mode.

The USBHS controls the output timing of the remote wakeup signals. When this bit is set to 1, the USBHS clears it to 0 after outputting the K-state for 10 ms. The USB 2.0 specification dictates that the USB bus idle state must be maintained for 5 ms or longer before a remote wakeup signal is sent. If the USBHS writes 1 to the WKUP bit immediately after detecting the Suspend state, the K-state is output after 2 ms.

Only write 1 to the WKUP bit when the device is in the Suspend state (the PL1CTRL1.DVSQ[3:0] flags are 01xxb) and the USB host enables the remote wakeup signal (RWUPE = 1). Do not stop the PHY clock while this bit is 1, even in the Suspend state (the LPSTS.SUSPENDM bit must be set to 1).

If the WKUP bit is set to 1 during transition to the L1 state, the USBHS outputs the K-state for 50 µs and then clears the bit to 0. Before writing 1 to the bit during the L1 state, check that the PL1CTRL1.DVSQ[3:0] flags are 10xxb.

Always set this bit to 0 in host controller mode.

HNPBTOA bit (Host Negotiation Protocol (HNP) Control)

The HNPBTOA bit is used when switching from device B to device A while in OTG mode.

If the HNPBTOA bit is 1, the internal function control maintains the Suspend state until HNP processing ends, even if the SYSCFG.DPRPU bit is set to 0 or the SYSCFG.DCFM bit is set to 1. Resume interrupts (RESM) are not generated even if a falling edge of D+ is detected.

The HNP processing ends when a host attach event is detected, because of a pull-up by the initiating party, or the HNPBTOA bit is cleared to 0 by software because the HNP processing times out.

38.2.6 TESTMODE : USB Test Mode Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x00C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3			0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	UTST[3:0]			
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

Bit	Symbol	Function	R/W
3:0	UTST[3:0]	Test Mode These bits output the USB test signals. See Table 38.5	R/W
15:4	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

UTST[3:0] bits (Test Mode)

Writing values to the UTST[3:0] bits allows the USBHS to output USB test signals in high-speed operation mode. Table 38.5 shows the test mode operation settings.

Table 38.5 Test mode operation settings

Test mode	UTST[3:0] bit setting	
	In device controller mode	In host controller mode
Normal operation	0x0	0x0
Test_J	0x1	0x9
Test_K	0x2	0xA
Test_SE0_NAK	0x3	0xB
Test_Packet	0x4	0xC
Test_Force_Enable	—	0xD
Reserved	0x5 to 0x7	0xE to 0xF

Host controller mode

In host controller mode, these bits can be set after setting the SYSCFG.DRPD bit to 1. After the UTST[3:0] bits are set, the USBHS outputs waveforms to the USB port by setting the DVSTCTR0.UACT bit to 1. The USBHS also performs high-speed termination for the USB port by setting these bits in host controller mode.

To set the UTST[3:0] bits in host controller mode:

1. Reset the hardware.
2. Start supplying the PHY clock, and then set the LPSTS.SUSPENDM bit to 1.
3. Set the SYSCFG.DCFM and SYSCFG.DRPD bits to 1. (Setting the SYSCFG.HSE bit to 1 is not required.)
4. Set the SYSCFG.USBE bit to 1.
5. Set the UTST[3:0] bits based on the test requirements.
6. Set the DVSTCTR0.UACT bit to 1.

Assuming the initial steps (1) to (6) are already complete, to change the UTST[3:0] bits in host controller mode:

1. Set the DVSTCTR0.UACT and SYSCFG.USBE bits to 0.
2. Set the SYSCFG.USBE bit to 1.
3. Set the UTST[3:0] bits based on the test requirements.
4. Set the DVSTCTR0.UACT bit to 1.

When the UTST[3:0] bits are set to 1011b (Test_SE0_NAK), the USBHS does not output SOF packets to ports for which the DVSTCTR0.UACT bit is set to 1.

When the UTST[3:0] bits are set to 1101b (Test_Force_Enable), the USBHS outputs SOF packets to ports for which the DVSTCTR0.UACT bit is set to 1. In this test mode, the USBHS does not control the hardware related to attach detection, even if it detects a high-speed detach event (DTCH interrupt).

Before setting the UTST[3:0] bits, set the PID[1:0] bits of all pipe control registers to 00b (NAK response). To return to normal USB communication after setting a test mode, issue a hardware reset.

Device controller mode

In device controller mode, set these bits using a SetFeature request from the USB host during high-speed communication. The USBHS does not enter the Suspend state while these bits are 0001b to 0100b. To return to normal USB communication after setting a test mode, issue a hardware reset.

38.2.7 CFIFO, DnFIFO : FIFO Port Register (n = 0, 1)

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x014 (CFIFO/CFIFOL/CFIFOLL)
0x016 (CFIFOH)
0x017 (CFIFOHH)
0x018 + 0x04 × n (DnFIFO/DnFIFOL/DnFIFOLL)
0x01A + 0x04 × n (DnFIFOH)
0x01B + 0x04 × n (DnFIFOHH)

Bit position:	31	...	0
Bit field:	FIFOPORT[31:0]		
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0		

Bit	Symbol	Function	R/W
31:0	FIFOPORT[31:0]*1	Read receive data from the FIFO buffer or write transmit data to the FIFO buffer by accessing these bits.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. The valid bits depend on the MBW[1:0] and BIGEND settings in the associated port selection register.

Three FIFO ports are provided:

- CFIFO
- D0FIFO
- D1FIFO

Each FIFO port is configured with:

- A port register (CFIFO, D0FIFO, or D1FIFO) that handles reading of data from the FIFO buffer and writing of data to the FIFO buffer
- A port selection register (CFIFOSEL, D0FIFOSEL, or D1FIFOSEL) that selects the pipe assigned to the FIFO port
- A port control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR)

Each FIFO port has the following constraints:

- Access to the FIFO buffer for DCP control transfers is through the CFIFO port
- Access to the FIFO buffer for DMAC or DTC transfers is through the D0FIFO or D1FIFO port
- The D0FIFO and D1FIFO ports can also be accessed by the CPU
- When using functions specific to the FIFO port, such as the DMAC or DTC transfer function, you cannot change the pipe number selected in the CURPIPE[3:0] bits of the Port Selection Register
- Registers configuring one FIFO port do not affect other FIFO ports
- The same pipe must not be assigned to two or more FIFO ports

- There are two FIFO buffer states, one giving access rights to the CPU and the other to the serial interface engine (SIE). When the SIE has access rights, the FIFO buffer cannot be accessed by the CPU

FIFOPORT bits (FIFOPORT[31:0])

When the FIFOPORT bits are accessed, the USBHS reads the received data from the FIFO buffer or writes the transmission data to the FIFO buffer. The FIFO port register can be accessed only when the FRDY flag in the associated port control register (CFIFOCTR, D0FIFOCTR, or D1FIFOCTR) is 1.

The valid bits in the FIFO port register depend on the MBW[1:0] and BIGEND settings in the port selection register (CFIFOSEL, D0FIFOSEL, or D1FIFOSEL). See Table 38.6 to Table 38.8.

Table 38.6 Endian operation in 32-bit access (MBW[1:0] = 10b)

BIGEND	CFIFO, D0FIFO, D1FIFO b31 to b24	CFIFO, D0FIFO, D1FIFO b23 to b16	CFIFO, D0FIFO, D1FIFO b15 to b8	CFIFO, D0FIFO, D1FIFO b7 to b0	Remarks
0	Located at N+3	Located at N+2	Located at N+1	Located at N+0	Transmit data is sent from the address N+0. Receive data is stored from the address N+0.
1	Located at N+0	Located at N+1	Located at N+2	Located at N+3	Transmission data is sent from the address N+3. Receive data is stored from the address N+3.

Table 38.7 Endian operation in 16-bit access (MBW[1:0] = 01b)

BIGEND	CFIFOL, D0FIFOL, D1FIFOL b15 to b8	CFIFOL, D0FIFOL, D1FIFOL b7 to b0	CFIFOH, D0FIFOH, D1FIFOH b15 to b8	CFIFOH, D0FIFOH, D1FIFOH b7 to b0	Remarks
0	Access prohibited[*1]		Located at N+1	Located at N+0	Transmit data is sent from the address N+0. Receive data is stored from the address N+0.
1	Located at N+0	Located at N+1	Access prohibited[*1]		Transmit data is sent from the address N+1. Receive data is stored from the address N+1.

Note 1. Writing to or reading from these areas is prohibited.

Table 38.8 Endian operation in 8-bit access (MBW[1:0] = 00b)

BIGEND	CFIFOLL, D1FIFOLL, D0FIFOLL	CFIFOHH, D1FIFOHH, D0FIFOHH
0	Access prohibited[*1]	Located at N+0
1	Located at N+0	Access prohibited[*1]

Note 1. Writing to or reading from these locations is prohibited.

38.2.8 CFIFOSEL : CFIFO Port Selection Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x020

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RCNT	REW	—	—	MBW[1:0]		—	BIGEND	—	—	ISEL	—	CURPIPE[3:0]			
Value after reset:	0	0	x	x	0	0	x	0	x	x	0	x	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CURPIPE[3:0]	FIFO Port Access Pipe Specification 0x0: DCP (default control pipe) 0x1: Pipe 1 0x2: Pipe 2 0x3: Pipe 3 0x4: Pipe 4 0x5: Pipe 5 0x6: Pipe 6 0x7: Pipe 7 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W
4	—	The read value is undefined. The write value should be 0.	R/W
5	ISEL	FIFO Port Access Direction when DCP Is Selected 0: Select reading from the FIFO buffer 1: Select writing to the FIFO buffer	R/W
7:6	—	The read values are undefined. The write value should be 0.	R/W
8	BIGEND	FIFO Port Endian Control 0: Little endian 1: Big endian	R/W
9	—	The read value is undefined. The write value should be 0.	R/W
11:10	MBW[1:0]	CFIFO Port Access Bit Width 0 0: 8-bit width 0 1: 16-bit width 1 0: 32-bit width 1 1: Setting prohibited	R/W
13:12	—	The read values are undefined. The write value should be 0.	R/W
14	REW	Buffer Pointer Rewind 0: Do not rewind buffer pointer (Writing 0 has no effect.) 1: Rewind buffer pointer	W
15	RCNT	Read Count Mode 0: Clear DTLN[11:0] flags in the FIFO port control register to 0x000 when all receive data is read from CFIFO 1: Decrement DTLN[11:0] flags each time receive data is read from CFIFO	R/W

Note: S-TYPE-3, P-TYPE-3

Do not specify the same pipe number in the CURPIPE[3:0] bits in the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers.

Do not change the pipe number while DMAC or DTC transfer is enabled.

CURPIPE[3:0] bits (FIFO Port Access Pipe Specification)

The CURPIPE[3:0] bits specify the pipe number used to read or write data through the CFIFO port. After writing to these bits, read them to check that the written value agrees with the read value before proceeding to the next process. Do not set the same pipe number to the CURPIPE[3:0] bits in CFIFOSEL, D0FIFOSEL, and D1FIFOSEL.

During FIFO buffer access, the pipe specification is maintained until the access is complete, even if the software attempts to change the CURPIPE[3:0] setting. Access continues after the current value is written back to the CURPIPE[3:0] bits.

ISEL bit (FIFO Port Access Direction when DCP Is Selected)

After writing a new value to the ISEL bit while the DCP is the selected pipe, read this bit to check that the written value agrees with the read value before proceeding to the next process. Set the ISEL and CURPIPE[3:0] bits simultaneously.

BIGEND bit (FIFO Port Endian Control)

Use the BIGEND bit to set the byte endian order of the CFIFO port.

MBW[1:0] bits (CFIFO Port Access Bit Width)

The MBW[1:0] bits specify the bit width for accessing the CFIFO port.

When the selected pipe is receiving, after a write to these bits starts a data read from the FIFO buffer, do not change the bits until all of the data is read. When reading the FIFO buffer, read with the access size set in MBW.

When the selected pipe is transmitting, set the CURPIPE[3:0] and MBW[1:0] bits simultaneously. The bit width cannot be changed from 8-bit to 16- or 32-bit, or from 16-bit to 32-bit while data is being written to the FIFO buffer.

An odd number of bytes can also be written through byte-access control even when 16- or 32-bit width is selected.

REW bit (Buffer Pointer Rewind)

The REW bit specifies whether or not to rewind the buffer pointer.

When the selected pipe is receiving, setting this bit to 1 while the FIFO buffer is being read allows re-reading of the FIFO buffer from the first data. In double-buffering when reading is already in progress, this setting enables reading either FIFO buffer from the first entry.

Do not set this bit to 1 while simultaneously changing the CURPIPE[3:0] bits. Before setting the bit to 1, always check that the FRDY flag is 1.

To rewrite to the FIFO buffer from the first data for the transmitting pipe, use the BCLR bit.

RCNT bit (Read Count Mode)

When the RCNT bit set to 0, the USBHS clears the CFIFOCTR.DTLN[11:0] flags to 0 on finishing reading all of the received data in the FIFO buffer assigned to the pipe specified in the CURPIPE[3:0] bits, or after reading a single plane in double buffer mode.

With this bit set to 1, the USBHS decrements the value in the CFIFOCTR.DTLN[11:0] flags each time it reads data received from the FIFO buffer assigned to the pipe specified in the CURPIPE[3:0] bits.

38.2.9 DnFIFOSEL : DnFIFO Port Selection Register (n = 0, 1)

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x028 + 0x4 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RCNT	REW	DCLRM	DREQE	MBW[1:0]		—	BIGEND	—	—	—	—	CURPIPE[3:0]			
Value after reset:	0	0	0	0	0	0	x	0	x	x	x	x	0	0	0	0

Bit	Symbol	Function	R/W
3:0	CURPIPE[3:0]	FIFO Port Access Pipe Specification 0x0: No pipe specification 0x1: Pipe 1 0x2: Pipe 2 0x3: Pipe 3 0x4: Pipe 4 0x5: Pipe 5 0x6: Pipe 6 0x7: Pipe 7 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W

Bit	Symbol	Function	R/W
7:4	—	The read values are undefined. The write value should be 0.	R/W
8	BIGEND	FIFO Port Endian Control 0: Little endian 1: Big endian	R/W
9	—	The read value is undefined. The write value should be 0.	R/W
11:10	MBW[1:0]	FIFO Port Access Bit Width 0 0: 8-bit width 0 1: 16-bit width 1 0: 32-bit width 1 1: Setting prohibited	R/W
12	DREQE	DMAC/DTC Transfer Request Enable 0: Disable DMAC/DTC transfer request 1: Enable DMAC/DTC transfer request	R/W
13	DCLRM	Auto FIFO Buffer Clear Mode after Specified Pipe is Read 0: Disable auto buffer clear mode 1: Enable auto buffer clear mode	R/W
14	REW	Buffer Pointer Rewind 0: Do not rewind buffer pointer (writing 0 has no effect) 1: Rewind buffer pointer	W
15	RCNT	Read Count Mode 0: Clear DTLN[11:0] flags in the FIFO port control register to 0x000 when all receive data is read from DnFIFO (after read of a single plane in double buffer mode) 1: Decrement DTLN[11:0] flags each time receive data is read from DnFIFO	R/W

Note: S-TYPE-3, P-TYPE-3

Do not specify the same pipe number in the CURPIPE[3:0] bits in the CFIFOSEL, D0FIFOSEL, and D1FIFOSEL registers. When the CURPIPE[3:0] bits in the D0FIFOSEL and D1FIFOSEL registers are set to 0000b, no pipe is selected.

Do not change the pipe number while DMAC or DTC transfer is enabled.

CURPIPE[3:0] bits (FIFO Port Access Pipe Specification)

The CURPIPE[3:0] bits specify the pipe number used to read or write data through the DnFIFO port. After writing to these bits, read them to check that the written value agrees with the read value before proceeding to the next process. Do not set the same pipe number to the CURPIPE[3:0] bits in CFIFOSEL, D0FIFOSEL, and D1FIFOSEL.

During FIFO buffer access, the pipe specification is maintained until the access is complete, even if the software attempts to change the CURPIPE[3:0] setting. Access continues after the current value is written back to the CURPIPE[3:0] bits.

BIGEND bit (FIFO Port Endian Control)

Use the BIGEND bit to set the byte endian order of the D0FIFO or D1FIFO port.

MBW[1:0] bits (FIFO Port Access Bit Width)

The MBW[1:0] bits specify the bit width for accessing the DnFIFO port.

When the selected pipe is receiving, after a write to these bits starts a data read from the FIFO buffer, do not change the bits until all of the data is read. When reading the FIFO buffer, read with the access size set in MBW.

When the selected pipe is transmitting, set the CURPIPE[3:0] and MBW[1:0] bits simultaneously. The bit width cannot be changed from 8-bit to 16- or 32-bit, or from 16-bit to 32-bit while data is being written to the FIFO buffer.

An odd number of bytes can also be written through byte-access control even when 16- or 32-bit width is selected.

DREQE bit (DMAC/DTC Transfer Request Enable)

The DREQE bit enables or disables issuing of DMAC or DTC transfer requests. Only change the settings of DREQE bit when the CURPIPE[3:0] bits are 0x0.

To enable DMAC or DTC transfer requests, set this bit to 1 after setting the CURPIPE[3:0] bits to 0x0, and then set the CURPIPE[3:0] bits to the PIPE number for the transfer.

DCLRM bit (Auto FIFO Buffer Clear Mode after Specified Pipe is Read)

The DCLRM bit enables or disables automatic FIFO buffer clearing after data in the selected pipe is read.

When this bit is set to 1, on receiving a zero-length packet while the FIFO buffer assigned to the selected pipe is empty, or when reading of a received short packet is complete while the PIPECFG.BFRE bit is 1, the USBHS sets the BCLR bit in the FIFO port control register to 1.

When using the USBHS with the SOFCFG.BRDYM bit set to 1, set this bit to 0.

REW bit (Buffer Pointer Rewind)

The REW bit specifies whether or not to rewind the buffer pointer.

When the selected pipe is receiving, setting this bit to 1 while the FIFO buffer is being read allows re-reading of the FIFO buffer from the first data. In double-buffering when reading is already in progress, this setting enables reading either FIFO buffer from the first entry.

Do not set this bit to 1 while simultaneously changing the CURPIPE[3:0] bits. Before setting the bit to 1, always check that the FRDY flag is 1.

To rewrite to the FIFO buffer from the first data for the transmitting pipe, use the BCLR bit.

RCNT bit (Read Count Mode)

When the RCNT bit set to 0, the USBHS clears the DnFIFOCTR.DTLN[11:0] flags (n = 0, 1) to 0 on finishing reading all of the received data in the FIFO buffer assigned to the pipe specified in the CURPIPE[3:0] bits, or after reading a single plane in double buffer mode.

With this bit set to 1, the USBHS decrements the value in the CFIFOCTR.DTLN[11:0] flags each time it reads data received from the FIFO buffer assigned to the pipe specified in the CURPIPE[3:0] bits. When accessing DnFIFO with the PIPECFG.BFRE bit set to 1, set the RCNT bit to 0.

38.2.10 CFIFOCTR, DnFIFOCTR : FIFO Port Control Register (n = 0, 1)

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x022 (CFIFOCTR)
0x02A + 0x4 × n (DnFIFOCTR)

Bit position:	15	14	13	12	11											0
Bit field:	BVAL	BCLR	FRDY	—	DTLN[11:0]											
Value after reset:	0	0	0	x	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	DTLN[11:0]	Receive Data Length Flag The meaning of the values differs depending on the RCNT bit setting in the port selection register. For details, see the description of the DTLN[11:0] bits.	R
12	—	The read value is undefined. The write value should be 0.	R/W
13	FRDY	FIFO Port Ready Flag 0: FIFO port access disabled 1: FIFO port access enabled	R
14	BCLR	CPU Buffer Clear 0: No operation (writing 0 has no effect) 1: Clear FIFO buffer on the CPU side	W
15	BVAL	FIFO Buffer Valid Flag Set this bit to 1 when data is completely written to the FIFO buffer on the CPU side for the selected pipe (CURPIPE[3:0] setting). 0: Invalid (writing 0 has no effect) 1: Writing ended	R/W

Note: S-TYPE-3, P-TYPE-3

The CFIFOCTR, D0FIFOCTR, and D1FIFOCTR registers correspond to the CFIFO, D0FIFO, and D1FIFO buffers.

DTLN[11:0] flags (Receive Data Length Flag)

The DTLN[11:0] flags indicate the length of the receive data.

While the FIFO buffer is being read, the DTLN[11:0] bits indicate different values depending on the DnFIFOSEL.RCNT bit (n = 0, 1), as follows:

- RCNT = 0:
  The USBHS sets the DTLN[11:0] flags to indicate the length of the receive data until the CPU or DMAC/DTC has read all of the received data in the FIFO buffer (or until it has read a single plane in double buffer mode).
  While the PIPECFG.BFRE bit is 1, the USBHS retains the length of the receive data until the BCLR bit is set to 1, even after all the data is read.
- RCNT = 1:
  The USBHS decrements the value indicated in the DTLN[11:0] flags each time the CPU or DMAC/DTC reads the receive data from the FIFO buffer. (The value is decremented by 1 when MBW[1:0] = 00b, by 2 when MBW[1:0] = 01b, and by 4 when MBW[1:0] = 10b.)
  The USBHS sets these flags to 0 when all the data is read from the FIFO buffer. In double buffer mode, if data is received in one FIFO buffer plane before all of the data is read from the other plane, the USBHS sets these bits to indicate the length of the receive data in the latter plane when all of the data is read from the former plane.
  When the RCNT bit is 1, reading the DTLN[11:0] flags while the FIFO buffer is being read returns the latest value within 150 ns after the FIFO port read cycle.

FRDY flag (FIFO Port Ready Flag)

The FRDY flag indicates whether the FIFO port can be accessed by the CPU or DMAC/DTC.

In the following cases, the USBHS sets the FRDY flag to 1 but data cannot be read through the FIFO port because there is no data to be read:

- A zero-length packet is received when the FIFO buffer assigned to the selected pipe is empty
- A short packet is received and the data is completely read while the PIPECFG.BFRE bit is 1

In these cases, set the BCLR bit to 1 to clear the FIFO buffer, and enable transmission and reception of the next data.

BCLR bit (CPU Buffer Clear)

Set the BCLR bit to 1 to clear the FIFO buffer on the CPU for the selected pipe.

When double buffer mode is set for the FIFO buffer assigned to the selected pipe, the USBHS clears only one plane of the FIFO buffer even when both planes are read-enabled.

When the DCP is the selected pipe, setting the BCLR bit to 1 allows the USBHS to clear both sets of FIFO buffers regardless of whether the CPU or SIE has access rights. To clear the buffer when the SIE has access rights, set the DCPCTR.PID[1:0] bits to 00b (NAK response) before setting the BCLR bit to 1.

When the selected pipe is not the DCP, only write 1 to the BCLR bit while the FRDY flag in the FIFO port control register is 1 (set by the USBHS).

BVAL bit (FIFO Buffer Valid Flag)

Set the BVAL bit to 1 when data is completely written to the FIFO buffer on the CPU for the pipe selected in CURPIPE[3:0].

When the selected pipe is transmitting, set this bit to 1 in the following cases:

- To transmit a short packet, set this bit to 1 after data is written
- To transmit a zero-length packet, set this bit to 1 before data is written to the FIFO buffer
- Set this bit to 1 after the specified number of data bytes is written for the pipe in continuous transfer mode, where the number is a natural integer multiple of the maximum packet size and less than the buffer size

The USBHS then switches the FIFO buffer from the CPU side to the SIE side, enabling transmission.

When the selected pipe is in use for transmission, simultaneously setting the BVAL flag and the BCLR bit to 1 causes the USBHS to clear the data that is already written and enables transmission of a zero-length packet. When data of the maximum packet size is written for the pipe in non-continuous transfer mode, the USBHS sets this bit to 1 and switches the FIFO buffer from the CPU side to the SIE side, enabling transmission.

Only write 1 to the BVAL flag while the FRDY bit is 1 (set by the USBHS). When the selected pipe is receiving, do not set the BVAL flag to 1.

38.2.11 INTENB0 : Interrupt Enable Register 0

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x030

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	VBSE	RSME	SOFE	DVSE	CTRE	BEMPE	NRDYE	BRDYE	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
7:0	—	The read values are undefined. The write value should be 0.	R/W
8	BRDYE	Buffer Ready Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
9	NRDYE	Buffer Not Ready Response Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
10	BEMPE	Buffer Empty Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
11	CTRE	Control Transfer Stage Transition Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
12	DVSE	Device State Transition Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
13	SOFE	Frame Number Update Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
14	RSME	Resume Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
15	VBSE	VBUS Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W

Note: S-TYPE-3, P-TYPE-3

Note: The RSME, DVSE, and CTRE bits can only be set to 1 in device controller mode. Do not set these bits to 1 in host controller mode.

When a status flag in the INTSTS0 register sets to 1 and the associated interrupt request enable bit setting in the INTENB0 register is 1, the USBHS issues a USBHS interrupt request.

Regardless of the INTENB0 register setting, the status flag in the INTSTS0 register sets to 1 in response to a state change that satisfies the associated condition.

When an interrupt request enable bit in the INTENB0 register is switched from 0 to 1 while the associated status flag in the INTSTS0 register is set to 1, a USBHS interrupt is requested.

38.2.12 INTENB1 : Interrupt Enable Register 1

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x032

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVRCRE	BCHGE	—	DTCHE	ATTCHE	—	L1RSMENDE	LPMENDE	—	EOFERRE	SIGNE	SACKE	—	—	—	PDDETINTE
Value after reset:	0	0	x	0	0	x	0	0	x	0	0	0	x	x	x	0

Bit	Symbol	Function	R/W
0	PDDETINTE	PDDETINT Detection Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
3:1	—	The read values are undefined. The write value should be 0.	R/W
4	SACKE	Setup Transaction Normal Response Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
5	SIGNE	Setup Transaction Error Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
6	EOFERRE	EOF Error Detection Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
7	—	The read values are undefined. The write value should be 0.	R/W
8	LPMENDE	LPM Transaction End Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
9	L1RSMENDE	L1 Resume End Interrupt Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
10	—	The read values are undefined. The write value should be 0.	R/W
11	ATTCHE	Connection Detection Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
12	DTCHE	Disconnection Detection Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
13	—	The read values are undefined. The write value should be 0.	R/W
14	BCHGE	USB Bus Change Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W
15	OVRCRE	OVRCRE Interrupt Request Enable 0: Disable interrupt request 1: Enable interrupt request	R/W

Note: S-TYPE-3, P-TYPE-3

When a status flag in the INTSTS1 register sets to 1 and the associated interrupt request enable bit setting in the INTENB1 register is 1, the USBHS issues a USBHS interrupt request.

Regardless of the INTENB1 register setting, the status flag in the INTSTS1 register sets to 1 in response to a state change that satisfies the associated condition.

When an interrupt request enable bit in the INTENB1 register is switched from 0 to 1 while the associated status flag in the INTSTS1 register is set to 1, a USBHS interrupt is requested.

38.2.13 BRDYENB : BRDY Interrupt Enable Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x036

Bit position:	15	14	13	12	11	10	9									0
Bit field:	—	—	—	—	—	—	PIPEBRDYE[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPEBRDYE[9:0]	BRDY Interrupt Request Enable for Pipes [9:0]*1 0: Disable interrupt request 1: Enable interrupt request	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Each bit number corresponds to the same pipe number.

The BRDYENB register enables or disables the INTSTS0.BRDY bit to be set to 1 when a BRDY interrupt is detected for each pipe.

When a status flag in the BRDYSTS register sets to 1 and the associated PIPEBRDYEn (n = 9 to 0) bit setting in the BRDYENB register is 1, the INTSTS0.BRDY flag sets to 1. In this case, if the BRDYE bit in INTENB0 is 1, the USBHS generates a BRDY interrupt request. While at least one PIPEBRDYEn flag indicates 1, the INTSTS0.BRDY flag sets to 1 when the associated interrupt request enable bit in the BRDYENB register is changed from 0 to 1 by software.

38.2.14 NRDYENB : NRDY Interrupt Enable Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x038

Bit position:	15	14	13	12	11	10	9									0
Bit field:	—	—	—	—	—	—	PIPENRDYE[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPENRDYE[9:0]	NRDY Interrupt Enable for Pipes [9:0]*1 0: Disable interrupt request 1: Enable interrupt request	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Each bit number corresponds to the same pipe number.

The NRDYENB register enables or disables the INTSTS0.NRDY bit to be set to 1 when a NRDY interrupt is detected for each pipe.

When a status flag in the NRDYSTS register sets to 1 and the associated PIPENRDYEn (n = 0 to 9) bit setting in the NRDYENB register is 1, the INTSTS0.NRDY flag sets to 1. In this case, if the NRDYE bit in INTENB0 is 1, the USBHS generates a NRDY interrupt request. While at least one PIPEBRDYEn flag indicates 1, the INTSTS0.NRDY flag sets to 1 when the associated interrupt request enable bit in the NRDYENB register is changed from 0 to 1 by software.

38.2.15 BEMPENB : BEMP Interrupt Enable Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x03A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PIPEBEMPE[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPEBEMPE[9:0]	BEMP Interrupt Enable for Pipes [9:0]*1 0: Disable interrupt request 1: Enable interrupt request	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Each bit number corresponds to the same pipe number.

The BEMPENB register enables or disables the INTSTS0.BEMP bit to be set to 1 when a BEMP interrupt is detected for each pipe.

When a status flag in the BEMPSTS register sets to 1 and the associated PIPEBEMPEn (n = 0 to 9) bit setting in the BEMPENB register is 1, the INTSTS0.BEMP flag sets to 1. In this case, if the BEMPE bit in INTENB0 is 1, the USBHS generates a BEMP interrupt request. While at least one PIPEBEMPEn flag indicates 1, the INTSTS0.BEMP flag sets to 1 when the associated interrupt request enable bit in the BEMPENB register is changed from 0 to 1 by software.

38.2.16 SOFCFG : SOF Output Configuration Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x03C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	TRNENSEL	—	BRDYM	INTL	EDGESTS	—	—	—	—
Value after reset:	x	x	x	x	x	x	x	0	x	0	0	0	x	x	x	x

Bit	Symbol	Function	R/W
3:0	—	The read values are undefined. The write value should be 0.	R/W
4	EDGESTS	Interrupt Edge Processing Status Flag*1 Indicates 1 during the edge processing of an edge interrupt output signal.	R
5	INTL	Interrupt Output Sense Select*2 0: Edge detection 1: Level detection	R/W
6	BRDYM	PIPEBRDY Interrupt Status Clear Timing*3 0: Clear BRDY flag through software 1: Clear BRDY flag by the USBHS through a data read from the FIFO buffer or data write to the FIFO buffer	R/W
7	—	The read values are undefined. The write value should be 0.	R/W
8	TRNENSEL	Transaction-Enabled Time Select*4 0: Not low-speed communication 1: Low-speed communication	R/W
15:9	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Confirm that the EDGESTS flag is 0 before stopping the clock supply to the USBHS.

Note 2. When the INTL bit is set to 0, to stop the PHY clock (LPSTS.SUSPENDM = 0) after clearing the interrupt status, write 0 to the LPSTS.SUSPENDM bit after confirming that the EDGESTS flag is cleared to 0.
Note 3. When setting the BRDYM bit to 1, set the INTL bit to 1.
Note 4. The setting in the TRNENSEL bit is only valid in host controller mode. Even in host controller mode, the setting of this bit has no effect on the transaction-enabled time during high-speed communication.

EDGESTS flag (Interrupt Edge Processing Status Flag)

The EDGESTS flag indicates 1 during the edge processing of an edge interrupt output signal. Confirm that this flag is 0 before stopping the PHY clock.

BRDYM bit (PIPEBRDY Interrupt Status Clear Timing)

The BRDYM bit specifies how the BRDY interrupt status flags for the pipes are cleared.

TRNENSEL bit (Transaction-Enabled Time Select)

When the USB port is in use for full-speed or low-speed communications, the TRNENSEL bit specifies the timing with which the USBHS issues tokens in a frame (transaction-enabled time).

Set this bit to 1 when a low-speed device is connected through a hub. The bit is only valid in host controller mode. Set this bit to 0 when the interface is in use as a device controller.

38.2.17 PHYSET : PHY Setting Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x03E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	HSEB	—	—	—	REPSTART	—	REPSEL[1:0]		—	—	CLKSEL[1:0]		CDPEN	—	PLLRESET	DIRPD
Value after reset:	x	x	x	x	0	x	0	0	x	x	1	1	0	x	1	1

Bit	Symbol	Function	R/W
0	DIRPD	Power-Down Control 0: Do not enter low power mode 1: Enter low power mode	R/W
1	PLLRESET	PLL Reset Control*1 0: Disable PLL reset control for UTMI_PHY 1: Enable PLL reset control for UTMI_PHY	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	CDPEN	Charging Downstream Port Enable 0: Disable downstream port charging 1: Enable downstream port charging	R/W
5:4	CLKSEL[1:0]	Input Clock Frequency Select 0 0: 12 MHz 0 1: 48 MHz 1 0: 20 MHz 1 1: 24 MHz	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
9:8	REPSEL[1:0]	Terminating Resistance Adjustment Cycle 0 0: No cycle is set 0 1: Adjust terminating resistance at 16-second intervals 1 0: Adjust terminating resistance at 64-second intervals 1 1: Adjust terminating resistance at 128-second intervals	R/W
10	—	This bit is read as 0. The write value should be 0.	R/W
11	REPSTART	Forcibly Start Terminating Resistance Adjustment 0: Force terminating resistance adjustment to start 1: Do not force terminating resistance adjustment to start	R/W
14:12	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
15	HSEB	CL-only mode 0: Disable CL-only mode 1: Enable CL-only mode	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Because the value of the PLLRESET bit is 1 after a reset, changing the setting after release from the reset state is not required. Do not set the PLLRESET bit to 1 after setting the PLLRESET bit to 0. Operation is not guaranteed.

CLKSEL[1:0] bits (Input Clock Frequency Select)

The CLKSEL[1:0] bits select the transfer clock source for the USBHS.

For the transfer clock generated in the USB-PHY internal PLL, these bits set the input clock frequency. To input the clock source from the EXTAL pin, the USB 2.0 clock specification must be strictly followed.

Writing to the CKSEL[1:0] bits is invalid in CL-only mode because the internal PLL is stopped (see the description for HSEB bit (CL-only mode)). For the clock settings, see section 38.3.3. Supplying the Clock.

HSEB bit (CL-only mode)

The HSEB bit selects whether the USBHS operates in CL-only mode. High-speed transfer by the USBHS requires the use of internal high-speed analog circuits including the PLL, clock, and data recovery (CDR) circuit in the USB-PHY block.

CL-only mode limits the transfer to the USB 1.1 specification (full-speed and low-speed transfer only). Power consumption can be reduced by stopping the internal PLL of the PHY module and other high-speed analog circuits.

In CL-only mode, the USBHS requires supply clocks of 48 MHz and 60 MHz, generated in the Clock Generation Circuit. For the clock supply method, see section 9, Clock Generation Circuit.

38.2.18 INTSTS0 : Interrupt Status Register 0

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x040

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	VBINT	RESM	SOFR	DVST	CTRT	BEMP	NRDY	BRDY	VBSTS	DVSQ[2:0]			VALID	CTSQ[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	x	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CTSQ[2:0]	Control Transfer Stage Flag*1 0 0 0: Idle or setup stage 0 0 1: Control read data stage 0 1 0: Control read status stage 0 1 1: Control write data stage 1 0 0: Control write status stage 1 0 1: Control write (no data) status stage 1 1 0: Control transfer sequence error	R
3	VALID	USB Request Reception Flag*1 0: Setup packet not received 1: Setup packet received	R/W*3
6:4	DVSQ[2:0]	Device State*1 Indicates the device state. 0 0 0: Powered state 0 0 1: Default state 0 1 0: Address state 0 1 1: Configured state Others: Suspend state	R
7	VBSTS	VBUS Input Status Flag 0: USBHS_VBUS pin is low 1: USBHS_VBUS pin is high	R

Bit	Symbol	Function	R/W
8	BRDY	BRDY Interrupt Status Flag 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R
9	NRDY	NRDY Interrupt Status Flag 0: No NRDY interrupt occurred 1: NRDY interrupt occurred	R
10	BEMP	BEMP Interrupt Status Flag 0: No BEMP interrupt occurred 1: BEMP interrupt occurred	R
11	CTRT	Control Transfer Stage Transition Interrupt Status Flag*2 0: No control transfer stage transition interrupt occurred 1: Control transfer stage transition interrupt occurred	R/W*3
12	DVST	Device State Transition Interrupt Status Flag*2 0: No device state transition interrupt occurred 1: Device state transition interrupt occurred	R/W*3
13	SOFR	Frame Number Refresh Interrupt Status Flag 0: No SOF interrupt occurred 1: SOF interrupt occurred	R/W*3
14	RESM	Resume Interrupt Status Flag*2 *4 0: No resume interrupt occurred 1: Resume interrupt occurred	R/W*3
15	VBINT	VBUS Interrupt Status Flag*4 0: No VBUS interrupt occurred on detecting a change in the USBHS_VBUS pin 1: VBUS interrupt occurred on detecting a change in the USBHS_VBUS pin	R/W*3

Note: S-TYPE-3, P-TYPE-3

Note 1. The CTSQ[2:0], VALID, and DVSQ[2:0] flags are only valid in device controller mode.

Note 2. The status of the CTRT, DVST, and RESM flags are changed only in device controller mode. Set the associated interrupt enable bits to 0 (disabled) in host controller mode.

Note 3. To clear the CTRT, DVST, SOFR, RESM, or VBINT flags, write 0 only to the flags to be cleared. Write 1 to the other flags. Do not write 0 to the status flags indicating 0.

Note 4. The USBHS detects a change in the status in the RESM or VBINT flag even while the clock supply is stopped (LPSTS.SUSPENDM = 0), and it requests the interrupt when the associated interrupt request bit is 1. Enable the clock supply before clearing the status by software.

BRDY flag (BRDY Interrupt Status Flag)

The BRDY flag indicates the BRDY interrupt state. For the conditions that cause the flag to be set, see section 38.2.13. BRDYENB : BRDY Interrupt Enable Register.

The USBHS clears the BRDY flag to 0 when 0 is written to the BRDYSTS.PIPEBRDYn (n = 0 to 9) flags for all pipes for which the BRDY interrupt is enabled (BRDYENB.PIPEBRDYEn bits). Writing 0 to the BRDY flag in the software does not clear the flag.

NRDY flag (NRDY Interrupt Status Flag)

The NRDY flag indicates the NRDY interrupt state. For the conditions that cause the flag to be set, see section 38.2.14. NRDYENB : NRDY Interrupt Enable Register.

The USBHS clears the NRDY flag to 0 when 0 is written to the NRDYSTS.PIPENRDYn (n = 0 to 9) flags for all pipes for which the NRDY interrupt is enabled (NRDYENB.PIPENRDYEn bits). Writing 0 to the NRDY flag in the software does not clear the flag.

BEMP flag (BEMP Interrupt Status Flag)

The BEMP indicates the BEMP interrupt state. For the conditions that cause the flag to be set, see section 38.2.15. BEMPENB : BEMP Interrupt Enable Register.

The USBHS clears the BEMP flag to 0 when 0 is written to the BEMPSTS.PIPEBEMPn (n = 0 to 9) flags for all pipes for which the BEMP interrupt is enabled (BEMPENB.PIPEBEMPEn bits). Writing 0 to the BEMP flag in the software does not clear the flag.

CTRT flag (Control Transfer Stage Transition Interrupt Status Flag)

In device controller mode, the USBHS updates the value of the CTSQ[2:0] bits and sets the CTRT flag to 1 on detecting a transition in the control transfer stage. When a control transfer stage transition interrupt occurs, clear the CTRT flag before the USBHS detects the next control transfer stage transition.

Values read from the CTRT flag in host controller mode are invalid.

DVST flag (Device State Transition Interrupt Status Flag)

In device controller mode, the USBHS updates the value of the PL1CTRL1.DVSQ[3:0] bits and sets the DVST flag to 1 on detecting a change in the device state. When a device state transition interrupt occurs, clear the DVST flag before the USBHS detects the next device state transition.

Values read from the DVST flag in host controller mode are invalid.

SOFR flag (Frame Number Refresh Interrupt Status Flag)

In host controller mode, the USBHS sets the SOFR flag to 1 on updating the frame number when the DVSTCTR0.UACT bit is set to 1 by software. An SOFR interrupt is detected every 1 ms.

In device controller mode, the USBHS sets the SOFR flag to 1 on updating the frame number. An SOFR interrupt is detected every 1 ms. The USBHS can detect an SOFR interrupt through the SOF complementation function even when a corrupted SOF packet is received from the USB host. See section 38.3.13. SOF Complementation Function.

RESM flag (Resume Interrupt Status Flag)

In device controller mode, the USBHS sets the RESM flag to 1 on detecting the falling edge of the signal on the USBHS_DP pin in the Suspend state (PL1CTRL1.DVSQ[3:0] = 01xxb).

Values read from the RESM flag in host controller mode are invalid.

VBINT flag (VBUS Interrupt Status Flag)

The USBHS sets the VBINT flag to 1 on detecting a level change (high to low or low to high) in the USBHS_VBUS pin input value. The USBHS sets the VBSTS flag to indicate the USBHS_VBUS pin input value. When a VBINT interrupt occurs, eliminate transient elements by reading the VBSTS flag at least three times through software processing and check that the values read are the same.

38.2.19 INTSTS1 : Interrupt Status Register 1

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x042

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVRCR	BCHG	—	DTCH	ATTCH	—	L1RSMEND	LPMEND	—	EOFERR	SIGN	SACK	—	—	—	PDDETINT
Value after reset:	0	0	x	0	0	x	0	0	x	0	0	0	x	x	x	0

Bit	Symbol	Function	R/W
0	PDDETINT	PDDET Detection Interrupt Status Flag*1 0: No PDDET interrupt occurred 1: PDDET interrupt occurred	R/W*2
3:1	—	The read values are undefined. The write value should be 0.	R/W
4	SACK	Setup Transaction Normal Response Interrupt Status Flag 0: No SACK interrupt occurred 1: SACK interrupt occurred	R/W*2
5	SIGN	Setup Transaction Error Interrupt Status Flag 0: No SIGN interrupt occurred 1: SIGN interrupt occurred	R/W*2
6	EOFERR	EOF Error Detection Interrupt Status Flag 0: No EOFERR interrupt occurred 1: EOFERR interrupt occurred	R/W*2

Bit	Symbol	Function	R/W
7	—	The read value is undefined. The write value should be 0.	R/W
8	LPMEND	LPM Transaction End Interrupt Status Flag 0: No LPMEND interrupt occurred 1: LPMEND interrupt occurred	R/W[*2]
9	L1RSMEND	L1 Resume End Interrupt Status Flag 0: No L1RSMEND interrupt occurred 1: L1RSMEND interrupt occurred	R/W[*2]
10	—	The read value is undefined. The write value should be 0.	R/W
11	ATTCH	USB Connection Detection Interrupt Status Flag 0: No ATTCH interrupt occurred 1: ATTCH interrupt occurred	R/W[*2]
12	DTCH	USB Disconnection Detection Interrupt Status Flag 0: No DTCH interrupt occurred 1: DTCH interrupt occurred	R/W[*2]
13	—	The read value is undefined. The write value should be 0.	R/W
14	BCHG	USB Bus Change Interrupt Status Flag[*1] 0: No BCHG interrupt occurred 1: BCHG interrupt occurred	R/W[*2]
15	OVRCR	OVRCR Interrupt Status Flag[*1] 0: No OVRCR interrupt occurred 1: OVRCR interrupt occurred	R/W[*2]

Note: S-TYPE-3, P-TYPE-3

Note: Only enable the status change interrupts indicated in the flags in INTSTS1 in host controller mode, except for the PDDET detection interrupt.

Note 1. The USBHS detects a change in the status in the PDDETINT, BCHG, or OVRCR flag even while the clock supply is stopped (LPSTS.SUSPENDM = 0), and it requests the interrupt when the associated interrupt request bit is 1. Enable the clock supply before clearing the status by software. No other interrupts can be detected while the clock supply is stopped (LPSTS.SUSPENDM = 0).

Note 2. To clear the flags in INTSTS1, write 0 only to the flags to be cleared. Write 1 to the other bits.

PDDETINT flag (PDDET Detection Interrupt Status Flag)

The USBHS sets the PDDETINT flag to 1 on detecting a level change (high to low or low to high) in the PDDET pin input value. When the PDDETINT interrupt is generated, perform debouncing by reading the PDDETSTS flag at least three times through software processing and checking that the values read are the same.

SACK flag (Setup Transaction Normal Response Interrupt Status Flag)

The SACK flag indicates the status of the setup transaction normal response interrupt in host controller mode.

The USBHS detects the SACK interrupt and sets this bit to 1 when an ACK response is returned from a peripheral device during the setup transactions issued by the USBHS. If the associated interrupt enable bit is set to 1 by software, the USBHS generates the interrupt.

Values read from the SACK flag in device controller mode are invalid.

SIGN flag (Setup Transaction Error Interrupt Status Flag)

The SIGN flag indicates the status of setup transaction error interrupts in host controller mode.

The USBHS detects the SIGN interrupt and sets this bit to 1 when an ACK response is not returned from a peripheral device three consecutive times during the setup transactions issued by the USBHS. If the associated interrupt enable bit is set to 1 by software, the USBHS generates the interrupt.

The USBHS detects the SIGN interrupt when any of the following response conditions occur for three consecutive setup transactions:

- Timeout is detected by the USBHS when the peripheral device has returned no response
- A corrupted ACK packet is received
- A handshake other than ACK (NAK, NYET, or STALL) is received

Values read from the SIGN flag in device controller mode are invalid.

EOFERR flag (EOF Error Detection Interrupt Status Flag)

The EOFERR flag indicates the status of EOF error detection interrupts in host controller mode.

The USBHS detects the EOFERR interrupt and sets this bit to 1 on detecting that communication did not complete at the EOF2 timing defined in the USB 2.0 specification. If the associated interrupt enable bit is set to 1 by software, the USBHS generates the interrupt.

After detecting the EOFERR interrupt, the USBHS controls the hardware as follows, regardless of the associated interrupt enable bit setting:

- Sets the DVSTCTR0.UACT bit for the port in which the EOFERR interrupt was detected to 0
- Puts the port in which the EOFERR interrupt occurred into the idle state

The software must terminate all pipes in which communications are being carried out and re-enumerate the USB port.

Values read from the EOFERR flag in device controller mode are invalid.

LPMEND flag (LPM Transaction End Interrupt Status Flag)

The PLPMEND flag indicates the status of LPM transaction end interrupts in host controller mode.

When the HL1CTRL1.L1REQ bit sets to 1, the USBHS sends an LPM token. When the LPM transaction is ended because a response from the function device or a timeout is detected, the USBHS sets this flag to 1.

Values read from the LPMEND flag in device controller mode are invalid.

L1RSMEND flag (L1 Resume End Interrupt Status Flag)

The L1RSMEND flag indicates the status of L1 resume end interrupts in host controller mode.

When performing resume processing after transitioning to the L1 state because an ACK was received in response to an LPM token, the USBHS sets this flag to 1.

Values read from the L1RSMEND flag in device controller mode are invalid.

ATTCH flag (USB Connection Detection Interrupt Status Flag)

The ATTCH flag indicates the status of USB attach detection interrupts in host controller mode.

The USBHS detects the ATTCH interrupt and sets this bit to 1 on detecting a J-state or K-state on the full-speed or low-speed signal level for 2.5 µs. If the associated interrupt enable bit is set to 1 by software, the USBHS generates the interrupt.

The USBHS detects the ATTCH interrupt on any of the following conditions:

- K-state, SE0, or SE1 changes to J-state, and J-state continues for 2.5 µs
- J-state, SE0, or SE1 changes to K-state, and K-state continues for 2.5 µs

Values read from the ATTCH flag in device controller mode are invalid.

DTCH flag (USB Disconnection Detection Interrupt Status Flag)

The DTCH flag indicates the status of USB detach detection interrupts in host controller mode.

The USBHS detects the DTCH interrupt and sets this bit to 1 on detecting a USB bus detach event. If the associated interrupt enable bit is set to 1 by software, the USBHS generates the interrupt.

The USBHS detects bus detach events based on the USB 2.0 specification.

After detecting the DTCH interrupt, the USBHS controls hardware as follows, regardless of the associated interrupt enable bit setting:

- Sets the DVSTCTR0.UACT bit for the port in which the DTCH interrupt was detected to 0
- Puts the port in which the DTCH interrupt occurred into the idle state

The software must terminate all pipes in which communications are being carried out and invoke the wait state for attaching to the USB port (waiting for ATTCH interrupt generation).

Values read from the DTCH flag in device controller mode are invalid.

BCHG flag (USB Bus Change Interrupt Status Flag)

The BCHG flag indicates the status of USB bus change interrupts in host controller mode.

The USBHS detects the BCHG interrupt and sets this bit to 1 when a change in the full-speed signal level occurs on the USB port. This includes any change from J-state, K-state, or SE0 to J-state, K-state, or SE0. If the associated interrupt enable bit is set to 1 by software, the USBHS generates the interrupt.

The USBHS sets the SYSSTS0.LNST[1:0] flags to indicate the input state of the USB port. When a BCHG interrupt occurs, eliminate transient elements by repeat reading the LNST[1:0] bits by software until the same value is read at least three times.

Changes in the USB bus state can be detected while the PHY clock is stopped.

Values read from the BCHG flag in device controller mode are invalid.

OVRCR flag (OVRCR Interrupt Status Flag)

The OVRCR flag indicates the input status on the USBHS_OVCUR0A pin or changes on the USBHS_OVCUR0B pin. If the INTENB1.OVRCRE bit sets to 1, the USBHS requests the interrupt.

The USBHS sets the SYSSTS0.OVCMON[1:0] flags to indicate the input state of the USBHS_OVCUR0A and USBHS_OVCUR0B pins.

These pins allow overcurrent detection by software in host controller mode. To implement this function, connect the overcurrent signal from the external power supply IC that supplies VBUS to connected USB devices to the OVCUR0A or OVCUR0B pin. On detection of an OVRCR interrupt, eliminate transients by repeatedly reading the OVCMON[1:0] flags through the software until the same value is read at least three times.

38.2.20 BRDYSTS : BRDY Interrupt Status Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x046

Bit position:	15	14	13	12	11	10	9									0
Bit field:	—	—	—	—	—	—	PIPEBRDY[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPEBRDY[9:0]	BRDY Interrupt Status Flag for Pipe[9:0]*1 0: No BRDY interrupt occurred 1: BRDY interrupt occurred	R/W*2
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Each bit number corresponds to the same pipe number.
Note 2. When the SOFCFG.BRDYM bit is set to 0, to clear the status indicated in the PIPEBRDY[9:0] flags, write 0 only to the bits to be cleared. Write 1 to the other bits.
When the SOFCFG.BRDYM bit is set to 0, clear BRDY interrupts before accessing the FIFO.

PIPEBRDY[9:0] flags (BRDY Interrupt Status Flag for Pipe[9:0])

When the BRDY interrupt is detected, the USBHS sets the associated bit in the PIPEBRDY[9:0] flags to 1. For details on BRDY interrupts, see section 38.3.6.1. BRDY Interrupt.

38.2.21 NRDYSTS : NRDY Interrupt Status Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x048

Bit position:	15	14	13	12	11	10	9									0
Bit field:	—	—	—	—	—	—	PIPENRDY[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPENRDY[9:0]	NRDY Interrupt Status Flag for Pipe[9:0]*1 0: No NRDY interrupt occurred 1: NRDY interrupt occurred.	R/W*2
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Each bit number corresponds to the same pipe number.
Note 2. To clear the status indicated in the PIPENRDY[9:0] flags, write 0 only to the bits to be cleared. Write 1 to the other bits.

PIPENRDY[9:0] flags (NRDY Interrupt Status Flag for Pipe[9:0])

If an internal NRDY interrupt is detected while the PID[1:0] bits in a pipe control register are 01b (BUF response), the USBHS sets the associated bit in the PIPENRDY[9:0] flags to 1. For details on NRDY interrupts, see section 38.3.6.2. NRDY Interrupt.

38.2.22 BEMPSTS : BEMP Interrupt Status Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x04A

Bit position:	15	14	13	12	11	10	9									0
Bit field:	—	—	—	—	—	—	PIPEBEMP[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PIPEBEMP[9:0]	BEMP Interrupt Status Flag for Pipe[9:0]*1 0: No BEMP interrupt occurred 1: BEMP interrupt occurred.	R/W*2
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Each bit number corresponds to the same pipe number.
Note 2. To clear the status indicated in the PIPEBEMP[9:0] flags, write 0 only to the bits to be cleared. Write 1 to the other bits.

PIPEBEMP[9:0] flags (BEMP Interrupt Status Flag for Pipe[9:0])

If an BEMP interrupt is detected while the PID[1:0] bits in a pipe control register are 01b (BUF response), the USBHS sets the associated bit in the PIPEBEMP[9:0] flags to 1. For details on BEMP interrupts, see section 38.3.6.3. BEMP Interrupt.

38.2.23 FRMNUM : Frame Number Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x04C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	OVRN	CRCE	—	—	—	FRNM[10:0]										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	FRNM[10:0]*1	Frame Number Flag	R
13:11	—	These bits are read as 0. The write value should be 0.	R/W
14	CRCE	CRC Error Detection Status Flag 0: No error occurred 1: Error occurred	R/W
15	OVRN	Overrun/Underrun Detection Status Flag 0: No error occurred 1: Error occurred.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. The OVRN flag is for debugging. Design the timing so that no overrun or underrun occurs in the system.

FRNM[10:0] flags (Frame Number Flag)

The USBHS sets the FRNM[10:0] flags to indicate the latest frame number, which is updated every 1 ms, when an SOF packet is issued or received.

CRCE flag (CRC Error Detection Status Flag)

The CRCE flag sets to 1 when a CRC error or bit stuffing error occurs during isochronous transfer. On detecting a CRC error, the USBHS generates an internal NRDY interrupt.

To clear the CRCE flag, write 0 to it while writing 1 to the other bits in the FRMNUM register.

OVRN flag (Overrun/Underrun Detection Status Flag)

The OVRN flag sets to 1 when an overrun or underrun error occurs during isochronous transfer. To clear the flag, write 0 to it while writing 1 to the other bits in the FRMNUM register.

In host controller mode, the OVRN flag sets to 1 on any of the following conditions:

- For a transmitting isochronous pipe, the time to issue an OUT token comes before all of the transmit data is written to the FIFO buffer
- For a receiving isochronous pipe, the time to issue an IN token comes when no FIFO buffer planes are empty

In device controller mode, the OVRN flag sets to 1 on any of the following conditions:

- For a transmitting isochronous pipe, the IN token is received before all of the transmit data is written to the FIFO buffer
- For a receiving isochronous pipe, the OUT token is received when no FIFO buffer planes are empty

38.2.24 UFRMNUM : μFrame Number Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x04E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DVCHG	—	—	—	—	—	—	—	—	—	—	—	—	UFRNM[2:0]		
Value after reset:	0	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

Bit	Symbol	Function	R/W
2:0	UFRNM[2:0]	Microframe number	R
14:3	—	These bits are read as 0. The write value should be 0.	R/W
15	DVCHG	Device State Change 0: Disable writes to the USBADDR.STSRECOV0[2:0] and USBADDR.USBADDR [6:0] bits 1: Enable writes to the USBADDR.STSRECOV0[2:0] and USBADDR.USBADDR[6:0] bits	R/W

Note: S-TYPE-3, P-TYPE-3

UFRNM[2:0] flags (Microframe number)

The USBHS sets the UFRNM[2:0] flags to indicate the microframe number during high-speed operation. When not in high-speed operation, the USBHS sets these bits to 00b.

Read these bits repeatedly until the same value is read twice.

38.2.25 USBADDR : USB Address Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x050

Bit position:	15	14	13	12	11	10		8	7	6						0
Bit field:	—	—	—	—	—	STSRECOV0[2:0]			—	USBADDR[6:0]						
Value after reset:	x	x	x	x	x	0	0	0	x	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	USBADDR[6:0]	USB Address Flag In device controller mode, these flags indicate the USB address assigned by the host when the USBHS processed the SET_ADDRESS request successfully.	R
7	—	The read value is undefined. The write value should be 0.	R/W
10:8	STSRECOV0[2:0]	Status Recovery [D]: In deice controller mode [H]: In host controller mode (settings other than 010b, 100b, or 110b are prohibited) 0 0 0: Reserved 0 0 1: [D] Return to the full-speed connection and Default state 0 1 0: [D] Return to the full-speed connection and Address state [H] Return to the low-speed state (bits DVSTCTR0.RHST[2:0] = 001b) 0 1 1: [D] Return to the full-speed connection and Configured state 1 0 0: [D] Return to the suspend connection and Suspend state [H] Return to the full-speed state (bits DVSTCTR0.RHST[2:0] = 010b) 1 0 1: [D] Return to the high-speed connection and Default state 1 1 0: [D] Return to the high-speed connection and Address state [H] Return to the high-speed state (bits DVSTCTR0.RHST[2:0] = 011b) 1 1 1: [D] Return to the high-speed connection and Configured state	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

USBADDR[6:0] flags (USB Address Flag)

In device controller mode, the USBADDR[6:0] flags indicate the USB address received when the USBHS processed a SetAddress request successfully. The USBHS sets the USBADDR[6:0] bits to 00b on detecting a USB bus reset.

In host controller mode, the USBADDR[6:0] bits are invalid.

STSRECOV0[2:0] bits (Status Recovery)

Use the STSRECOV[3:0] bits to resume the state of the internal sequencer on recovering from USB power shut-off. For details, see section 38.3.17. Deep Software Standby Mode 1 Because of USB Suspend/Resume Interrupts.

Writing to these bits is enabled while the UFRMNUM.DVCHG bit is set to 1.

38.2.26 USBREQ : USB Request Type Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x054

Bit position:	15								7							0
Bit field:	BREQUEST[7:0]								BMREQUESTTYPE[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	BMREQUESTTYPE [7:0]	USB request bmRequestType value	R/W*1
15:8	BREQUEST[7:0]	USB request bRequest value	R/W*1

Note: S-TYPE-3, P-TYPE-3
Note 1. In device controller mode, these bits can be read, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

BMREQUESTTYPE[7:0] bits (USB request bmRequestType value)

The BMREQUESTTYPE[7:0] bits hold the bmRequestType value of USB requests.

- In host controller mode:
  Set these bits to the value of the USB request data in transmission setup transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.
- In device controller mode:
  These bits indicate the value of the USB request data in reception setup transactions. Writing to the bits has no effect.

BREQUEST[7:0] bits (USB request bRequest value)

The BREQUEST[7:0] bits hold the bRequest value of USB requests.

- In host controller mode:
  Set these bits to the value of the USB request data in transmission setup transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.
- In device controller mode:
  These bits indicate the value of the USB request data in reception setup transactions. Writing to the bits has no effect.

38.2.27 USBVAL : USB Request Value Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x056

Bit position:	15															0
Bit field:	WVALUE[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WVALUE[15:0]	USB request wValue value	R/W*1

Note: S-TYPE-3, P-TYPE-3
Note 1. In device controller mode, these bits are readable, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

WVALUE[15:0] bits (USB request wValue value)

The WVALUE[15:0] bits hold the wValue value of USB requests.

- In host controller mode:

Set these bits to the wValue value for USB requests in transmission setup transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.

- In device controller mode:
  These bits indicate the wValue value of USB requests in reception setup transactions. Writing to the bits has no effect.

38.2.28 USBINDX : USB Request Index Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x058

Bit position:	15															0
Bit field:	WINDEX[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WINDEX[15:0]	USB request wIndex value	R/W*1

Note: S-TYPE-3, P-TYPE-3
Note 1. In device controller mode, these bits are readable, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

WINDEX[15:0] bits (USB request wIndex value)

- In host controller mode:
  Set these bits to the wIndex value of USB requests in transmission setup transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.
- In device controller mode:
  These bits indicate the wIndex value of USB requests received in reception setup transactions. Writing to the bits has no effect.

38.2.29 USBLENG : USB Request Length Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x05A

Bit position:	15															0
Bit field:	WLENTUH[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	WLENTUH[15:0]	USB request wLength value	R/W*1

Note: S-TYPE-3, P-TYPE-3
Note 1. In device controller mode, these bits are readable, but writing to them has no effect. In host controller mode, these bits are both read/write bits.

WLENTUH[15:0] bits (USB request wLength value)

The WLENTUH[15:0] bits hold the wLength value of USB requests.

- In host controller mode:
  Set the wLength value of USB requests in transmission setup transactions. Do not change the value of the bits while the DCPCTR.SUREQ bit is 1.
- In device controller mode:
  These bits indicate the wLength value of USB requests in reception setup transactions. Writing to the bits has no effect.

38.2.30 DCPCFG : DCP Configuration Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x05C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CNTMD	SHTNAK	—	—	DIR	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	DIR	Transfer Direction 0: Data receiving direction 1: Data transmitting direction	R/W
6:5	—	The read values are undefined. The write value should be 0.	R/W
7	SHTNAK	Pipe Blocking on End of Transfer 0: Keep pipe open after transfer ends 1: Disable pipe after transfer ends	R/W
8	CNTMD	Continuous Transfer Mode 0: Non-continuous transfer mode 1: Continuous transfer mode	R/W
15:9	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Only set the bits in the DCPCFG register while the PID is NAK. Before setting the bits, check that the DCPCTR.PBUSY bit is 0, and then change the DCPCTR.PID[1:0] bits for the DCP from BUF to NAK. If the PID[1:0] bits are changed to NAK by the USBHS, checking the PBUSY bit through software is not necessary.

DIR bit (Transfer Direction)

In host controller mode, the DIR bit sets the transfer direction of the data stage and status stage for control transfers. In device controller mode, set the DIR bit to 0.

SHTNAK bit (Pipe Blocking on End of Transfer)

The SHTNAK bit specifies whether to change PID to NAK on transfer end when the selected pipe is receiving. It is only valid when the selected pipe is receiving.

When the SHTNAK bit is 1, the USBHS changes the DCPCTR.PID[1:0] bits for the DCP to NAK on determining that a transfer has ended. The USBHS determines transfer end on the following condition:

- A short packet, including a zero-length packet, is successfully received

CNTMD bit (Continuous Transfer Mode)

The CNTMD bit indicates whether transfer through the default control pipe is in continuous transfer mode.

38.2.31 DCPMAXP : DCP Maximum Packet Size Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x05E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DEVSEL[3:0]				—	—	—	—	—	MXPS[6:0]						
Value after reset:	0	0	0	0	x	x	x	x	x	1	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	MXPS[6:0]	Maximum Packet Size*1 Maximum data payload specification (maximum packet size) for the DCP	R/W
11:7	—	The read values are undefined. The write value should be 0.	R/W
15:12	DEVSEL[3:0]	Device Select*2 0x0: Address 0x0 0x1: Address 0x1 0x2: Address 0x2 0x3: Address 0x3 0x4: Address 0x4 0x5: Address 0x5	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set the MXPS[6:0] bits while PID is NAK. Before setting this bit, check that the CSSTS and PBUSY bits are 0, and then change the DCPCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK), and the CFIFOSEL.CURPIPE[3:0] bits to 0000b. If the DCPCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the CSSTS and PBUSY bits through software is not necessary. After the MXPS[6:0] bits are set and the DCP is set to the CURPIPE[3:0] bits in a port select register, clear the buffer by setting the BCLR bit the port control register to 1.

Note 2. Only set the DEVSEL[3:0] bits while PID is NAK and the DCPCTR.SUREQ bits are 0. Before setting these bits, check that the CSSTS and PBUSY flags are 0, and then change the DCPCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK), and the DCPCTR.SUREQ[3:0] bits to 0. If the DCPCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the CSSTS and PBUSY bits through software is not necessary.

MXPS[6:0] bits (Maximum Packet Size)

The MXPS[6:0] bits specify the maximum data payload (maximum packet size) for the DCP. The initial value is 0x40 (64 bytes). Set the bits to a USB 2.0-compliant value. Do not write to the FIFO buffer or set PID = BUF while MXPS[6:0] is set to 0.

DEVSEL[3:0] bits (Device Select)

In host controller mode, the DEVSEL[3:0] bits specify the address of the target peripheral device for a control transfer. Set up the device address in the associated DEVADDn (n = 0 to A) register first, and then set these bits to the corresponding value. To set the DEVSEL[3:0] bits to 0010b, for example, first set the address in the DEVADD2 register. In device controller mode, set these bits to 0000b.

38.2.32 DCPCTR : DCP Control Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x060

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BSTS	SUREQ	CSCLR	CSSTS	SUREQCLR	—	—	SQCLR	SQSET	SQMON	PBUSY	PINGE	—	CCPL	PID[1:0]	
Value after reset:	0	0	0	0	x	x	x	0	0	1	0	0	x	0	0	0

Bit	Symbol	Function	R/W
1:0	PID[1:0]	Response PID 0 0: NAK response 0 1: BUF response (depends on buffer state) 1 x: STALL response	R/W
2	CCPL	Control Transfer End Enable 0: Disable control transfer completion 1: Enable control transfer completion	R/W
3	—	The read value is undefined. The write value should be 0.	R/W
4	PINGE	PING Token Issue Enable*1 0: Disable PING token 1: Enable normal PING operation	R/W

Bit	Symbol	Function	R/W
5	PBUSY	Pipe Busy Flag 0: DCP not used for the USB bus 1: DCP in use for the USB bus	R
6	SQMON	Sequence Toggle Bit Monitor Flag 0: DATA0 1: DATA1	R
7	SQSET	Sequence Toggle Bit Set*1 0: Invalid (writing 0 has no effect) 1: Set the expected value for the next transaction to DATA1	W
8	SQCLR	Sequence Toggle Bit Clear*1 0: Invalid (writing 0 has no effect) 1: Clear the expected value for the next transaction to DATA0	W
10:9	—	The read values are undefined. The write value should be 0.	R/W
11	SUREQCLR	SUREQ Bit Clear 0: Invalid (writing 0 has no effect) 1: Clear SUREQ to 0	W
12	CSSTS	CSSTS Status Flag 0: Start-split (SSPLIT) transaction, or processing for devices that are not using split transactions, in progress 1: Complete-split (CSPLIT) transaction in progress	R
13	CSCLR	CSSTS Status Flag Clear 0: (writing 0 has no effect) 1: Clear CSSTS to 0	W
14	SUREQ	SETUP Token Transmission 0: Invalid (writing 0 has no effect) 1: Transmit setup packet	R/W
15	BSTS	Buffer Status Flag 0: Buffer access disabled 1: Buffer access enabled	R

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set the SQSET, SQCLR, and PINGE bits while PID is NAK. Before setting these bits, check that the CSSTS and PBUSY bits are 0, and then change the DCPCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the DCPCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the CSSTS and PBUSY bits through the software is not necessary.

PID[1:0] bits (Response PID)

The PID[1:0] bits control the USB response type during control transfers.

In host controller mode, to change the PID[1:0] setting from NAK to BUF:

- When the transmitting direction is set:
  a. Write all of the transmit data to the FIFO buffer while the DVSTCTR0.UACT bit is 1 and PID is NAK.
  b. Set PID[1:0] bits to 01b (BUF).
     The USBHS then executes the OUT transaction (or PING transaction)
- When the receiving direction is set:
  a. Check that the FIFO buffer is empty (or empty the buffer) while the DVSTCTR0.UACT bit is 1 and PID is NAK.
  b. Set PID[1:0] bits to 01b (BUF).
     The USBHS then executes the IN transaction.

The USBHS changes the PID[1:0] setting as follows:

- When the PID[1:0] bits are set to BUF (01b) by software and the USBHS has received data exceeding MaxPacketSize, the USBHS sets PID[1:0] to STALL (11b)
- When a reception error, such as a CRC error, is detected three times consecutively, the USBHS sets PID[1:0] to NAK (00b)
- On receiving the STALL handshake, the USBHS sets PID[1:0] to STALL (11b)

In device controller mode, the USBHS changes the PID[1:0] setting as follows:

- On receiving a setup packet, the USBHS sets PID[1:0] to NAK (00b). The USBHS then sets the INTSTS0.VALID flag to 1, and the PID[1:0] setting cannot be changed until the software clears the VALID flag to 0.
- When the PID[1:0] bits are set to BUF (01b) by software and the USBHS has received data exceeding MaxPacketSize, the USBHS sets PID[1:0] to STALL (11b)
- On detecting a control transfer sequence error, the USBHS sets PID[1:0] to STALL (1xb)
- On detecting a USB bus reset, the USBHS sets PID[1:0] to NAK

The USBHS does not check the PID[1:0] setting while processing a SET_ADDRESS request.

CCPL bit (Control Transfer End Enable)

In device controller mode, setting the CCPL bit to 1 enables the status stage of the control transfer to be completed. When the bit is set to 1 by software while the associated PID[1:0] bits are set to BUF, the USBHS completes the control transfer status stage.

During control read transfers, the USBHS transmits the ACK handshake in response to the OUT transaction from the USB host. During control write or no-data control transfers, it transmits the zero-length packet in response to the IN transaction from the USB host. On detecting a SET_ADDRESS request, the USBHS operates in auto response mode from the setup stage up to status stage completion regardless of the CCPL bit setting.

The USBHS changes the CCPL bit from 1 to 0 on receiving a new setup packet. The software cannot write 1 to the bit while the INTSTS0.VALID bit is 1. The bit is initialized by a USB bus reset.

In host controller mode, always write 0 to the CCPL bit.

PINGE bit (PING Token Issue Enable)

In host controller mode, when the software sets the PINGE bit to 1, the USBHS issues a PING token for transfer in the transmitting direction, which triggers the transfer to start. If an ACK handshake is detected in the PING transaction, the OUT transaction is executed in the next transaction. If a NAK or NYET handshake is detected in the OUT transaction, the PING transaction is executed in the next transaction.

If the software sets this bit to 0, the USBHS issues no PING token for transfer in the transmitting direction. All transfers in the transmitting direction are executed in the OUT transaction.

PBUSY flag (Pipe Busy Flag)

The PBUSY bit indicates whether DCP is used for the transaction when USBHS changes the PID[1:0] bits from BUF to NAK. The USBHS changes the PBUSY flag from 0 to 1 on start of a USB transaction for the selected pipe. It changes the PBUSY flag from 1 to 0 on completion of one transaction.

After PID is set to NAK by software, the value in the PBUSY flag indicates whether changes to pipe settings can proceed.

For details, see section 38.3.7.1. Pipe Control Register Switching Procedures.

SQMON flag (Sequence Toggle Bit Monitor Flag)

The SQMON bit indicates the expected value of the sequence toggle bit for the next transaction during a DCP transfer.

The USBHS toggles the bit on normal completion of the transaction. It does not toggle the bit, however, when a DATAPID mismatch occurs during a transfer in the receiving direction.

In device controller mode, the USBHS sets the SQMON bit to 1 (specifies DATA1 as the expected value) on successful reception of the setup packet.

In device controller mode, the USBHS does not reference this bit during IN or OUT transactions at the status stage, and it does not toggle the bit on normal completion.

SQSET bit (Sequence Toggle Bit Set)

The SQSET bit specifies DATA1 as the expected value of the sequence toggle bit for the next transaction during a DCP transfer.

Do not set the SQCLR and SQSET bits to 1 simultaneously.

SQCLR bit (Sequence Toggle Bit Clear)

The SQCLR bit specifies DATA0 as the expected value of the sequence toggle bit for the next transaction during a DCP transfer. It is read as 0.

Do not set the SQCLR and SQSET bits to 1 simultaneously.

SUREQCLR bit (SUREQ Bit Clear)

In host controller mode, setting the SUREQCLR bit to 1 clears the SUREQ bit to 0. The bit is read as 0.

If transfer stops while the SUREQ bit is set to 1 in a setup transaction, set the SUREQCLR bit to 1 through software. This is not necessary at the end of a normal setup transaction, because the USBHS automatically clears the SUREQ bit to 0.

Only control the SUREQ bit through the SUREQCLR bit while the DVSTCTR0.UACT bit is 0. When UACT is 0, communication is halted or no transfer is occurring because a bus disconnection was detected.

In device controller mode, always write 0 to the SUREQCLR bit.

CSSTS flag (CSSTS Status Flag)

In host controller mode, the CSSTS flag indicates the complete-split state in split transactions for pipes that are not isochronous. The USBHS sets the CSSTS flag to 1 at the beginning of a complete-split transaction and sets the flag back to 0 when it detects transaction completion.

Values read from the CSSTS flag in device controller mode are invalid.

CSCLR bit (CSSTS Status Flag Clear)

In host controller mode, setting the CSCLR bit to 1 clears the CSSTS bit to 0.

Set this bit to 1 through software when forcing the next transfer to restart from start-split in transfers using split transactions. This is not necessary at the end of a successful complete-split transaction in a normal split transaction, because the USBHS automatically clears the CSSTS flag to 0.

Only control the CSSTS flag through the CSCLR bit while the DVSTCTR0.UACT bit is 0. When UACT is 0, communication is halted or no transfer is occurring because a port disconnection was detected. Writing 1 to this bit while the CSSTS flag is 0 has no effect; the flag remains 0.

In device controller mode, always write 0 to this bit.

SUREQ bit (SETUP Token Transmission)

In host controller mode, setting the SUREQ bit to 1 triggers the USBHS to transmit the setup packet. After completing the setup transaction process, the USBHS generates either the SACK or SIGN interrupt and clears the SUREQ bit to 0. The USBHS also clears the SUREQ bit to 0 when the software sets the SUREQCLR bit to 1.

Before setting the SUREQ bit to 1, set the DCPMAXP.DEVSEL[3:0] bits, USBREQ, USBVAL, USBINDX, and USBLENG appropriately to transmit the wanted USB request in the setup transaction. Also check that the PID[1:0] bits for the DCP are set to NAK. After setting the SUREQ bit to 1, do not change the DCPMAXP.DEVSEL[3:0] bits, USBREQ, USBVAL, USBINDX, or USBLENG until the setup transaction is complete (SUREQ bit = 1). Write 1 to the SUREQ bit only when transmitting the setup token. Otherwise, write 0.

In device controller mode, always write 0 to this bit.

BSTS flag (Buffer Status Flag)

The BSTS flag indicates the status of access to the DCP FIFO buffer. The meaning of this flag varies as follows depending on the CFIFOSEL.ISEL setting:

- When ISEL = 0, the bit indicates whether receive data can be read from the buffer
- When ISEL = 1, the bit indicates whether transmit data can be written to the buffer

38.2.33 PIPESEL : Pipe Window Select Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x064

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3			0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	PIPESEL[3:0]			
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0	0

Bit	Symbol	Function	R/W
3:0	PIPESEL[3:0]	Pipe Window Select 0x0: No pipe selected 0x1: Pipe 1 0x2: Pipe 2 0x3: Pipe 3 0x4: Pipe 4 0x5: Pipe 5 0x6 Pipe 6 0x7: Pipe 7 0x8: Pipe 8 0x9: Pipe 9 Others: Setting prohibited	R/W
15:4	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Set pipes 1 to 9 using the PIPESEL, PIPECFG, PIPEMAXP, PIPEPERI, PIPEnCTR, PIPEnTRE, and PIPEnTRN registers (n = 0 to 9).

After selecting the pipe in the PIPESEL register, pipe functions must be set in the associated PIPECFG, PIPEMAXP, and PIPEPERI registers. PIPEnCTR, PIPEnTRE, and PIPEnTRN can be set independently of the pipe selection in this register.

PIPESEL[3:0] bits (Pipe Window Select)

The PIPESEL[3:0] bits select the pipe number associated with the PIPECFG, PIPEMAXP, and PIPEPERI registers used for data writing and reading. Selecting a pipe number in the PIPESEL[3:0] bits allows writing to and reading from PIPECFG, PIPEMAXP, and PIPEPERI associated with the selected pipe number.

When PIPESEL[3:0] = 0x0, 0 is read from all of the bits in PIPECFG, PIPEMAXP, and PIPEPERI. Writing to these bits has no effect.

38.2.34 PIPECFG : Pipe Configuration Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x068

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3			0
Bit field:	TYPE[1:0]		—	—	—	BFRE	DBLB	CNTMD	SHTNAK	—	—	DIR	EPNUM[3:0]			
Value after reset:	0	0	x	x	x	0	0	0	0	x	x	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	EPNUM[3:0]	Endpoint Number*1 Specifies the endpoint number for the selected pipe. Setting 0x0 indicates the pipe is not used.	R/W
4	DIR	Transfer Direction*2 *3 0: Receiving direction 1: Transmitting direction	R/W

Bit	Symbol	Function	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	SHTNAK	Pipe Disabled at End of Transfer*1 0: Continue pipe operation after transfer ends 1: Disable pipe after transfer ends	R/W
8	CNTMD	Continuous Transfer Mode*2 *3 0: Discontinuous transfer mode 1: Continuous transfer mode	R/W
9	DBLB	Double Buffer Mode*2 *3 0: Single buffer 1: Double buffer	R/W
10	BFRE	BRDY Interrupt Operation Specification*2 *3 0: Generate BRDY interrupt on transmitting or receiving data 1: Generate BRDY interrupt on completion of reading data	R/W
13:11	—	These bits are read as 0. The write value should be 0.	R/W
15:14	TYPE[1:0]	Transfer Type*1 0 0: Pipe not used 0 1: (Pipe 1 to 5) Bulk transfer (Pipe 6 to 9) Setting prohibited 1 0: (Pipe 1 to 5) Setting prohibited (Pipe 6 to 9) Interrupt transfer 1 1: (Pipe 1 to 2) Isochronous transfer (Pipe 3 to 9) Setting prohibited	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set the TYPE[1:0], SHTNAK, and EPNUM[3:0] bits while PID is NAK. Before setting these bits, check that the PIPEnCTR.CSSTS and PIPEnCTR.PBUSY flags are 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PIPEnCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the CSSTS and PBUSY flags through the software is not necessary.

Note 2. Only set the BFRE, DBLB, CNTMD, and DIR bits while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting these bits, check that the PIPEnCTR.CSSTS and PIPEnCTR.PBUSY flags are 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PIPEnCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the PBUSY flag through the software is not necessary.

Note 3. To change the BFRE, DBLB, CNTMD, or DIR bit after completing USB communication on the selected pipe, in addition to the constraints described in note 2, write 1 and 0 to the PIPEnCTR.ACLRM bit continuously through software and clear the FIFO buffer assigned to the pipe.

EPNUM[3:0] bits (Endpoint Number)

The EPNUM[3:0] bits specify the endpoint number for the selected pipe. Setting 0000b indicates the pipe not used.

Set these bits so that the combination of the DIR and EPNUM[3:0] settings is different from those for other pipes. (The EPNUM[3:0] bits can be set to 0000b for all pipes.)

DIR bit (Transfer Direction)

The DIR bit specifies the transfer direction for the selected pipe.

When the software sets this bit to 0, the USBHS uses the selected pipe for receiving. When the software sets this bit to 1, the USBHS uses the selected pipe for transmitting.

SHTNAK bit (Pipe Disabled at End of Transfer)

The SHTNAK bit specifies whether to change the PIPEnCTR.PID[1:0] bits to 00b (NAK) at the end of transfer when the selected pipe is set in the receiving direction. The bit is valid for pipes 1 to 5 in the receiving direction.

When the software sets this bit to 1 for a receiving pipe, the USBHS changes the associated PIPEnCTR.PID[1:0] bits to 00b (NAK) on determining the transfer end. The USBHS determines that the transfer has ended on the following conditions:

- Short packet data (including a zero-length packet) was successfully received
- The transaction counter is used and the number of packets specified for the transaction counter were successfully received

CNTMD bit (Continuous Transfer Mode)

The CNTMD bit specifies whether to operate the selected pipe in continuous transfer mode. The bit is valid for pipes 1 to 5 of the bulk transfer type.

Based on this bit setting, the USBHS determines the completion of transmission or reception for the FIFO buffer allocated to the selected pipe as shown in Table 38.9.

Table 38.9 Relationship between the CNTMD setting and methods for determining completion of FIFO buffer transmission or reception

CNTMD bit setting	Methods for determining readable state and transmittable state
0	Condition for FIFO buffer readable state in receiving direction (DIR = 0): ● The USBHS received one packet
	Conditions for FIFO buffer transmittable state in transmitting direction (DIR = 1): When one of the following is satisfied: 1. CPU or DMAC/DTC wrote data of the maximum packet size to the FIFO buffer 2. CPU or DMAC/DTC wrote data of the short packet size (including 0 bytes) to the FIFO buffer and set the BVAL flag in the port control register to 1
1	Condition for FIFO buffer readable state in receiving direction (DIR = 0): 1. The byte count of data received in the FIFO buffer allocated to the selected pipe is equal to the allocated byte count ((BUFSIZE + 1) × 64). 2. The USBHS received a short packet, other than a zero-length packet. 3. The USBHS received a zero-length packet when data was already contained in the FIFO buffer allocated to the selected pipe. 4. Software received the number of packets specified for the transaction counter set for the selected pipe.
	Conditions for FIFO buffer transmittable state in transmitting direction (DIR = 1): When one of the following is satisfied. 1. The amount of data written by CPU or DMAC/DTC is equal to the size of the FIFO buffer allocated to the selected pipe. 2. CPU or DMAC/DTC wrote data of smaller size than that of the FIFO buffer allocated to the selected pipe (including 0 bytes) and set the BVAL flag in the port control register to 1. 3. CPU or DMAC/DTC wrote data of smaller size than that of one FIFO buffer allocated to the selected pipe (including 0 bytes) and asserted the DENDx_N signal on the last write.

DBLB bit (Double Buffer Mode)

The DBLB bit selects either single or double buffer mode for the FIFO buffer used by the selected pipe. The bit is valid for pipes 1 to 5.

When the software sets this bit to 1, the USBHS allocates twice the FIFO buffer size specified in the PIPEBUF.BUFSIZE[5:0] bits for the selected pipe. The FIFO buffer size that the USBHS allocates to the selected pipe is as follows:

$$(\mathrm{BUFSIZE} + 1) \times 64 \times (\mathrm{DBLB} + 1)\ [\mathrm{bytes}]$$

BFRE bit (BRDY Interrupt Operation Specification)

The BFRE bit specifies the BRDY interrupt generation timing from the USBHS to the CPU for the selected pipe.

When the software sets the BFRE bit to 1 and the selected pipe is in the receiving direction, the USBHS detects the transfer completion and generates the BRDY interrupt on reading the packet.

When a BRDY interrupt is generated with this setting, the software must write 1 to the BCLR bit in the port control register. The FIFO buffer assigned to the selected pipe is not enabled for reception until 1 is written to the BCLR bit.

When the BFRE bit is set to 1 by software and the selected pipe is in the transmitting direction, the USBHS does not generate the BRDY interrupt. For details, see section 38.3.6.1. BRDY Interrupt.

TYPE[1:0] bits (Transfer Type)

The TYPE[1:0] bits specify the transfer type for the pipe selected in the PIPESEL.PIPESEL[3:0] bits. Before setting PID to BUF and starting USB communication on the selected pipe, set the TYPE[1:0] bits to a value other than 00b.

38.2.35 PIPEBUF : Pipe Buffer Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x06A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	BUFSIZE[4:0]					—	—	BUFNMB[7:0]							
Value after reset:	x	0	0	0	0	0	x	x	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	BUFNMB[7:0]	Buffer Number Specifies the FIFO buffer number of the selected pipe (0x04 to 0x87).	R/W
9:8	—	The read values are undefined. The write value should be 0.	R/W
14:10	BUFSIZE[4:0]	Buffer Size 0x00: 64 bytes 0x01: 128 bytes : 0x1F: 2 KB	R/W
15	—	The read value is undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Only set the bits in the PIPEBUF register while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting these bits, check that the PIPEnCTR.CSSTS and PIPEnCTR.PBUSY flags are 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PIPEnCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the CSSTS and PBUSY flags through the software is not necessary.

BUFNMB[7:0] bits (Buffer Number)

The BUFNMB[7:0] bits specify the first block number of the FIFO buffer to be allocated to the selected pipe.

The USBHS allocates the FIFO buffer blocks to the selected pipe as follows:

Block number: BUFNMB to block number: BUFNMB + (BUFSIZE + 1) × (DBLB + 1) - 1

Set a value within the memory size range for these bits (0 [0x00] to 8640 [0x87] for 8.5 KB), while observing the following conditions:

- 0x00 is for DCP only
- 0x04 is for pipe 6 only, but is available for other pipes when pipe 6 is not used. When pipe 6 is selected, writes to these bits are disabled. The USBHS automatically allocates 0x04 to the BUFNMB bits for pipe 6.
- 0x05 is for pipe 7 only, but is available for other pipes when pipe 7 is not used. When pipe 7 is selected, writes to these bits are disabled. The USBHS automatically allocates 0x05 to the BUFNMB bits for pipe 7.
- 0x06 is for pipe 8 only, but is available for other pipes when pipe 8 is not used. When pipe 8 is selected, writes to these bits are disabled. The USBHS automatically allocates 0x06 to the BUFNMB bits for pipe 8.
- 0x07 is for pipe 9 only, but is available for other pipes when pipe 9 is not used. When pipe 9 is selected, writes to these bits are disabled. The USBHS automatically allocates 0x07 to the BUFNMB bits for pipe 9.

BUFSIZE[4:0] bits (Buffer Size)

The BUFSIZE[4:0] bits specify the FIFO buffer size (number of blocks) to be allocated to the selected pipe. One block is 64 bytes.

When the software sets the DBLB bit to 1, the USBHS allocates twice the FIFO buffer size specified in these bits to the selected pipe. The DBLB = 1 setting is valid for pipes 1 to 5.

The USBHS allocates the FIFO buffer blocks to the selected pipe as follows:

(BUFSIZE + 1) × 64 × (DBLB + 1) [bytes]

Set the value within the following range:

- For pipes 1 to 5, set a value from 0x00 to 0x1F (up to 2 KB)

- For pipes 6 to 9, only set a value of 0x00 (64 bytes)

38.2.36 PIPEMAXP : Pipe Maximum Packet Size Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x06C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DEVSEL[3:0]				—	MXPS[10:0]										
Value after reset:	0	0	0	x	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
10:0	MXPS[10:0]*1 *2	Maximum Packet Size • Pipes 1 and 2 1 byte (0x001) to 1024 bytes (0x400) • Pipes 3 to 5 8 bytes (0x008), 16 bytes (0x010), 32 bytes (0x020), 64 bytes (0x040), 512 bytes (0x200) (Bits 2 to 0 not supported.) • Pipes 6 to 9 1 byte (0x001) to 64 bytes (0x040) (Bits 10 to 7 not supported.)	R/W
11	—	This bit is read as 0. The write value should be 0.	R/W
15:12	DEVSEL[3:0]*3	Device Select 0x0: Address 0x0 0x1: Address 0x1 ⋮ 0x9: Address 0x9 0xA: Address 0xA Others: Reserved	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. The initial value of the MXPS[10:0] bits is 0x00 when no pipe is selected in the PIPESEL.PIPESEL[3:0] bits and 0x40 when a pipe is selected.

Note 2. Only set the MXPS[10:0] bits while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting these bits, check that the PIPEnCTR.CSSTS and PIPEnCTR.PBUSY flags are 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PIPEnCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the CSSTS and PBUSY flags through the software is not necessary.

Note 3. Only set the DEVSEL[3:0] bits while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting these bits, check that the PIPEnCTR.CSSTS and PIPEnCTR.PBUSY flags are 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PIPEnCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the PBUSY flag through the software is not necessary.

MXPS[10:0] bits (Maximum Packet Size)

The MXPS[10:0] bits specify the maximum data payload (maximum packet size) for the selected pipe.

Set these bits to the appropriate value for each transfer type based on the USB 2.0 specification. When MXPS[10:0] = 0, do not write to the FIFO buffer or set PID to BUF. These writes have no effect.

To communicate on an isochronous pipe using a split transaction, set the value in the MXPS[10:0] bits to 188 bytes or less.

DEVSEL[3:0] bits (Device Select)

In host controller mode, the DEVSEL[3:0] bits specify the address of the target device for USB communication. Set up the device address in the associated DEVADDn (n = 0 to A) register first, and then set these bits to the corresponding value. To set the DEVSEL[3:0] bits to 0x2, for example, first set the address in the DEVADD2 register.

In device controller mode, set these bits to 0x0.

38.2.37 PIPEPERI : Pipe Cycle Control Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x06E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	IFIS	—	—	—	—	—	—	—	—	—	IITV[2:0]		
Value after reset:	x	x	x	0	x	x	x	x	x	x	x	x	x	0	0	0

Bit	Symbol	Function	R/W
2:0	IITV[2:0]*1	Interval Error Detection Interval Specifies the interval error detection timing for the selected pipe as the n-th power of 2 of the frame timing.	R/W
11:3	—	These bits are read as 0. The write value should be 0.	R/W
12	IFIS	Isochronous IN Buffer Flush 0: Do not flush buffer 1: Flush buffer	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set the IITV[2:0] bits while PID is NAK. Before setting these bits, check that the PIPEnCTR.CSSTS and PIPEnCTR.PBUSY flags are 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PIPEnCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the PBUSY flag through the software is not necessary.

PIPEPERI selects whether the buffer is flushed or not when an interval error occurred during isochronous IN transfers, and sets the interval error detection interval for pipes 1 to 9.

IITV[2:0] bits (Interval Error Detection Interval)

To change the IITV[2:0] bits to another value after they are set and USB communication is performed, set the PIPEnCTR.PID[1:0] bits to 00b (NAK) and then set the PIPEnCTR.ACLRM bit to 1 to initialize the interval timer.

The IITV[2:0] bits are not provided for pipes 3 to 5. Write 000b to bit positions of the IITV[2:0] bits associated with pipes 3 to 5.

IFIS bit (Isochronous IN Buffer Flush)

The IFIS bit specifies whether to flush the buffer when the pipe specified in the PIPESEL.PIPESEL[3:0] bits is used for isochronous IN transfers.

In device controller mode when the selected pipe is for isochronous IN transfers, the USBHS automatically clears the FIFO buffer if the USBHS fails to receive the IN token from the USB host within the interval set in the IITV[2:0] bits in terms of frames.

When double buffering is specified (PIPECFG.DBLB = 1), the USBHS only clears the data in the previously used plane.

The USBHS clears the FIFO buffer on receiving the SOF packet immediately after the frame in which the USBHS expected to receive the IN token. Even if the SOF packet is corrupted, the FIFO buffer is cleared at the time the SOF packet is expected to be received by using the internal complementation function.

In host controller mode, set the IITV[2:0] bits to 000b.

Set the IITV[2:0] bits to 000b when the selected pipe is not used for isochronous transfers.

38.2.38 PIPEnCTR : Pipe n Control Register (n = 1 to 9)

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x070 + 0x2 × (n - 1)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BSTS	INBUFM	CSCLR	CSSTS	—	ATREPM	ACLRM	SQCLR	SQSET	SQMON	PBUSY	—	—	—	PID[1:0]	
Value after reset:	0	0	0	0	x	0	0	0	0	0	0	x	x	x	0	0

Bit	Symbol	Function	R/W
1:0	PID[1:0]	Response PID 0 0: NAK response 0 1: BUF response (depends on buffer state) 1 0: STALL response 1 1: STALL response	R/W
4:2	—	The read values are undefined. The write value should be 0.	R/W
5	PBUSY	Pipe Busy Flag 0: Pipe n not in use for the transaction 1: Pipe n in use for the transaction	R
6	SQMON	Sequence Toggle Bit Monitor Flag 0: DATA0 1: DATA1	R
7	SQSET	Sequence Toggle Bit Set*1 0: Invalid (writing 0 has no effect) 1: Set the expected value for the next transaction to DATA1. This bit is read as 0.	R/W
8	SQCLR	Sequence Toggle Bit Clear*1 0: Invalid (writing 0 has no effect) 1: Clear the expected value for the next transaction to DATA0	W
9	ACLRM	Auto Buffer Clear Mode*2 0: Disable 1: Enable (initialize all buffers)	R/W
10	ATREPM	Auto Response Mode*1 *3 0: Disable auto response mode 1: Enable auto response mode	R/W
11	—	The read value is undefined. The write value should be 0.	R/W
12	CSSTS	CSSTS Status Flag 0: Start-split (SSPLIT) transaction, or processing for devices that are not using split transactions, in progress. 1: Complete-split (CSPLIT) transaction in progress.	R
13	CSCLR	CSPLIT Status Clear 0: Invalid (writing 0 has no effect) 1: Clear CSSTS to 0	W
14	INBUFM	Transmit Buffer Monitor Flag*3 0: No data to be transmitted is in the FIFO buffer 1: Data to be transmitted is in the FIFO buffer	R
15	BSTS	Buffer Status Flag 0: Buffer access disabled 1: Buffer access enabled	R

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set the ATREPM, SQCLR, and SQSET bits while PID is NAK. Before setting this bit, check that the PIPEnCTR.CSSTS and PIPEnCTR.PBUSY flags are 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PIPEnCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the PBUSY flag through the software is not necessary.

Note 2. Only set the ACLRM bit while PID is NAK and before the pipe is selected in the CURPIPE[3:0] bits in the port select register. Before setting this bit, check that the PIPEnCTR.CSSTS and PIPEnCTR.PBUSY flags are 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PIPEnCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the PBUSY flag through the software is not necessary.

Note 3. The ATREPM bit and the INBUFM flag in the PIPE6CTR to PIPE9CTR registers are reserved. The read value is undefined. The write value must be 0.

PID[1:0] bits (Response PID)

The PID[1:0] bits specify the response type for the next transaction on the selected pipe.

The default PID[1:0] setting is NAK. Change the PID[1:0] setting to BUF to use the associated pipe for USB transfer. Table 38.10 and Table 38.11 show the basic operations of the USBHS (when there are no errors in the communication packets) based on the PID[1:0] bit setting.

After changing the PID[1:0] setting from BUF to NAK through the software during USB communication on the selected pipe, check that the PBUSY bit is 1 to see if USB transfer on the selected pipe has actually entered the NAK state. If the USBHS changes the PID[1:0] bits to NAK, checking the PBUSY bit through the software is not necessary.

The USBHS changes the PIPEnCTR.PID[1:0] setting in the following cases:

- The USBHS sets PID to NAK on recognizing completion of the transfer when the selected pipe is in the receiving direction and the PIPECFG.SHTNAK bit for the selected pipe is set to 1 by software
- The USBHS sets PID to STALL (11b) on receiving a data packet with a payload exceeding the maximum packet size of the selected pipe
- The USBHS sets PID to NAK on detecting a USB bus reset in device controller mode
- The USBHS sets PID to NAK on detecting a reception error, such as a CRC error, three consecutive times in host controller mode
- The USBHS sets PID to STALL (11b) on receiving the STALL handshake in host controller mode

To specify the response type, set the PID[1:0] bits as follows:

- To transition from NAK (00b) to STALL, set 10b
- To transition from BUF (01b) to STALL, set 11b
- To transition from STALL (11b) to NAK, set 10b and then 00b
- To transition from STALL to BUF, set 00b (NAK) and then 01b (BUF)

Table 38.10 Operation of the USBHS based on the PIPEnCTR.PID[1:0] setting in host controller mode

PID[1:0] value	Transfer type (TYPE[1:0] value)	Transfer direction (DIR value)	USBHS operation
00b (NAK)	Does not depend on the setting	Does not depend on the setting	Does not issue tokens
01b (BUF)	Bulk or Interrupt	Does not depend on the setting	Issues tokens when the DVSTCTR0.UACT bit is 1 and the FIFO buffer associated with the selected pipe is ready for transmission and reception. Does not issue tokens when the DVSTCTR0.UACT bit is 0 or the FIFO buffer associated with the selected pipe is not ready for transmission or reception.
	Isochronous	Does not depend on the setting	Issues tokens when the DVSTCTR0.UACT bit is 1, regardless of the state of the FIFO buffer associated with the selected pipe. Does not issue tokens when UACT = 0.
10b (STALL) or 11b (STALL)	Does not depend on the setting	Does not depend on the setting	Does not issue tokens.

Table 38.11 Operation of the USBHS based on the PIPEnCTR.PID[1:0] setting in device controller mode

PID[1:0] value	Transfer type (TYPE[1:0] value)	Transfer direction (DIR value)	USBHS operation
00b (NAK)	Bulk or Interrupt	Does not depend on the setting	Returns NAK in response to the token from the USB host
	Isochronous	Receiving direction (DIR = 0)	Returns nothing in response to the token from the USB host
		Transmitting direction (DIR = 1)	Transmits a zero-length packet in response to the token from the USB host
01b (BUF)	Bulk	Receiving direction (DIR = 0)	Receives data and returns ACK or NYET in response to the OUT token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception. Otherwise, returns NAK. Returns ACK in response to the PING token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception. Otherwise, returns NAK.
	Interrupt	Receiving direction (DIR = 0)	Receives data and returns ACK response in response to the OUT token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception. Otherwise, returns NAK.
	Bulk or Interrupt	Transmitting direction (DIR = 1)	Transmits data in response to the token from the USB host if the FIFO buffer associated with the selected pipe is ready for transmission. Otherwise, returns NAK.
	Isochronous	Receiving direction (DIR = 0)	Receives data in response to the OUT token from the USB host if the FIFO buffer associated with the selected pipe is ready for reception. Otherwise, discards the data.
		Transmitting direction (DIR = 1)	Transmits data in response to the token from the USB host if the associated FIFO buffer is ready for transmission. Otherwise, transmits a zero-length packet.
10b (STALL) or 11b (STALL)	Bulk or Interrupt	Does not depend on the setting.	Returns STALL in response to the token from the USB host
	Isochronous	Does not depend on the setting.	Returns nothing in response to the token from the USB host

PBUSY flag (Pipe Busy Flag)

The PBUSY flag indicates whether the selected pipe is being used for the current transaction.

The USBHS changes the PBUSY bit from 0 to 1 on start of the USB transaction for the selected pipe, and changes the PBUSY bit from 1 to 0 on completion of one transaction.

Reading the PBUSY bit by software after PID is set to NAK allows you to check whether changing the pipe setting is possible. For details, see section 38.3.7.1. Pipe Control Register Switching Procedures.

SQMON flag (Sequence Toggle Bit Monitor Flag)

The SQMON flag indicates the expected value of the sequence toggle bit for the next transaction of the selected pipe.

When the selected pipe is not the isochronous transfer type, the USBHS toggles the SQMON flag on successful completion of the transaction. However, the USBHS does not toggle the SQMON flag when a DATA-PID mismatch occurs during transfer in the receiving direction.

SQSET bit (Sequence Toggle Bit Set)

Setting the SQSET bit to 1 through the software causes the USBHS to set DATA1 as the expected value of the sequence toggle bit for the next transaction on the selected pipe.

SQCLR bit (Sequence Toggle Bit Clear)

Setting the SQCLR bit to 1 through the software causes the USBHS to clear the expected value of the sequence toggle bit for the next transaction on the selected pipe to DATA0.

In host controller mode, when this bit is set to 1 for a bulk OUT transfer pipe, the USBHS starts the next transfer for the selected pipe from a PING token.

ACLRM bit (Auto Buffer Clear Mode)

The ACLRM bit enables or disables auto buffer clear mode for the selected pipe. To completely clear the data in the FIFO buffer allocated to the selected pipe, write 1 and then 0 to the ACLRM bit continuously.

Table 38.12 shows the data cleared by writing 1 and 0 continuously to the ACLRM bit and the cases in which this processing is required.

Table 38.12 Data cleared by the USBHS when ACLRM = 1

Number	Data cleared by setting the ACLRM bit	Situations requiring data clear
1	All data in the FIFO buffer allocated to the selected pipe (two FIFO buffers in double buffer mode)	When clearing all data in the FIFO buffer allocated to the selected pipe
2	Interval count value when the selected pipe is the isochronous transfer type	When resetting the interval count value

ATREPM bit (Auto Response Mode)

The ATREPM bit enables or disables auto response mode for the selected pipe.

This bit can be set to 1 in device controller mode when the selected pipe is the bulk transfer type. When the bit is set to 1, the USBHS responds to the token from the USB host as follows:

- When the selected pipe is set for bulk IN transfers (PIPECFG.TYPE[1:0] = 01b and PIPECFG.DIR = 1):
  - When the ATREPM bit = 1 and PID = BUF, the USBHS transmits a zero-length packet in response to the IN token.
  - The USBHS updates (allows toggling of) the sequence toggle bit (DATA-PID) each time the USBHS receives ACK from the USB host. In a single transaction, the IN token is received, a zero-length packet is transmitted, and then ACK is received. The USBHS does not generate the BRDY or BEMP interrupt.
- When the selected pipe is set for bulk OUT transfers (PIPECFG.TYPE[1:0] = 01b and PIPECFG.DIR = 0):
  When the ATREPM bit = 1 and PID = BUF, the USBHS returns NAK in response to the OUT token or PING token and generates an NRDY interrupt.

For USB communication in auto response mode, set the ATREPM bit to 1 while the FIFO buffer is empty. Do not write to the FIFO buffer during USB communication in auto response mode. When the selected pipe uses isochronous transfer, always set this bit to 0.

In host controller mode, always set the ATREPM bit to 0.

CSSTS flag (CSSTS Status Flag)

In host controller mode, the CSSTS flag indicates the complete-split status of a split transaction. It is valid for pipes that are not the isochronous transfer type.

The USBHS sets the CSSTS flag to 1 at the beginning of the complete-split transaction, and sets the CSSTS flag to 0 on detecting completion of the complete-split transaction. If a detach event is detected during the transaction, the CSSTS flag might stay set to 1. In this case, clear the CSSTS flag by setting the CSCLR bit to 1.

Values read from the CSSTS flag in device controller mode are invalid.

CSCLR bit (CSPLIT Status Clear)

In host controller mode, if the software sets the CSCLR bit to 1, the USBHS clears the CSSTS flag to 0. In split transactions, set the CSCLR bit to 1 by software to force the next transfer to restart from start-split. Because the USBHS automatically clears the CSSTS flag to 0 at the end of a successful complete-split transaction in a normal split transaction, clearing the flag through software is not required. Only clear the CSSTS flag using the CSCLR bit when the DVSTCTR0.UACT bit is set to 0 or when no transfer was made after a detach detect. If the CSCLR bit is set to 1 while the CSSTS flag is 0, the CSSTS flag remains 0.

In device controller mode, always write 0 to the CSCLR bit.

INBUFM flag (Transmit Buffer Monitor Flag)

The INBUMFM flag indicates the FIFO buffer status for the selected pipe in the transmitting direction.

When the selected pipe is set in the transmitting direction (PIPECFG.DIR = 1), the USBHS sets this bit to 1 when the CPU or DMAC/DTC completes writing data to at least one FIFO buffer plane.

The USBHS sets this bit to 0 when the USBHS completes transmission of data from the FIFO buffer plane to which all the data is written. In double buffer mode (PIPECFG.DBLB = 1), the USBHS sets the INBUFM flag to 0 when the USBHS completes transmission of data from the two FIFO buffer planes before the CPU or DMAC/DTC completes writing data to one FIFO buffer plane.

The INBUFM flag indicates the same value as the BSTS flag when the selected pipe is in the receiving direction (PIPECFG.DIR = 0).

BSTS flag (Buffer Status Flag)

The BSTS flag indicates the FIFO buffer status for the selected pipe. The meaning of the BSTS flag depends on the PIPECFG.DIR, PIPECFG.BFRE, and DnFIFOSEL.DCLRM settings, as shown in Table 38.13.

Table 38.13 BSTS flag operation

DIR value	BFRE value	DCLRM value	Meaning of BSTS flag
0	0	0	Sets to 1 when receive data can be read from the FIFO buffer, and clears to 0 on completion of data read
		1	Setting prohibited
	1	0	Sets to 1 when receive data can be read from the FIFO buffer, and clears to 0 when the software sets the BCLR bit in the port control register to 1 after the data read is complete
		1	Sets to 1 when receive data can be read from the FIFO buffer, and clears to 0 on completion of data read
1	0	0	Sets to 1 when transmit data can be written to the FIFO buffer, and clears to 0 on completion of data write
		1	Setting prohibited
	1	0	Setting prohibited
		1	Setting prohibited

38.2.39 PIPEnTRE : Pipe n Transaction Counter Enable Register (n = 1 to 5)

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x090 + 0x4 × (n - 1)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	TRENB	TRCLR	—	—	—	—	—	—	—	—
Value after reset:	x	x	x	x	x	x	0	0	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
7:0	—	The read values are undefined. The write value should be 0.	R/W
8	TRCLR	Transaction Counter Clear 0: Invalid (writing 0 has no effect) 1: Clear current counter value	R/W
9	TRENB	Transaction Counter Enable 0: Disable transaction counter 1: Enable transaction counter	R/W
15:10	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Only change the PIPEnTRE register settings while the PIPEnCTR.CSSTS flag is 0 and the PIPEnCTR.PID[1:0] bits are 00b (NAK response). Only change the PIPEnCTR.PID[1:0] bits of the selected pipe from 01b (BUF response) to 00b (NAK response) after confirming that the value of the PIPEnCTR.PBUSY and PIPEnCTR.CSSTS flags is 0. However, software processing to check the PIPEnCTR.PBUSY flag is not required if the USBHS has changed the PID[1:0] bits to 00b (NAK response).

TRCLR bit (Transaction Counter Clear)

When the TRCLR bit sets to 1, the USBHS clears the count value of the transaction counter associated with the selected pipe and then clears the TRCLR bit to 0.

TRENB bit (Transaction Counter Enable)

The TRENB bit enables or disables the transaction counter.

For receiving pipes, setting the TRENB bit to 1 after setting the total number of the packets to be received in the PIPEnTRN.TRNCNT[15:0] bits through the software allows the USBHS to control hardware on having received the number of packets equal to the TRNCNT[15:0] setting as follows:

- When the PIPECFG.SHTNAK bit is 1, the USBHS changes the PID bits to NAK for the associated pipe on having received the number of packets equal to the TRNCNT[15:0] setting
- When the PIPECFG.BFRE bit is 1, the USBHS asserts the BRDY interrupt on having received the number of packets equal to the TRNCNT[15:0] setting and then reading the last received data.

For transmitting pipes, set the TRENB bit to 0.

When the transaction counter is not used, set this bit to 0. When the transaction counter is used, set the TRNCNT[15:0] bits before setting this bit to 1. Set this bit to 1 before receiving the first packet to be counted by the transaction counter.

38.2.40 PIPEnTRN : Pipe n Transaction Counter Register (n = 1 to 5)

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x092 + 0x4 × (n - 1)

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TRNCNT[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	TRNCNT[15:0]	Transaction Counter[*1] • When written to: Specifies the total packets (number of transactions) to be received by pipe n. • When read from: When PIPEnTRE.TRENB is 0, indicates the specified number of transactions. When PIPEnTRE.TRENB is 1, indicates the current transaction count.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Only set the TRNCNT[15:0] bits while PID is NAK and PIPEnTRE.TRENB is 0. Before setting these bits, check that the PIPEnCTR.CSSTS and PIPEnCTR.PBUSY flags are 0, and then change the PIPEnCTR.PID[1:0] bits from 01b (BUF) to 00b (NAK). If the PIPEnCTR.PID[1:0] bits are changed to 00b (NAK) by the USBHS, checking the PBUSY flag through the software is not necessary.

The PIPEnTRN registers retain their settings during a USB bus reset.

TRNCNT[15:0] bits (Transaction Counter)

The USBHS increments the value of the TRNCNT[15:0] bits by one when all of the following conditions are satisfied on receiving the packet:

- The PIPEnTRE.TRENB bit is 1
- (TRNCNT[15:0] set value ≠ current counter value + 1) on receiving the packet
- The payload of the received packet agrees with the PIPEMAXP.MXPS[8:0] setting

The USBHS clears the value of the TRNCNT[15:0] bits to 0 when any of the following conditions is satisfied.

All of the following conditions are satisfied:

- The PIPEnTRE.TRENB bit = 1
- (TRNCNT[15:0] set value = current counter value + 1) on receiving the packet
- The payload of the received packet agrees with the PIPEMAXP.MXPS[8:0] setting

Both the following conditions are satisfied:

- The PIPEnTRE.TRENB bit = 1
- The USBHS received a short packet

Both the following conditions are satisfied:

- The PIPEnTRE.TRENB bit = 1
- The PIPEnTRE.TRCLR bit was set to 1 by software

For transmitting pipes, set the TRNCNT[15:0] bits to 0. When the transaction counter is not used, set the TRNCNT[15:0] bits to 0.

Setting the number of transactions to be transferred to the TRNCNT[15:0] bits is enabled only when the PIPEnTRE.TRENB bit is 0. To set the number of transactions to be transferred, set the TRCLR bit to 1 to clear the current counter value before setting the PIPEnTRE.TRENB bit to 1.

38.2.41 DEVADDn : Device Address n Configuration Register (n = 0 to 9, A)

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x0D0 + 0x2 × n (n = 0 to 9)
0x0E4 (n = A)

Bit position:	15	14				10			7		5	4	3	2	1	0
Bit field:	—	UPPHUB[3:0]				HUBPORT[2:0]			USBSPD[1:0]		—	—	—	—	—	—
Value after reset:	x	0	0	0	0	0	0	0	0	0	x	x	x	x	x	x

Bit	Symbol	Function	R/W
5:0	—	The read values are undefined. The write value should be 0.	R/W
7:6	USBSPD[1:0]	Transfer Speed of Communication Target Device 0 0: Do not use DEVADDn 0 1: Low speed 1 0: Full speed 1 1: High speed	R/W
10:8	HUBPORT[2:0]	Communication Target Connecting Hub Port 0 0 0: Connect directly to the USBHS port Others: Port number of the hub	R/W
14:11	UPPHUB[3:0]	Communication Target Connecting Hub Register 0x0: Connect directly to the USBHS port Others: USB address of the hub. The value as 0xB or more is reserved.	R/W
15	—	The read value is undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The DEVADDn register specifies the transfer speed of the peripheral device that is the communication target for pipes 0 to 9.

In host controller mode, set all DEVADDn bits before starting communication to any pipes. Only change the bits in DEVADDn when no valid pipes are using the bit settings. A valid pipe is defined as one that satisfies both of the following conditions:

- DEVADDn is selected in the DEVSEL[3:0] bits
- The PID[1:0] bits are set to BUF for the selected pipe, or the selected pipe is the DCP with the DCPCTR.SUREQ bit set to 1

In device controller mode, set all bits in this register to 0.

USBSPD[1:0] bits (Transfer Speed of Communication Target Device)

The USBSPD[1:0] bits specify the USB transfer speed of the target peripheral device. In host controller mode, the USBHS generates packets based on the USBSPD[1:0] setting. In device controller mode, set these bits to 00b.

HUBPORT[2:0] bits (Communication Target Connecting Hub Port)

In host controller mode, the USBHS generates packets based on the HUBPORT[2:0] setting when performing a split transaction.

UPPHUB[3:0] bits (Communication Target Connecting Hub Register)

In host controller mode, the USBHS generates packets based on the UPPHUB[3:0] setting when performing a split transaction.

38.2.42 LPCTRL : Low Power Control Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x100

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	HWUPM	—	—	—	—	—	—	—
Value after reset:	x	x	x	x	x	x	x	0	0	x	x	x	x	x	x	0

Bit	Symbol	Function	R/W
6:0	—	The read values are undefined. The write value should be 0.	R/W
7	HWUPM	Resume Return Mode Setting 0: Hardware does not recover while CPU clock inactive 1: Hardware recovers while CPU clock inactive	R/W
8	—	This bit is read as 0. The write value should be 0.	R/W
15:9	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

HWUPM bit (Resume Return Mode Setting)

The HWUPM bit specifies whether to enable hardware processing for return from low power mode even while the CPU clock is inactive.

In device controller mode, processing for return from low power mode on detecting Resume is enabled even while the CPU clock is inactive.

This bit specifies whether to detect Resume while the CPU clock is inactive. The PL1CTRL1.L1EXTMD bit controls whether to make a hardware return. To make a hardware return from the LPM L1 low power state while the CPU clock is inactive, set this bit and the PL1CTRL1.L1EXTMD bit to 1.

38.2.43 LPSTS : Low Power Status Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x102

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	SUSP ENDM	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	x	0	x	0	x	x	x	0	x	x	x	x	0	x	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
2	—	The read value is undefined. The write value should be 0.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
7:4	—	The read values are undefined. The write value should be 0.	R/W
8	—	This bit is read as 0. The write value should be 0.	R/W
11:9	—	The read values are undefined. The write value should be 0.	R/W
12	—	This bit is read as 0. The write value should be 0.	R/W
13	—	The read value is undefined. The write value should be 0.	R/W
14	SUSPENDM	UTMI SuspendM Control 0: UTMI suspension mode 1: UTMI normal mode	R/W
15	—	The read value is undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

SUSPENDM bit (UTMI SuspendM Control)

The SUSPENDM bit controls the SuspendM signal to be sent to the PHY designed under the UTMI specification. The initial value is 0 with the UTMI is in suspension mode.

Set this bit to 1 to supply the PHY clock to operate the USB2.0 host or device controller.

In compliance with the UTMI specification, clock output is normally controlled by the SuspendM signal. When the SUSPENDM bit is 0, the clock to LINK is stopped. Because the PHY in this MCU follows the UTMI specification, setting the SUSPENDM bit to 1 is required to supply the PHY clock. For the clock settings, see section 38.3.3. Supplying the Clock.

When the SUSPENDM bit is 0, the USBHS cannot be written to but can be read from. The registers listed in Table 38.14 are writable even when the SUSPENDM bit is 0.

Table 38.14 Registers that can be written to by software when SUSPENDM = 0

Address	Register or bit name
0x4006_0000	SYSCFG register
0x4006_0002	BUSWAIT register
0x4006_0032	INTENB1.PDDETINTE bit
0x4006_0100	LPCTRL register
0x4006_0102	LPSTS register
0x4006_0140	BCCTRL register

The value written to the SYSCFG register while the PHY clock is inactive is updated only after the PHY clock begins oscillating. The PHY clock oscillates in the following cases described in this section.

When SUSPENDM bit is set to 1, the PLLSTA.PLLLOCK flag is set to 1 after the predetermined time has passed. The USB-PHY internal PLL is stopped when the SUSPENDM bit is set to 0.

For details on CL-only mode, see section 38.2.17. PHYSET : PHY Setting Register.

If the PL1CTRL1.L1EXTMD bit is 0, setting or clearing of this bit is controlled by software. If the PL1CTRL1.L1EXTMD bit is 1, transitions to the L1 or L2 state of this bit are controlled by software and recovery from the L1 or L2 state is controlled by hardware.

38.2.44 BCCTRL : Battery Charging Control Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x140

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PDDETSTS	CHGDETSTS	—	—	DCPMODE	VDMSRCE	IDPSINKE	VDPSRCE	IDMSINKE	IDPSRCE
Value after reset:	x	x	x	x	x	x	0	0	x	x	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IDPSRCE	IDPSRC Control*2 0: Disable IDP_SRC circuit 1: Enable IDP_SRC circuit	R/W
1	IDMSINKE	IDMSINK Control*2 0: Disable IDM_SINK circuit 1: Enable IDM_SINK circuit	R/W*1
2	VDPSRCE	VDPSRC Control*2 0: Disable VDP_SRC circuit 1: Enable VDP_SRC circuit	R/W
3	IDPSINKE	IDPSINK Control*2 0: Disable IDP_SINK circuit 1: Enable IDP_SINK circuit	R/W
4	VDMSRCE	VDMSRC Control*2 0: Disable VDM_SRC circuit 1: Enable VDM_SRC circuit	R/W
5	DCPMODE	DCP Mode Control 0: Disable RDCP_DAT resistor 1: Enable RDCP_DAT resistor	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	CHGDETSTS	CHGDET Status Flag 0: The CHGDET pin is at low level 1: The CHGDET pin is at high level	R
9	PDDETSTS	PDDET Status Flag 0: The PDDET pin is at low level 1: The PDDET pin is at high level	R
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. All bits in the BCCTRL register can be changed while the UTMI clock is inactive.
Note 2. In device controller mode, set the IDPSRCE, IDMSINKE, VDPSRCE, IDPSINKE, and VDMSRCE bits to 1 after setting the SYSCFG.DRPD bit to 0.

IDPSRCE bit (IDPSRC Control)

In device controller mode, set the IDPSRCE bit to 1 to perform data contact detection.

The Battery Charging Standard provides two ways to handle data contact detection, one through the software and one using hardware to contact the data line. The IDPSRE bit uses the hardware method.

When the IDPSRE bit is set to 1, the USBHS enables the IDP_SRC circuit and, at the same time, controls D- pull-down.

(D- pull-down is controlled with the VUH_DMPULLDOWN signal.)

IDMSINKE bit (IDMSINK Control)

In device controller mode, set the IDMSINKE bit to 1 to perform primary detection.

VDPSRCE bit (VDPSRC Control)

In device controller mode, set the VDPSRCE bit to 1 to perform primary detection.

IDPSINKE bit (IDPSINK Control)

In device controller mode, set the IDPSINKE bit to 1 to perform secondary detection. In host controller mode, set this bit to 1 to enable the portable device detection circuit.

VDMSRCE bit (VDMSRC Control)

In device controller mode, set the VDMSRCE bit to 1 to perform secondary detection. Setting this bit to 1 enables the DCP detection circuit. In host controller mode, set this bit to 1 when a portable device is detected. Setting this bit to 1 allows the device that is performing primary detection to determine the charger detection method.

DCPMODE bit (DCP Mode Control)

Set the DCPMODE bit to 1 to operate as a dedicated charging port (DCP). Setting this bit to 1 disables USB communication.

CHGDETSTS flag (CHGDET Status Flag)

The CHGDETSTS flag indicates the charger port detection state.

PDDETSTS flag (PDDET Status Flag)

The PDDETSTS flag indicates the following states based on the controller mode:

- In host controller mode: PD detection state
- In device controller mode: DCP detection state

38.2.45 PL1CTRL1 : Function L1 Control Register 1

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x144

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	L1EXTMD	—		HIRDTHR[3:0]				DVSQ[3:0]				L1NEGOMD	L1RESPMD[1:0]		L1RESPEN
Value after reset:	x	0	x	x	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	L1RESPEN	L1 Response Enable 0: Do not support LPM 1: Support LPM	R/W
2:1	L1RESPMD[1:0]	L1 Response Mode 0 0: NYET response 0 1: ACK response 1 0: STALL response 1 1: Response based on L1NEGOMD setting	R/W
3	L1NEGOMD	L1 Response Negotiation Control This bit is only valid when the L1RESPMD[1:0] value is 11b. 0: Return ACK when received HIRD is larger than HIRDTHR[3:0]. Otherwise (including when HIRD = HIRDTHR[3:0]), return NYET 1: Return ACK when received HIRD is smaller than HIRDTHR[3:0]. Otherwise (including when HIRD = HIRDTHR[3:0]), return NYET	R/W

Bit	Symbol	Function	R/W
7:4	DVSQ[3:0]	DVSQ Extension Flag 0 0 0 0: Powered state 0 0 0 1: Default state 0 0 1 0: Address state 0 0 1 1: Configured state 0 1 x x: Suspend state 1 0 x x: L1 state	R
11:8	HIRDTHR[3:0]	L1 Response Negotiation Threshold Value HIRD threshold value used when the L1RESPMD[1:0] bits are 11b. The format is the same as the HIRD field in HL1CTRL.	R/W
13:12	—	The read values are undefined. The write value should be 0.	R/W
14	L1EXTMD	PHY Control Mode at L1 Return 0: Do not set LPSTS.SUSPENDM bit through hardware when Host K is received 1: Set LPSTS.SUSPENDM bit through hardware when Host K is received	R/W
15	—	The read value is undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

L1RESPEN bit (L1 Response Enable)

If the USBHS receives an LPM token while the L1RESPEN bit is 0, it returns no response. If the USBHS receives an LPM token while this bit is 1, it returns a response based on the L1RESPMD[1:0] setting.

L1RESPMD[1:0] bits (L1 Response Mode)

When the L1RESPEN bit is set to 1, the USBHS returns a response to the LPM token based on the setting in the L1RESPMD[1:0] bits.

L1NEGOMD bit (L1 Response Negotiation Control)

The L1NEGOMD bit specifies the negotiation function for the HIRD value.

HIRDTHR[3:0] bits (L1 Response Negotiation Threshold Value)

The HIRDTHR[3:0] bits specify the HIRD threshold value used for L1NEGOMD. The format of the set value is the same as the HIRD field in HL1CTRL.

L1EXTMD bit (PHY Control Mode at L1 Return)

The L1EXTMD bit specifies the LPSTS.SUSPENDM bit control method when a host K signal is received in the L1 state while the LPSTS.SUSPENDM bit is 0 and the PHY is inactive.

Similar to the Suspend constraints, because the minimum host K period is 50 µs, the PHY might not recover within the host K period specified for software settings on return. The initial value is within software control, so set this bit to 1 during the initialization process when the L1 state is supported.

The LPSTS.SUSPENDM bit is controlled by software on transition to the L1 state regardless of the setting in this bit. It is not cleared by hardware.

When this bit is set to 1, the LPSTS.SUSPENDM bit is also set to 1 on return from L2.

38.2.46 PL1CTRL2 : Function L1 Control Register 2

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x146

Bit position:	15	14	13	12	11				7	6	5	4	3	2	1	0
Bit field:	—	—	—	RWEMON	HIRDMON[3:0]				—	—	—	—	—	—	—	—
Value after reset:	x	x	x	0	0	0	0	0	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
7:0	—	The read values are undefined. The write value should be 0.	R/W
11:8	HIRDMON[3:0]	HIRD Value Monitor When set, indicates that the HIRD field value reflects the last received LPM token.	R
12	RWEMON	RWE Value Monitor When set, indicates that the RWE bit value reflects the last received LPM token.	R
15:13	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

HIRDMON[3:0] bits (HIRD Value Monitor)

Access the HIRDMON[3:0] bits when monitoring the HIRD field value of the received LPM token. The bits reflect the HIRD field value of the last received LPM token.

RWEMON bit (RWE Value Monitor)

Access the RWEMON bit when monitoring the RWE field value of the received LPM token. The bits reflect the RWE field value of the last received LPM token.

38.2.47 HL1CTRL1 : Host L1 Control Register 1

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x148

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2		0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	L1STATUS[1:0]		L1REQ
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0	0

Bit	Symbol	Function	R/W
0	L1REQ	L1 Transition Request Set this bit to 1 when requesting a transition to the L1 state. This bit is cleared to 0 by the hardware when the LPM transaction is complete.	R/W
2:1	L1STATUS[1:0]	L1 Request Completion Status Indicates the result of the LPM transaction made by the L1REQ bit: 0 0: ACK received 0 1: NYET received 1 0: STALL received 1 1: Transaction error	R
15:3	—	The read value is undefined. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

L1REQ bit (L1 Transition Request)

Set the L1REQ bit to 1 to transition to the L1 state. When the USBHS detects that this bit is 1, it starts the LPM transaction. The USBHS clears this bit to 0 through hardware on completion of the transaction.

L1STATUS[1:0] bits (L1 Request Completion Status)

The L1STATUS[1:0] bits indicate the result of the LPM transaction initiated by the L1REQ bit.

38.2.48 HL1CTRL2 : Host L1 Control Register 2

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x14A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BESL	—	—	L1RWE	HIRD[3:0]				—	—	—	—	L1ADDR[3:0]			
Value after reset:	0	x	x	0	0	0	0	0	x	x	x	x	0	0	0	0

Bit	Symbol	Function	R/W
3:0	L1ADDR[3:0]	LPM Token Device Address Specify the value to be set in the ADDR field of the LPM token	R/W
7:4	—	The read values are undefined. The write value should be 0.	R/W
11:8	HIRD[3:0]	LPM Token HIRD Specify the value to be set in the HIRD field of the LPM token	R/W
12	L1RWE	LPM Token L1 Remote Wake Enable Specify the value to be set in the RWE field of the LPM token	R/W
14:13	—	The read values are undefined. The write value should be 0.	R/W
15	BESL	BESL & Alternate HIRD Selects the K-State drive period on L1 Resume	R/W

Note: S-TYPE-3, P-TYPE-3

L1ADDR[3:0] bits (LPM Token Device Address)

The L1ADDR[3:0] bits specify the value to be set in the ADDR field of the LPM token that the USBHS transmits when the HL1CTRL1.L1REQ bit is set to 1.

HIRD[3:0] bits (LPM Token HIRD)

The HIRD[3:0] bits specify the value to be set in the HIRD field of the LPM token that the USBHS transmits when the HL1CTRL1.L1REQ bit is set to 1. Table 38.15 shows the relationship between the HIRD settings and the HIRD field values.

Table 38.15 Relationship between the HIRD bit settings and the HIRD field values (1 of 2)

HIRD[3:0] setting	When BESL = 0	When BESL = 1
0x0	50 μs (setting prohibited)	75 μs
0x1	125 μs	100 μs
0x2	200 μs	150 μs
0x3	275 μs	250 μs
0x4	350 μs	350 μs
0x5	425 μs	450 μs
0x6	500 μs	950 μs
0x7	575 μs	1950 μs
0x8	650 μs	2950 μs
0x9	725 μs	3950 μs
0xA	800 μs	4950 μs
0xB	875 μs	5950 μs
0xC	950 μs	6950 μs
0xD	1025 μs (setting prohibited)	7950 μs
0xE	1100 μs (setting prohibited)	8950 μs

Table 38.15 Relationship between the HIRD bit settings and the HIRD field values (2 of 2)

HIRD[3:0] setting	When BESL = 0	When BESL = 1
0xF	1175 μs (setting prohibited)	9950 μs

Note: The set value of the HIRD bit is used for the host K drive period on host resume and for the host K period on remote wakeup.

L1RWE bit (LPM Token L1 Remote Wake Enable)

The L1RWE bit specifies the value to be set in the RWE field of the LPM token that the USBHS transmits when the HL1CTRL1.L1REQ bit is set to 1.

The USBHS does not control detection of the remote wakeup signal in the L1 state with this bit. The remote wakeup signal is controlled by the DVSTCTR0.RWUPE bit, as with Suspend.

BESL bit (BESL & Alternate HIRD)

The BESL bit selects the K-state drive period on L1 Resume. For details, see the description of the HIRD bits.

38.2.49 DPUSR0R : Deep Software Standby USB Transceiver Control/Pin Monitor Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x160

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	DVBSTSHM	—	DOVCBHM	DOVCAHM	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	x	0	x	x	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
19:0	—	These bits are read as 0. The write value should be 0.	R
20	DOVCAHM	OVRCURA Input Flag Indicates OVRCURA input signal on the USBHS side	R
21	DOVCBHM	OVRCURB Input Flag Indicates OVRCURB input signal on the USBHS side	R
22	—	This bit is read as 0. The write value should be 0.	R
23	DVBSTSHM	VBUS Input Flag Indicates VBUS input signal on the USBHS side	R
31:24	—	These bits are read as 0. The write value should be 0.	R

Note: S-TYPE-3, P-TYPE-3
Note: This register is accessed by the clock of PCLKA/64 to reduce current consumption in Deep Software Standby mode 1.

38.2.50 DPUSR1R : Deep Software Standby USB Suspend/Resume Interrupt Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x164

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	DVBSTSH	—	DOVCBH	DOVCAH	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	DVBSTSHE	—	DOVCBHE	DOVCAHE	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	—	These bits are read as 0. The write value should be 0.	R/W
4	DOVCAHE	OVRCURA Interrupt Enable Clear 0: Disable recovery from Deep Software Standby mode 1 1: Enable recovery from Deep Software Standby mode 1	R/W
5	DOVCBHE	OVRCURB Interrupt Enable Clear 0: Disable recovery from Deep Software Standby mode 1 1: Enable recovery from Deep Software Standby mode 1	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	DVBSTSHE	VBUS Interrupt Enable/Clear 0: Disable recovery from Deep Software Standby mode 1 1: Enable recovery from Deep Software Standby mode 1	R/W
19:8	—	These bits are read as 0. The write value should be 0.	R/W
20	DOVCAH	OVRCURA Interrupt Source Return Status Flag 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1	R
21	DOVCBH	OVRCURB Interrupt Source Return Status Flag 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1	R
22	—	This bit is read as 0. The write value should be 0.	R/W
23	DVBSTSH	VBUS Interrupt Source Return Status Flag 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1	R
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note: This register is accessed by the clock of PCLKA/64 to reduce current consumption in Deep Software Standby mode 1.

38.2.51 DPUSR2R : Deep Software Standby USB Suspend/Resume Interrupt Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x168

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	DPINTE	DMINTE	—	—	DMVAL	DPVAL	—	—	DPINT	DMINT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DMINT	Indication of Return from DM Interrupt Source 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1	R
1	DPINT	Indication of Return from DP Interrupt Source 0: System has not recovered from Deep Software Standby mode 1 1: System recovered from Deep Software Standby mode 1	R
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	DPVAL	DP Input Indicates DP input signal on the USBHS side	R
5	DMVAL	DM Input Indicates DM input signal on the USBHS side	R
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	DMINTE	DM Interrupt Enable Clear 0: Disable recovery from Deep Software Standby mode 1 1: Enable recovery from Deep Software Standby mode 1	R/W
9	DPINTE	DP Interrupt Enable Clear 0: Disable recovery from Deep Software Standby mode 1 1: Enable recovery from Deep Software Standby mode 1	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note: This register is accessed by the clock of PCLKA/64 to reduce current consumption in Deep Software Standby mode 1.

38.2.52 DPUSRCR : Deep Software Standby USB Suspend/Resume Command Register

Base address: USBHS = 0x4035_1000
USBHS_NS = 0x5035_1000

Offset address: 0x16A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FIXPHYPD	FIXPHY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FIXPHY	USB Transceiver Control Fix 0: Normal mode 1: Invoke/recover from Deep Software Standby mode 1	R/W
1	FIXPHYPD	USB Transceiver Control Fix for PLL 0: Normal mode 1: Invoke/recover from Deep Software Standby mode 1	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note: This register is accessed by the clock of PCLKA/64 to reduce current consumption in Deep Software Standby mode 1.

38.3 Operation

38.3.1 System Control

This section describes register settings required for initializing the USBHS and controlling power consumption.

38.3.1.1 Setting Data to the USBHS Registers

Setting the SYSCFG.USBE bit to 1 after starting the PHY clock supply enables and starts USBHS operation. For information on how to supply the PHY clock, see section 38.3.3. Supplying the Clock.

38.3.1.2 Selecting the Controller Function

The USBHS can operate as a host or device controller.

Use the SYSCFG.DCFM bit to select one of these USBHS functions. The DCFM bit must be changed in the initial settings immediately after a reset or in the D+ pull-up-disabled state (SYSCFG.DPRPU bit = 0) and D+ and D- pull-down- disabled state (SYSCFG.DRPD bit = 0).

38.3.2 Controlling the USB Data Bus Using Resisters

The USBHS provides pull-up and pull-down resistors for the D+ and D- lines. Pull these lines up or down by setting the SYSCFG.DPRPU and DRPD bits.

In device controller mode, confirm that connection to the USB host is made, and then set the SYSCFG.DPRPU bit to 1 and pull up the D+ line (in full-speed communication).

When the SYSCFG.DPRPU bit is set to 0 during communication with a PC, the USBHS disables the pull-up resistor for the USB data line, thereby notifying the USB host of disconnection.

Table 38.16 shows the settings for controlling the resistors for the USB data bus. Control the USB data bus appropriately for your system using the DRPD and DPRPU bit settings.

Table 38.16 Control settings for the USB data bus resistors (excluding OTG operation)

SYSCFG register settings		USB data bus control		
DRPD bit	DPRPU bit	D-Line	D+Line	Function
0	0	Open	Open	When resistors not used
0	1	Open	Pull-Up	When operating as a device controller at full-speed
1	0	Pull-Down	Pull-Down	When operating as a host controller
1	1	—	—	Setting prohibited except during OTG operation

38.3.3 Supplying the Clock

Table 38.17 shows the two input clocks required for the USBHS.

Table 38.17 Input clocks

Input clock name	Description
PCLKA	Peripheral module clock A input. There is no constraint on the frequency of the PCLKA input.
PHY clock	PHY clock generated from external input or internal supply ● External input: The clock is generated by the USB-PHY internal PLL based on a 12-MHz, 20-MHz, 24-MHz, or 48-MHz clock supplied to the EXTAL pin from outside the MCU. For the external clock specifications, especially the jitter characteristics, strictly follow the specifications of ±50 ppm. ● Internal supply: The clock is generated by supplying 48 MHz and 60 MHz to the USB-PHY module and selecting CL-only mode (PHYSET.HSEB). High-speed operation is not supported with this mode.

Figure 38.2 illustrates the PHY clock settings.

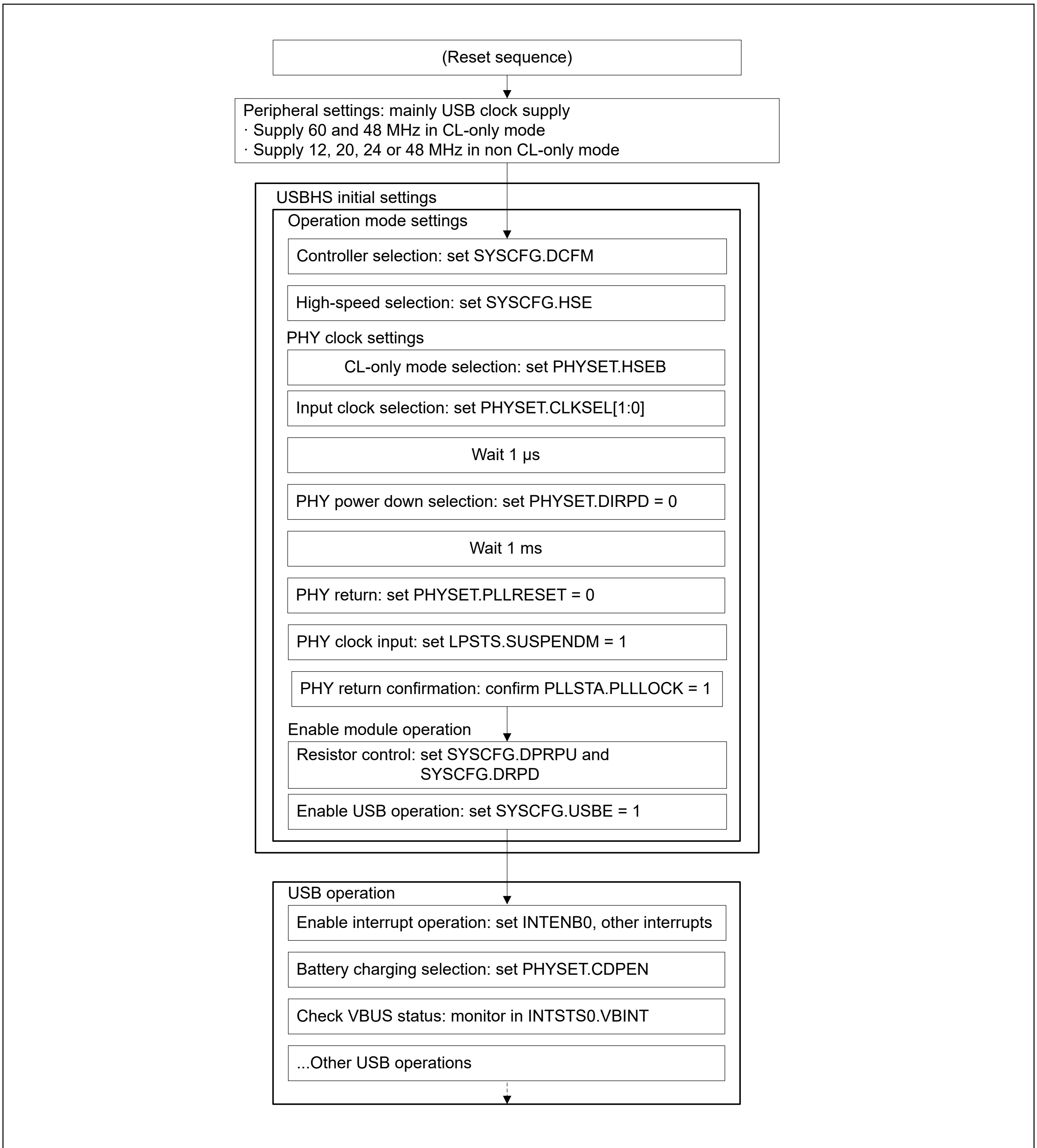


Figure 38.2 PHY clock settings

38.3.4 Constraints on Stopping the Clock

PCLKA and PHY clock can be stopped during disconnection or suspension. However, to stop any of these clocks while the USB is suspended in device controller mode, the stopped clock must be resupplied using the resume interrupt. The PHY clock must be resupplied within 5.5 ms after the resume interrupt is generated.

38.3.5 Interrupts

Table 38.18 lists the interrupt sources in the USBHS. When an interrupt generation condition is satisfied and the interrupt output is enabled using the associated interrupt enable register, the USBHS issues a USBHS interrupt request to the Interrupt Controller Unit (ICU) and a USBHS interrupt is generated.

Table 38.18 Interrupt sources (1 of 2)

Flag to be set to 1	Interrupt name	Interrupt source	Applicable controller function	Status flag
VBINT	VBUS interrupt	• A change in the state of the USB_VBUS input pin is detected (low to high or high to low)	Host or function*1	INTSTS0.VBSTS
RESM	Resume interrupt	• A change in the state of the USB bus is detected in the Suspend state (J-state to K-state or J-state to SE0)	Function	—
SOFR	Frame number update interrupt	In host controller mode: • An SOF packet with a different frame number is transmitted In device controller mode: • When SOFRM is 0: An SOF packet with a different frame number is received • When SOFRM is 1: Failed to receive an SOF packet with the μ frame number 0 because the packet is corrupted.	Host or function	—
DVST	Device state transition interrupt	• A device state transition is detected because of one of the following: – USB bus reset is detected – Suspend state is detected – SET_ADDRESS request is received – SET_CONFIGURATION request is received	Function	PL1CTRL.DVSQ[3:0]
CTRT	Control transfer stage transition interrupt	• A control transfer stage transition is detected because of one of the following: – Setup stage completed – Control write transfer status stage transition occurred – Control read transfer status stage transition occurred – Control transfer completed – Control transfer sequence error occurred	Function	INTSTS0.CTSQ[2:0]
BEMP	Buffer empty interrupt	• The buffer is empty after all FIFO buffer data is transmitted • A packet larger than the maximum packet size is received	Host or function	BEMPSTS.PIPEBEMP
NRDY	Buffer not ready interrupt	In host controller mode: • A STALL response is received from the peripheral device in response to the issued token • The response from the peripheral device in response to the issued token is not received successfully (no response three times consecutively or packet reception error three times consecutively) • An overrun or underrun error occurred during isochronous transfer In device controller mode: • NAK is returned for an IN or OUT token while the PID [1:0] bits were set to 01b (BUF) • A CRC error or bit stuffing error occurred during data reception in isochronous transfer • An interval error occurred during data reception in isochronous transfer	Host or function	NRDYSTS.PIPENRDY
BRDY	Buffer ready interrupt	• The buffer is ready (read or write state)	Host or function	BRDYSTS.PIPEBRDY
OVRCR	Overcurrent input change interrupt	• USBHS_OVRCURA, USBHS_OVRCURA-DS, USBHS_OVRCURB or USBHS_OVRCURAB-DS pin state change is detected (low to high or high to low)	Host or function	SYSSTS0.OVCMON [1:0]

Table 38.18 Interrupt sources (2 of 2)

Flag to be set to 1	Interrupt name	Interrupt source	Applicable controller function	Status flag
BCHG	Bus change interrupt	• USB bus state change is detected	Host	—
DTCH	Device disconnect detection interrupt	• Peripheral device disconnect is detected	Host	—
ATTCH	Device connect detection interrupt	• J-state or K-state is detected on the USB bus for 2.5 µs continuously This interrupt can be used to check whether peripheral devices are connected.	Host	—
EOFERR	EOF error detection interrupt	• An EOF error is detected for a peripheral device	Host	—
SACK	Setup normal interrupt	• A setup transaction normal response (ACK) is received	Host	—
SIGN	Setup error interrupt	• A setup transaction error (no response or ACK packet corruption) is detected three consecutive times	Host	—
PDDETINT	PDDETSTS change detect interrupt	• PDDET pin change is detected	Host or function	BCCTRL.PDDETSTS
LPMEND	LPM transaction end interrupt	• LPM transaction is complete	Host	PL1CTRL.DVSQ[3:0]
L1RSMEND	L1 resume end interrupt	• Resume (from L1 state) processing is complete	Host	PL1CTRL.DVSQ[3:0]

Note 1. Although this interrupt can be generated in host controller mode, it is not usually used in this mode.

38.3.5.1 Selecting the USBHS Interrupt Detection Method

Table 38.19 shows operations for an USBHS interrupt output from the USBHS. In case two or more interrupt sources are generated, the USBHS interrupt output method can be set in the SOFCFG.INTL bit. Set the USBHS interrupt output operation based on your system.

Table 38.19 USBHS interrupt operation

USBHS interrupt output (INTL setting)	When one interrupt source is generated	When two or more interrupt sources are generated
Edge detection (SOFCFG.INTL bit = 0)	Low level output until the source is cleared	When one source is cleared, the USBHS interrupt is negated for 32 clocks at 48 MHz (high pulse output)
Level detection (SOFCFG.INTL bit = 1)	Low level output until the source is cleared	Low level output until all sources are cleared

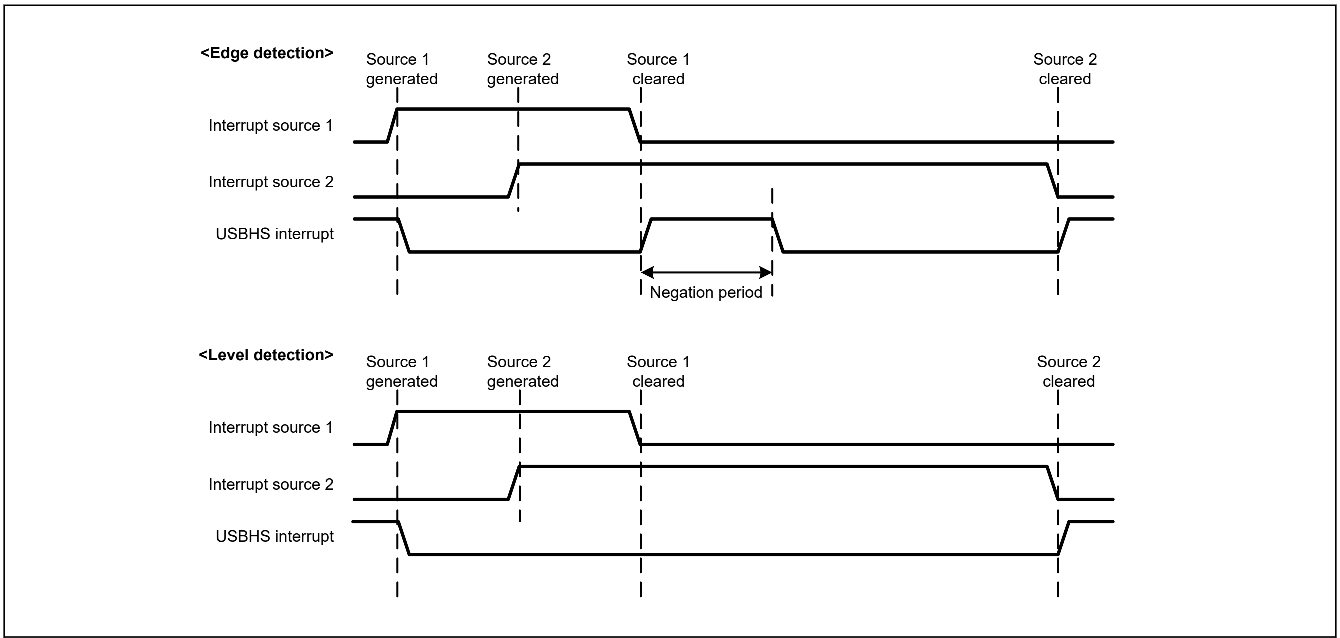


Figure 38.3 USBHS interrupt operation

Figure 38.4 shows an interrupt association chart of the USBHS.

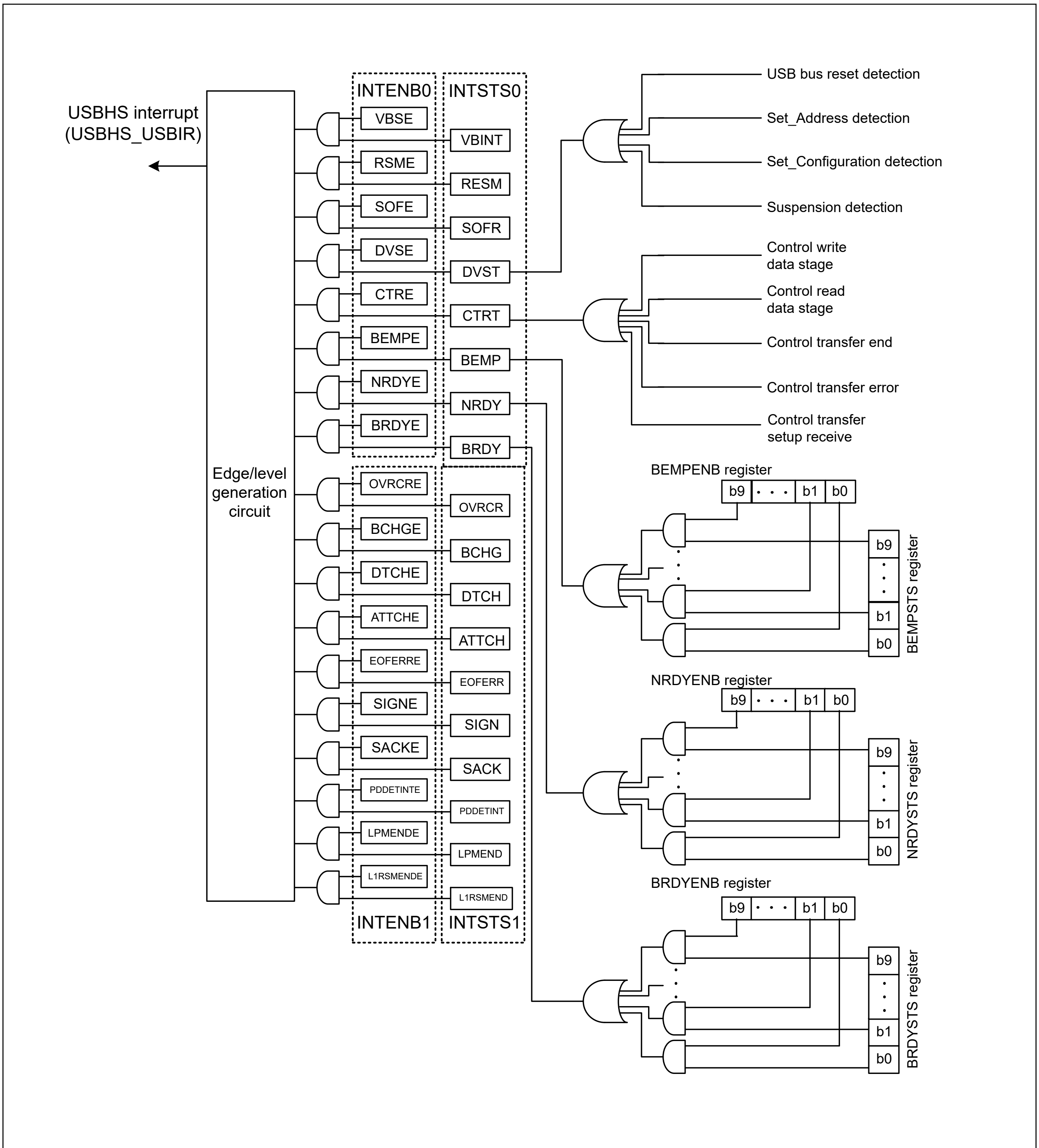


Figure 38.4 USBHS interrupt-related circuits

Table 38.20 shows the interrupts generated by the USBHS.

Table 38.20 USBHS interrupts (1 of 2)

Interrupt name	Interrupt status flag	DTC activation	DMAC activation
USBHS_D0FIFO	DMA transfer request 0	Possible	Possible
USBHS_D1FIFO	DMA transfer request 1	Possible	Possible

Table 38.20 USBHS interrupts (2 of 2)

Interrupt name	Interrupt status flag	DTC activation	DMAC activation
USBHS_USBIR	VBUS interrupt, resume interrupt, frame number update interrupt, device state transition interrupt, control transfer stage transition interrupt, buffer empty interrupt, buffer not ready interrupt, buffer ready interrupt, overcurrent interrupt, bus change interrupt, device disconnect detection interrupt, device connect detection interrupt, EOF error detection interrupt, normal setup operation interrupt, setup error interrupt, PDDETSTS change detection interrupt, LPM transaction end interrupt, and L1 resume end interrupt	Not possible	Not possible

38.3.6 Interrupt Descriptions

38.3.6.1 BRDY Interrupt

The BRDY interrupt is generated in both host and device controller modes. This section describes the conditions in which the USBHS sets the associated bit in BRDYSTS to 1. Under these conditions, the USBHS generates a BRDY interrupt if the software sets the bit in BRDYENB associated with the given pipe to 1 and INTENB0.BRDYE bit to 1.

The conditions for generating and clearing the BRDY interrupt depend on the SOFCFG.BRDYM and PIPECFG.BFRE settings for each pipe as follows:

(1) When SOFCFG.BRDYM = 0 and PIPECFG.BFRE = 0

With these settings, the BRDY interrupt indicates that the FIFO port is accessible.

On any of the following conditions, the USBHS generates an internal BRDY interrupt request trigger and sets the BRDYSTS.PIPEBRDY flag associated with the pipe to 1.

For transmitting pipes

- When the DIR bit is changed from 0 to 1 by software
- When writing by the CPU to the FIFO buffer is disabled for a pipe (when the BSTS flag is read as 0) and the USBHS has completed packet transmission. In continuous transfer, a BRDY interrupt is generated on completion of the transmission of data from one FIFO buffer.
- When one FIFO buffer is empty on completion of writing data to the other FIFO buffer in double buffer mode
- No request trigger is generated until completion of writing data to the currently-written FIFO buffer even if transmission to the other FIFO buffer is complete
- When the hardware flushes the buffer of the pipe for isochronous transfers
- When 1 is written to the PIPEnCTR.ACLRM bit, which causes the FIFO buffer to transition from the write-disabled to write-enabled state.

No request trigger is generated for the DCP, that is, during data transmission for control transfers.

For receiving pipes

- When packet reception is successfully complete, enabling the FIFO buffer to be read while read-access from the CPU to the FIFO buffer for the given pipe is disabled (when the BSTS flag is read as 0). No request trigger is generated for transactions in which DATA-PID mismatch has occurred. In continuous transmission or reception mode, the request trigger is not generated when the data is of the specified maximum packet size and the buffer has available space. When a short packet is received, the request trigger is generated even if the FIFO buffer has available space. When the transaction counter is used, the request trigger is generated on receiving the specified number of packets. In this case, the request trigger is generated even if the FIFO buffer has available space.
- When one FIFO buffer is read-enabled on completion of reading data from the other FIFO buffer in double buffer mode. No request trigger is generated until completion of reading data from the currently-read FIFO buffer, even if reception by the other FIFO buffer is complete.
  In device controller mode, the BRDY interrupt is not generated in the status stage of control transfers. The PIPEBRDY interrupt status of the selected pipe can be set to 0 by writing 0 to the associated PIPEBRDY flag through the software. In this case, 1s must be written to the associated bits for the other pipes. Clear the BRDY status before accessing the FIFO buffer.

(2) When SOFCFG.BRDYM = 0 and PIPECFG.BFRE = 1

With these settings, the USBHS generates a BRDY interrupt on completion of reading all data for a single transfer using the receiving pipe, and sets the bit in BRDYSTS associated with the pipe to 1.

On any of the following conditions, the USBHS determines that the last data for a single transfer was received:

- When a short packet including a zero-length packet is received
- When the PIPEnTRN register is used and the number of packets specified in the PIPEnTRN.TRNCNT[15:0] bits are completely received

When the data is completely read after any of these conditions is satisfied, the USBHS determines that all data for a single transfer is completely read.

When a zero-length packet is received while the FIFO buffer is empty, the USBHS determines that all data for a single transfer is completely read when the FRDY flag in the FIFO port control register is 1 and the DTLN[11:0] flags are 0. In this case, to start the next transfer, write 1 to the BCLR bit in the associated port control register through software. With these settings, the USBHS does not detect a BRDY interrupt for the transmitting pipe.

The PIPEBRDY interrupt status of a pipe can be set to 0 by writing 0 to the associated BRDYSTS.PIPEBRDY flag through the software. In this case, 1s must be written to the PIPEBRDY bits for the other pipes.

In this mode, do not change the PIPECFG.BFRE bit setting until all data for a single transfer is processed. When it is necessary to change the PIPECFG.BFRE bit before completion of processing, all FIFO buffers for the pipe must be cleared using the PIPEnCTR.ACLRM bit.

(3) When SOFCFG.BRDYM = 1 and PIPECFG.BFRE = 0

With these settings, the BRDYSTS.PIPEBRDY flag values are linked to the BSTS flag setting for each pipe. In other words, the BRDY interrupt status bits are set to 1 or 0 by the USBHS depending on the FIFO buffer status.

For transmitting pipes

The BRDY interrupt status bits are set to 1 when the FIFO port is ready for write access, and are set to 0 when it is not ready. The BRDY interrupt is not generated for the DCP in the transmitting direction even when it is ready for write access.

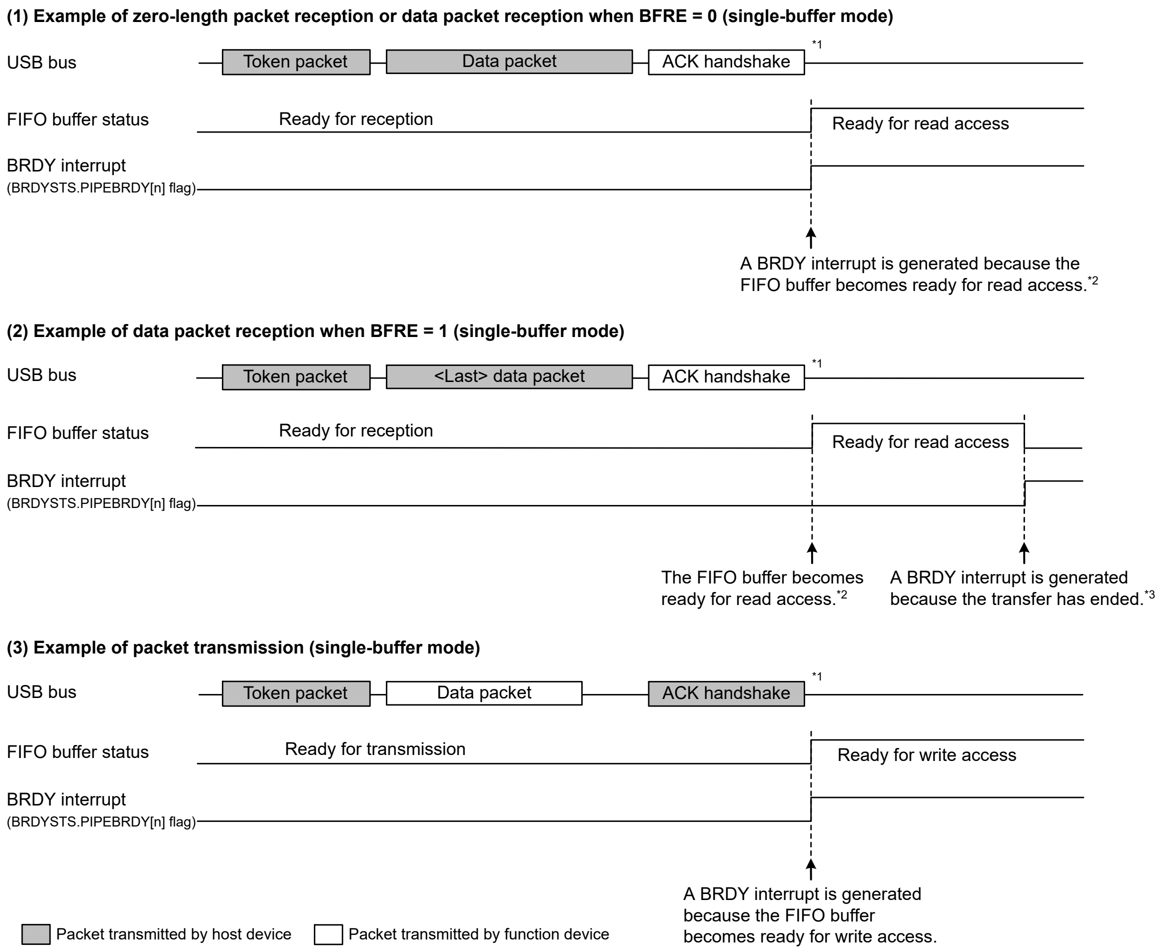
For receiving pipes

The BRDY interrupt status bits are set to 1 when the FIFO buffer is ready for read access, and are set to 0 when all data is read (not ready for read access).

When a zero-length packet is received while the FIFO buffer is empty, the associated bit is set to 1 and the BRDY interrupt is continuously generated until the software writes 1 to BCLR. With this setting, the PIPEBRDY flag cannot be set to 0 by software.

When the SOFCFG.BRDYM bit is set to 1, set the PIPECFG.BFRE bit for all pipes to 0, and the SOFCFG.INTL bit to 1 for level detection.

Figure 38.5 shows the timing of BRDY interrupt generation.



Note 1. The ACK handshake is not used in isochronous transfers.

Note 2. The FIFO buffer becomes ready for read access under the following condition:
When a packet is received while no unread data remains in the FIFO buffer in the CPU.

Note 3. A transfer ends under either of the following conditions:
1. When a short packet including a zero-length packet is received
2. When the number of packets specified in the transaction counter are received

Figure 38.5 Timing of BRDY interrupt generation

The condition for clearing the INTSTS0.BRDY flag depends on the SOFCFG.BRDYM bit setting value, as shown in Table 38.21

Table 38.21 Conditions for clearing the BRDY flag

BRDYM bit	Condition for clearing BRDY flag
0	The USBHS clears the BRDY flag to 0 when all bits in BRDYSTS are set to 0 by software
1	The USBHS clears the BRDY flag to 0 when the BSTS flags for all pipes have cleared to 0

38.3.6.2 NRDY Interrupt

On generating an internal NRDY interrupt request for the pipe whose PID[1:0] bits are set to 01b (BUF response) by software, the USBHS sets the associated NRDYSTS.PIPENRDY flag to 1. If the associated bit in NRDYENB is set to 1 by software, the USBHS sets the INTSTS0.NRDY flag to 1 and generates a USBHS interrupt.

This section describes the conditions in which the USBHS generates the internal NRDY interrupt request for a given pipe.

The internal NRDY interrupt request is not generated during setup transaction execution in host controller mode. During setup transactions in host controller mode, the SACK or SIGN interrupt is detected.

The internal NRDY interrupt request is not generated during status stage execution of the control transfer in device controller mode

(1) In host controller mode when no split transactions occur in the connection

For transmitting pipes

On any of the following conditions, the USBHS detects an NRDY interrupt:

- For isochronous transfer pipes, when the time to issue an OUT token comes while there is no data to be transmitted in the FIFO buffer. In this case, the USBHS transmits a zero-length packet following the OUT token and sets the associated NRDYSTS.PIPENRDY flag and the FRMNUM.OVRN flag to 1.
- During communications other than setup transactions on pipes not used for isochronous transfers, when any combination of the following two conditions occurs three consecutive times:
  - No response is returned from the peripheral device (when timeout is detected before detection of the handshake packet from the peripheral device
  - An error is detected in the packet from the peripheral device. In this case, the USBHS sets the associated PIPENRDY flag to 1 and changes the PID[1:0] setting for the associated pipe to 00b (NAK response)
- During communications other than setup transactions, when the STALL handshake is received from the peripheral device (includes STALL for both OUT and PING). In this case, the USBHS sets the associated PIPENRDY flag to 1 and changes the PID[1:0] setting for the associated pipe to 11b (STALL response).

For receiving pipes

- For isochronous transfer pipes, when the time to issue an IN token comes but there is no space available in the FIFO buffer. In this case, the USBHS discards the received data for the IN token and sets the associated PIPENRDY flag and the OVRN flag to 1. When a packet error is detected in the received data for the IN token, the USBHS also sets the FRMNUM.CRCE flag to 1.
- For non-isochronous transfer pipes, when any combination of the following two cases occur three consecutive times:
  - No response is returned from the peripheral device for the IN token issued by the USBHS (when timeout is detected before detection of the DATA packet from the peripheral device)
  - An error is detected in the packet from the peripheral device. In this case, the USBHS sets the associated PIPENRDY flag to 1 and changes the associated PID[1:0] setting for the pipe to 00b (NAK response).
- For isochronous transfer pipes, when no response is returned from the peripheral device for the IN token (when timeout is detected before detection of the DATA packet from the peripheral device) or an error is detected in the packet from the peripheral device. In this case, the USBHS sets the associated NRDYSTS.PIPENRDY flag for each pipe to 1. The PID[1:0] setting for the pipe is not changed.
- For isochronous transfer pipes, when a CRC error or a bit stuffing error is detected in the received data packet. In this case, the USBHS sets the associated NRDYSTS.PIPENRDY flag for each pipe and the CRCE flag to 1.
- When the STALL handshake is received. In this case, the USBHS sets the associated NRDYSTS.PIPENRDY flag for each pipe to 1 and changes the PID[1:0] setting for the associated pipe to STALL.

(2) In host controller mode when split transactions occur in the connection

For transmitting pipes

On any of the following conditions, the USBHS detects an NRDY interrupt:

- For isochronous transfer pipes, when the time to issue an OUT token comes while there is no data to be transmitted in the FIFO buffer. In this case, the USBHS sets the associated RDYSTS.PIPENRDY flag for the given pipes to 1 on issuing a start-split transaction and sets the FRMNUM.OVRN flag to 1. The USBHS also transmits a zero-length packet following the OUT token.
- For non-isochronous transfer pipes, when any combination of the following two cases occurs three consecutive times:
  - No response is returned from the hub for start-split and complete-split transactions (when timeout is detected before detection of the handshake packet from the hub)

  - An error is detected in the packet from the hub. In this case, the USBHS sets the associated NRDYSTS.PIPENRDY flag for the pipe to 1 and changes the associated PID[1:0] setting for the pipe to 00b (NAK response). When an NRDY interrupt is detected on complete-split issuance, the USBHS clears the CSSTS flag to 0.
- When a STALL handshake is received for the complete-split transaction. In this case, the USBHS sets the associated NRDYSTS.PIPENRDY flag for the pipe to 1, changes the associated PID[1:0] setting for the pipe to 11b (STALL response), and clears the CSSTS flag to 0. An interrupt is not detected during setup transaction.

For receiving pipes

- For isochronous transfer pipes, when the time to issue an IN token comes but there is no space available in the FIFO buffer. In this case, the USBHS sets the associated NRDYSTS.PIPENRDY flag for the given pipe and the FRMNUM.OVRN flag to 1 on start-split issuance. The USBHS discards the received data for the IN token.
- During bulk-pipe transfers or transfers other than setup transactions with the DCP, when any combination of the following two cases occurs three consecutive times:
  - No response is returned from the hub for the IN token the USBHS issued on issuance of the start-split or complete-split transactions (when timeout is detected before detection of the data packet from the hub)
  - An error is detected in the packet from the hub. In this case, the USBHS sets the associated NRDYSTS.PIPENRDY flag for the pipe to 1 and changes the associated PID[1:0] setting for the pipe to 00b (NAK response). When this condition occurs during complete-split, the USBHS clears the CSSTS flag to 0.
- During a complete-split transaction for isochronous transfer or interrupt transfer pipes, when any combination of the following two cases occurs three consecutive times:
  - No response is returned from the hub for the IN token issued by the USBHS (when a timeout is detected before detection of the DATA packet from the hub)
  - An error is detected in the packet from the hub. On generating this condition for an interrupt transfer pipe, the USBHS sets the associated NRDYSTS.PIPENRDY flag to 1, changes the associated PID[1:0] setting for the pipe to 00b (NAK response), and clears the CSSTS flag to 0. On generating this condition for the pipe for isochronous transfers, the USBHS sets the associated NRDYSTS.PIPENRDY flag for the pipe to 1, CRCE flag to 1, and clears the CSSTS bit to 0. It does not change the PID[1:0] setting.
- During a complete-split transaction, when the STALL handshake is received for a non-isochronous transfer pipe. In this case, the USBHS sets the associated NRDYSTS.PIPENRDY flag for the pipe to 1, changes the associated PID[1:0] setting for the pipe to 11b (STALL response), and clears the CSSTS flag to 0.
- During a complete-split transaction, when the NYET handshake is received for an isochronous transfer or interrupt transfer pipe for the microframe number = 4. In this case, the USBHS sets the associated NRDYSTS.PIPENRDY flag for each pipe to 1 and the CRCE flag to 1, and clears the CSSTS flag to 0. It does not change the PID[1:0] setting.

(3) In device controller mode

For transmitting pipes

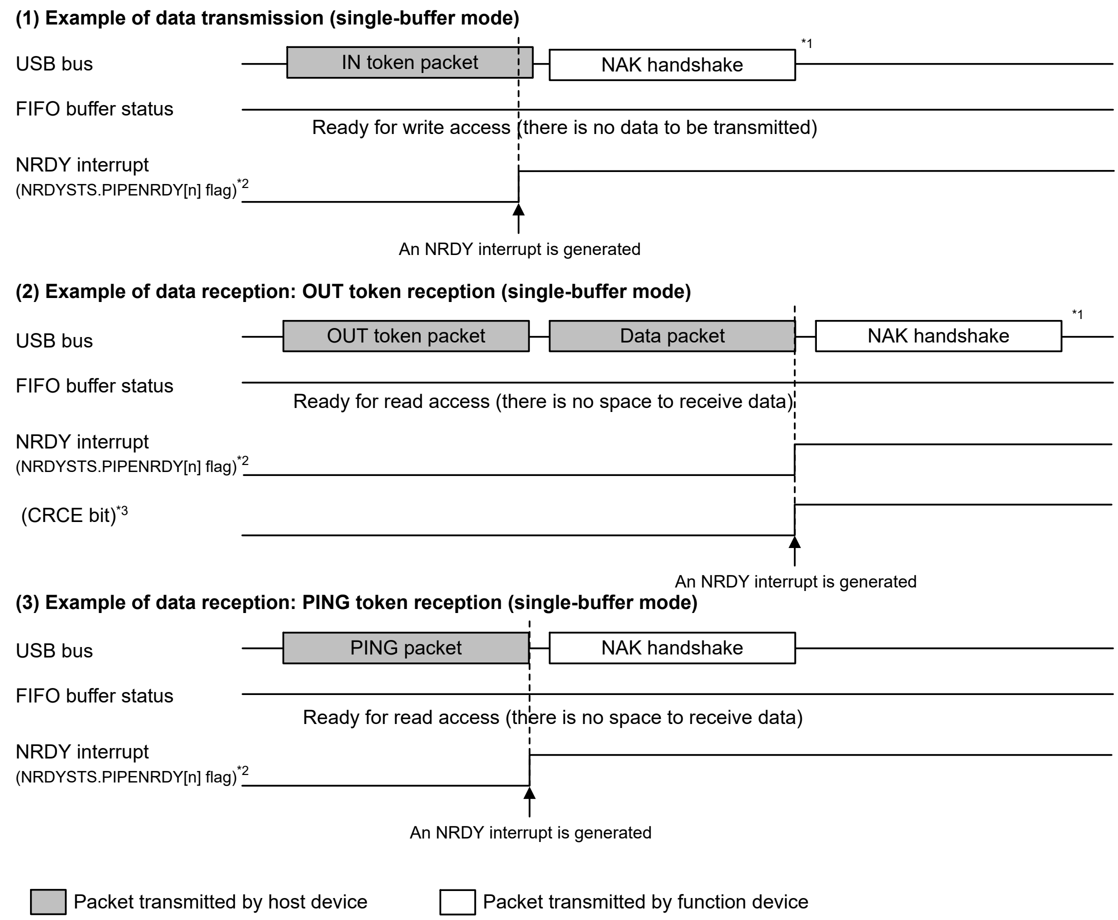
- When an IN token is received while there is no data to be transmitted in the FIFO buffer. In this case, the USBHS generates an NRDY interrupt request on reception of the IN token and sets the NRDYSTS.PIPENRDY flag to 1. For an isochronous transfer pipe in which an interrupt is generated, the USBHS transmits a zero-length packet and sets the FRMNUM.OVRN flag to 1.

For receiving pipes

- When an OUT token is received but there is no space available in the FIFO buffer. For an isochronous transfer pipe in which an interrupt is generated, the USBHS generates an NRDY interrupt request on reception of the OUT token and sets the NRDYSTS.PIPENRDY flag to 1 and the FRMNUM.OVRN flag to 1. For a non-isochronous transfer pipe in which an interrupt is generated, the USBHS generates an NRDY interrupt request when an NAK handshake is transferred after the data following the OUT token is received, and sets the NRDYSTS.PIPENRDY flag to 1. The NRDY interrupt request is not generated during retransmission because of a DATA-PID mismatch. In addition, the NRDY interrupt request is not generated if an error occurs in the DATA packet.
- On receiving a PING token when there is no space available in the FIFO buffer. The USBHS generates an NRDY interrupt request on reception of the PING token, setting the NRDYSTS.PIPENRDY flag to 1.

- For isochronous transfer pipes, when a token is not received successfully within an interval frame. In this case, the USBHS generates an NRDY interrupt request when the SOF is received, and sets the NRDYSTS.PIPENRDY flag to 1.

Figure 38.6 shows the timing of NRDY interrupt generation in device controller mode.



Note 1. The handshake is not used in isochronous transfers.
Note 2. The PIPENRDY[n] flag is changed to 1 only when PIPEnCTR.PID[1:0] bits are set to 01b (BUF response).
Note 3. The CRCE and OVRN flags change only while the target pipe is set to isochronous transfers.

Figure 38.6 Timing of NRDY interrupt generation in device controller mode

38.3.6.3 BEMP Interrupt

On detecting a BEMP interrupt for the pipe whose PID[1:0] bits in the pipe control register are set to 01b (BUF response) by software, the USBHS sets the associated BEMPSTS.PIPEBEMP flag to 1. If the associated BEMPENB bit is set to 1 by software, the USBHS sets the INTSTS0.BEMP flag to 1 and generates a USB interrupt. This section describes the conditions in which the USBHS generates an internal BEMP interrupt request.

(1) For transmitting pipes

When the FIFO buffer of the associated pipe is empty on completion of transmission, including zero-length packet transmission, and in single buffer mode, an internal BEMP interrupt request is generated simultaneously with the BRDY interrupt for a non-DCP pipe. The internal BEMP interrupt request is not generated in any of the following conditions:

- When the CPU or DMAC/DTC has already started writing data to the FIFO buffer of the CPU on completion of transmitting data from one FIFO buffer in double buffer mode

- When the buffer is cleared (emptied) by setting 1 to the PIPEnCTR.ACLRM or the BCLR bit in the port control register
- When an IN transfer (zero-length packet transmission) is performed during the control transfer status stage in device controller mode

(2) For receiving pipes

When a successfully-received data packet size exceeds the specified maximum packet size. In this case, the USBHS generates a BEMP interrupt request, sets the associated BEMPSTS.PIPEBEMP flag to 1, discards the received data, and changes the associated PID[1:0] setting for the pipe to STALL (11b). The USBHS returns no response in host controller mode, and returns STALL response in device controller mode.

The internal BEMP interrupt request is not generated in any of the following conditions:

- When a CRC error or a bit stuffing error is detected in the received data
- When a setup transaction is being performed:
  - Writing 0 to the BEMPSTS.PIPEBEMP flag clears the status
  - Writing 1 to the BEMPSTS.PIPEBEMP flag has no effect

Figure 38.7 shows the timing of BEMP interrupt generation in device controller mode.

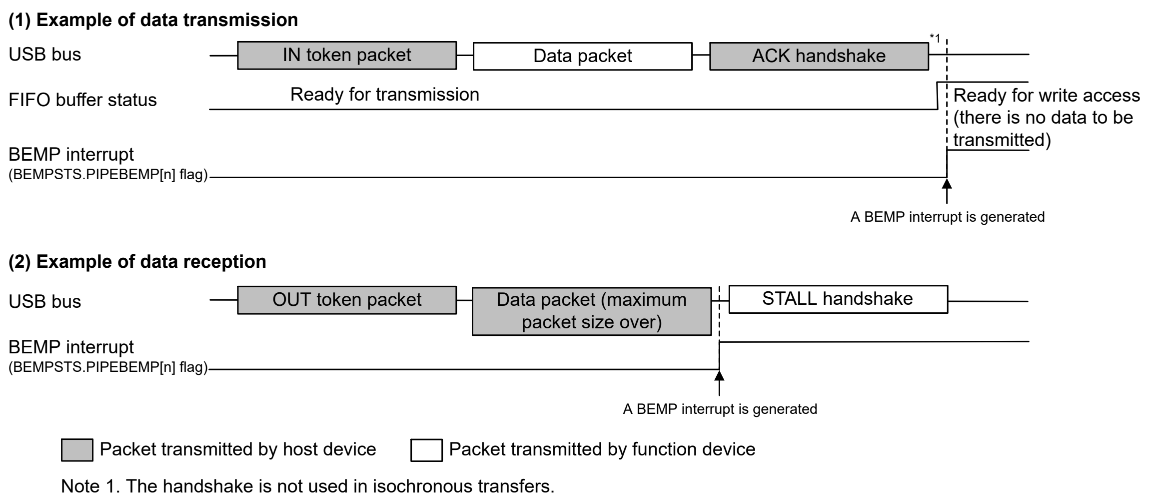


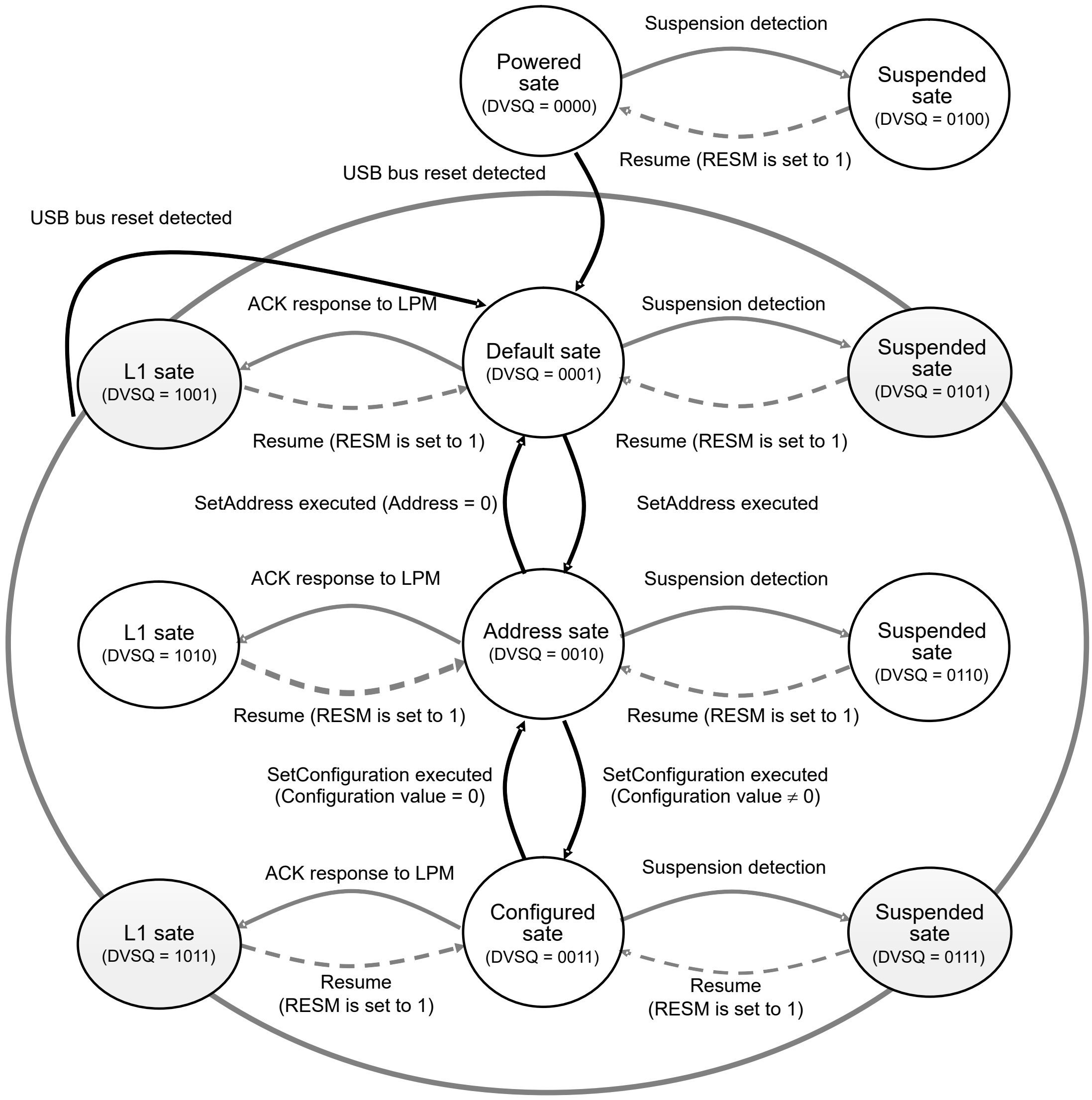
Figure 38.7 Timing of BEMP interrupt generation in device controller mode

38.3.6.4 Device State Transition Interrupt (Device Controller Mode)

Figure 38.8 shows a diagram of the USBHS device state transitions. The USBHS controls device states and generates device state transition interrupts. However, recovery from the Suspend state (resume signal detection) is detected by means of the resume interrupt. Device state transition interrupts can be enabled or disabled independently in INTENB0. Devices whose states have changed can be checked in the PL1CTRL.DVSQ[3:0] flags.

When a transition is made to the default state, a device state transition interrupt is generated after a USB bus reset is detected.

The USBHS controls device states, and device state transition interrupts can be generated, only in device controller mode.



Note 1. For a transition, indicated by a solid line, the DVST bit is set to 1. For a resume, indicated by a dashed line, the RESM bit is set to 1.

Figure 38.8 Device state transitions

38.3.6.5 Control Transfer Stage Transition Interrupt (Device Controller Mode)

Figure 38.9 shows a diagram of the control transfer stage transitions of the USBHS. The USBHS controls the control transfer sequence and generates control transfer stage transition interrupts. Control transfer stage transition interrupts can be enabled or disabled independently in INTENB0. Transfer stages that have transitioned can be checked in the INTSTS0.CTSQ[2:0] bits.

Control transfer stage transition interrupts are generated only in device controller mode. This section describes control transfer sequence errors. When an error occurs, the DCPCTR.PID[1:0] bits are set to 1xb (STALL response).

- Control read transfer errors
  - An OUT token or PING token is received, but no data is transferred in response to the IN token at the data stage
  - An IN token is received at the status stage

– A data packet with DATAPID = DATA0 is received at the status stage

- Control write transfer errors
  – An IN token is received, but no ACK returned in response to the OUT token in the data stage
  – A data packet with DATAPID = DATA0 is received as the first data packet at the data stage
  – An OUT token or PING token is received in the status stage
- Control write no data transfer errors
  – An OUT token or PING token is received at the status stage

At the control write transfer data stage, if the receive data length exceeds the wLength value of the USB request, it is not recognized as a control transfer sequence error. At the control read transfer status stage, packets other than zero-length packets are received by an ACK response and the transfer ends normally.

When a CTRT interrupt occurs in response to a sequence error (INTSTS0.CTRT flag = 1), the CTSQ[2:0] = 110b value is saved until CTRT flag clears to 0, clearing the interrupt status. While CTSQ[2:0] bits = 110b is being saved, no CTRT interrupt for ending the setup stage is generated, even if a new USB request is received. The USBHS saves the setup stage completion status, and it generates a CTRT interrupt after the interrupt status is cleared by software.

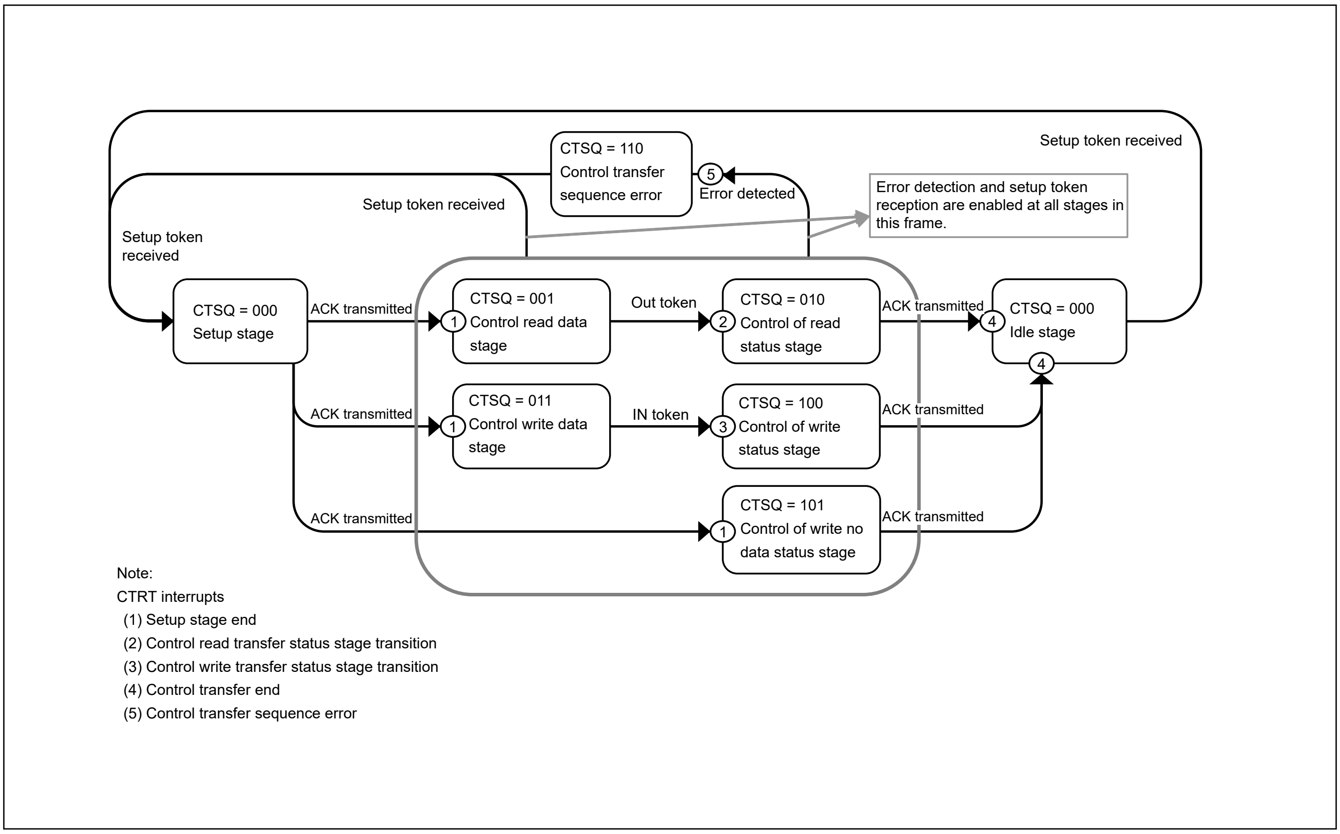


Figure 38.9 Control transfer stage transitions

38.3.6.6 Frame Update Interrupt

In host controller mode, an interrupt is generated when the frame number is updated.

In device controller mode, an SOFR interrupt is generated when the frame number is updated. The USBHS updates the frame number and generates an SOFR interrupt if it detects a new SOF packet during full-speed operation.

38.3.6.7 VBUS Interrupt

When the USBHS_VBUS pin level changes, a VBUS interrupt is generated. The level of the USBHS_VBUS pin can be checked in the INTSTS0.VBSTS flag. Whether the host controller is connected or disconnected can be confirmed using the VBUS interrupt. If the system is activated with the host controller connected, the first VBUS interrupt is not generated, because there is no change in the USBHS_VBUS pin level.

38.3.6.8 Resume Interrupt

In device controller mode, a resume interrupt is generated when the device is in the Suspend state and the USB bus state has changed (from J-state to K-state, or from J-state to SE0). Recovery from the Suspend state is detected by means of the resume interrupt.

In host controller mode, no resume interrupt is generated. Use the BCHG interrupt to detect a change in the USB bus state.

38.3.6.9 OVRCR Interrupt

An OVRCR interrupt is generated when the USBHS_OVRCURA, USBHS_OVRCURA-DS, USBHS_OVRCURB or USBHS_OVRCURB-DS pin level has changed. The levels of the USBHS_OVRCURA, USBHS_OVRCURA-DS, USBHS_OVRCURB or USBHS_OVRCURB-DS pins can be checked in the SYSSTS0.OVCMON[1:0] flags. The external power supply IC can check whether overcurrent is detected using the OVRCR interrupt.

For OTG connections, the OVRCR interrupt allows you to check whether a change is detected in the VBUS comparator.

38.3.6.10 BCHG Interrupt

A BCHG interrupt is generated when the USB bus state has changed. The BCHG interrupt can be used to detect whether a peripheral device is connected. It can also be used to detect a remote wakeup in host controller mode. The BCHG interrupt is generated in both host and device controller modes.

38.3.6.11 DTCH Interrupt

A DTCH interrupt occurs when a USB bus disconnect is detected in host controller mode. The USBHS detects bus disconnects in compliance with the USB 2.0 specification.

On interrupt detection, all pipes in which communications are being carried out for the relevant port must be terminated by software. The pipes enter the wait state for a bus connection to the port, waiting for an ATTCH interrupt to occur. Regardless of the value set in the associated interrupt enable bit, the USBHS hardware:

- Sets the DVSTCTR0.UACT bit for the port in which the DTCH interrupt is detected to 0
- Puts the port in which the DTCH interrupt occurred into the idle state

38.3.6.12 SACK Interrupt

A SACK interrupt is generated when an ACK response for the transmitted setup packet is received from the peripheral device in host controller mode. The SACK interrupt can be used to confirm that the setup transaction is successfully complete.

38.3.6.13 SIGN Interrupt

A SIGN interrupt is generated when an ACK response for the transmitted setup packet is not correctly received from the peripheral device three consecutive times in host controller mode. The SIGN interrupt can be used to detect no ACK response transmitted from the peripheral device or corruption of an ACK packet.

38.3.6.14 ATTCH Interrupt

An ATTCH interrupt is generated when J-state or K-state of the full-speed signal level is detected on the USB port for 2.5 µs in host controller mode. To be more specific, an ATTCH interrupt is detected in any of the following conditions:

- When K-state, SE0, or SE1 changes to J-state, and J-state continues 2.5 µs
- When J-state, SE0, or SE1 changes to K-state, and K-state continues 2.5 µs

38.3.6.15 EOFERR Interrupt

An EOFERR interrupt occurs when the USBHS detects that communication is not complete at the EOF2 timing defined in the USB 2.0 specification.

On interrupt detection, all pipes in which communications are being carried out for the relevant port must be terminated by software, and the port must be re-enumerated. Regardless of the value set in the associated interrupt enable bit, the USBHS hardware:

- Sets the DVSTCTR0.UACT bit for the port in which the EOFERR interrupt is detected to 0
- Puts the port in which the EOFERR interrupt is generated into the idle state

38.3.6.16 PDDETINT Interrupt

The USBHS sets the INTSTS1.PDDETINT flag to 1 on detecting a level change (high to low or low to high) in the PDDET pin input value and generates the PDDETINT interrupt. When the PDDETINT interrupt is generated, use software to repeatedly read the BCCTRL.PDDETSTS flag until the same value is read three or more times, and perform debounce processing.

38.3.6.17 LPMEND Interrupt

When the LPM transaction ends because a response from the peripheral device or a timeout is detected, the INTSTS1.LPMEND flag sets to 1 and the LPMEND interrupt is generated.

38.3.6.18 L1RSMEND Interrupt

When performing resume processing when the USBHS has transitioned to the L1 state because an ACK is received in response to an LPM token, the USBHS sets the INTSTS1.L1RSMEND flag to 1 on completion of the resume processing.

38.3.7 Pipe Control

Table 38.22 lists the pipe settings for the USBHS. USB data transfer is performed through logical pipes that the software associates with endpoints. The USBHS provides 10 pipes for data transfer. Set up the pipes based on your system specifications.

Table 38.22 Pipe settings (1 of 2)

Register name	Bit name	Setting	Notes
DCPCFG PIPECFG	TYPE[1:0]	Transfer type	Pipes 1 to 9: Settable
	BFRE	BRDY interrupt mode	Pipes 1 to 5: Settable
	DBLB	Double buffer select	Pipes 1 to 5: Settable
	CNTMD	Selection of continuous transfer or discontinuous transfer	Pipes 1, 2: Settable only for bulk transfers Pipes 3 to 5: Settable
	DIR	Transfer direction select	IN or OUT settable
	EPNUM[3:0]	Endpoint number	Pipes 1 to 9: Settable Set this number to a value other than 0000 when one or more pipes are used.
	SHTNAK	Selects disabled state for pipe when transfer ends	Pipes 1, 2: Settable only for bulk transfers Pipes 3 to 5: Settable
PIPEBUF	BUFSIZE	FIFO buffer size	DCP: Setting disabled (fixed to 256 bytes) Pipes 1 to 5: Settable up to 2 KB Pipes 6 to 9: Setting disabled (fixed to 64 bytes)
	BUFNMB	FIFO buffer number	DCP: Setting disabled (fixed to 0x0-0x3 area) Pipes 1 to 5: Setting disabled (0x8-0x87 area specifiable) Pipes 6 to 9: Setting disabled (fixed to 0x4-0x7 area)
DCPMAXP PIPEMAXP	DEVSEL[3:0]	Device select	Viewable only in host controller mode
	MXPS	Maximum packet size	Setting compliant with USB specification

Table 38.22 Pipe settings (2 of 2)

Register name	Bit name	Setting	Notes
PIPEPERI	IFIS	Buffer flush	Pipes 1, 2: Settable only for isochronous transfers Pipes 3 to 5: Setting disabled Pipes 6 to 9: Setting disabled
	IITV[2:0]	Interval counter	Pipes 1, 2: Settable only for isochronous transfers Pipes 3 to 5: Setting disabled Pipes 6 to 9: Settable only in host controller mode
DCPCTR PIPEnCTR	BSTS	Buffer status	For the DCP, receive buffer status and transmit buffer status are switched with the ISEL bit
	INBUFM	IN buffer monitor	Available only for pipes 1 to 5
	SUREQ	Setup request	Settable only for the DCP and controlled in host controller mode
	SUREQCLR	SUREQ clear	Settable only for the DCP and controlled in host controller mode
	CSCLR	CSSTS clear	Controllable only in host controller mode
	CSSTS	Split status check	Viewable only in host controller mode
	ATREPM	Auto response mode	Pipes 1 to 5: Settable only in device controller mode
	ACLRM	Auto buffer clear	Pipes 1 to 9: Settable
	SQCLR	Sequence clear	Clears the data toggle bit
	SQSET	Sequence set	Sets the data toggle bit
	SQMON	Sequence check	Monitors the data toggle bit
	PBUSY	PIPE busy check	—
	PID[1:0]	Response PID	—
PIPEnTRE	TRENB	Transaction count enable	Pipes 1 to 5: Settable
	TRCLR	Current transaction counter clear	Pipes 1 to 5: Settable
PIPEnTRN	TRNCNT	Transaction counter	Pipes 1 to 5: Settable

38.3.7.1 Pipe Control Register Switching Procedures

The following bits in the pipe control registers can be changed only when USB communication is prohibited (PID[1:0] bits are 00b (NAK response)). Figure 38.10 shows pipe control register switching procedures when USB communication is enabled (PID[1:0] bits are 00b (BUF response)).

Do not change the following registers and bits when USB communication is enabled (PID[1:0] bits are 01b (BUF response)):

- Bits in DCPCFG and DCPMAXP
- SQCLR and SQSET bits in DCPCTR
- Bits in PIPECFG, PIPEBUF, PIPEMAXP, and PIPEPERI
- ATREPM, ACLRM, SQCLR, and SQSET bits in PIPEnCTR
- Bits in PIPEnTRE and PIPEnTRN
- Bits in DEVADDn (n = 0 to A)

To set the CSCLR bits and bits in DEVADDn (n = 0 to A), follow the procedures described in section 38.2. Register Descriptions.

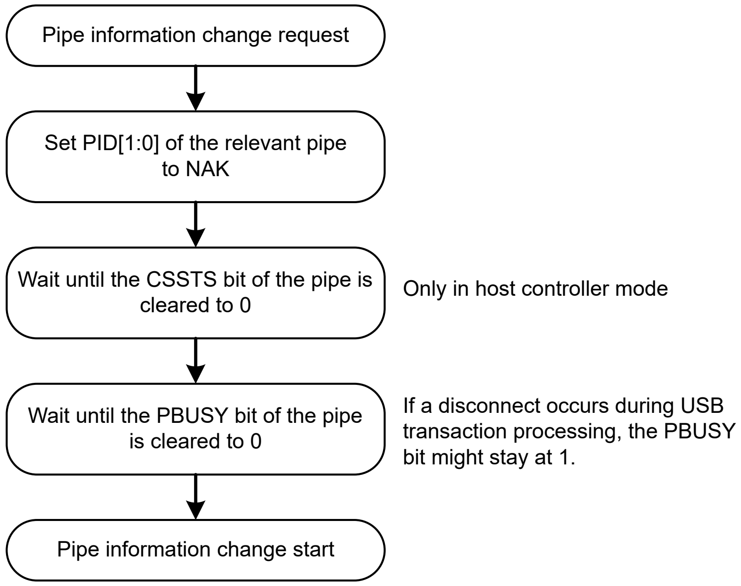


Figure 38.10 Procedure for changing pipe information when USB communication is enabled and PID[1:0] bits are 01b (BUF response)

The following bits in the pipe control registers can be changed only when the selected pipe information is not set in the CURPIPE[3:0] bits in CFIFOSEL, D0FIFOSEL, and D1FIFOSEL.

Do not set the following registers while the CURPIPE[3:0] bits are set:

- Bits in DCPCFG and DCPMAXP
- Bits in PIPECFG, PIPEBUF, PIPEMAXP, and PIPEPERI
- PIPEnCTR and ACLRM bits

To change pipe information, you must set the CURPIPE[3:0] bits to a pipe other than the one to be changed. For the DCP, the buffer must be cleared using the BCLR bit after the pipe information is changed.

38.3.7.2 Transfer Types

The PIPECFG.TYPE[1:0] bits specify the following transfer types for each pipe:

- DCP: No setting necessary (fixed at control transfer)
- Pipes 1 and 2: Set to bulk transfer or isochronous transfer
- Pipes 3 to 5: Set to bulk transfer
- Pipes 6 to 9: Set to interrupt transfer

38.3.7.3 Endpoint Number

The PIPECFG.EPNUM[3:0] bits are used to set the endpoint number for each pipe. The DCP is fixed at endpoint 0. The other pipes can be set from endpoint 1 to 15.

- DCP: No setting is necessary (fixed at endpoint 0)
- Pipes 1 to 9: Select and set the endpoint numbers from 1 to 15 so that the combination of the PIPECFG.DIR and EPNUM[3:0] bits is unique

38.3.7.4 Maximum Packet Size Setting

Specify the maximum packet size for each pipe in the MXPS bits in DCPMAXP and PIPEMAXP. The DCP and pipes 1 to 5 can be set to any of the maximum pipe sizes defined in the USB 2.0 specification. For pipes 6 to 9, the maximum packet size is 64 bytes. Set the maximum packet size as follows before starting a transfer (PID[1:0] bits are set to 01b (BUF response)):

- DCP: Set to 64 for high-speed operation
- DCP: Set to 8, 16, 32, or 64 for full-speed operation
- Pipes 1 to 5: Set to 512 for high-speed bulk transfers
- Pipes 1 to 5: Set to 8, 16, 32, or 64 for full-speed bulk transfers
- Pipes 1, 2: Set between 1 and 1024 for high-speed isochronous transfers
- Pipes 1, 2: Set between 1 and 1023 for full-speed isochronous transfers
- Pipes 6 to 9: Set between 1 and 64

High-bandwidth interrupt transfers and isochronous transfers are not supported.

38.3.7.5 Transaction Counter for Pipes 1 to 5 in the Receiving Direction

When the specified number of transactions is complete in the data packet receiving direction, the USBHS recognizes that the transfer has ended. Two transaction counters are provided. One is the PIPEnTRN register, which specifies the number of transactions to be executed, and the other is the current counter, which internally counts the number of executed transactions. If the PIPECFG.SHTNAK bit is set to 1, when the current counter value matches the specified number of transactions, the associated PIPEnCTR.PID[1:0] bits are set to 00b (NAK response) and the subsequent transfer is disabled. The transactions can be counted again from the beginning by initializing the current counter of the transaction counter function through the PIPEnTRE.TRCLR bit. The data read from PIPEnTRN differs depending on the PIPEnTRE.TRENB setting as follows:

- The TRENB bit = 0: Specified transaction counter value can be read
- The TRENB bit = 1: Current counter value indicating the internally counted number of executed transactions can be read.

The following constraints apply when working with the TRCLR bit:

- If the transactions are being counted and the PIPEnCTR.PID[1:0] bits are set to 01b (BUF response), the current counter cannot be cleared
- If there is any data left in the buffer, the current counter cannot be cleared

38.3.7.6 Response PID

Specify the response PID for each pipe in the PID[1:0] bits in DCPCTR and PIPEnCTR. This section describes the USBHS operation with different response PID settings.

Software response PID settings in host controller mode

Select the response PID to specify the execution of transactions as follows:

- NAK setting: Using pipes is disabled and no transactions are executed
- BUF setting: Transactions are executed based on the FIFO buffer state:
  OUT direction: An OUT token is issued if the FIFO buffer contains transmit data.
  IN direction: An IN token is issued if the FIFO buffer is not full and can receive data.
- STALL setting: Using pipes is disabled and no transactions are executed

Note: Use the SUREQ bit to execute setup transactions for the DCP.

Software response PID settings in device controller mode

Select the response PID to respond as follows to transactions from the host:

- NAK setting: A NAK response is returned to all generated transactions

- BUF setting: A response is returned to transactions based on the FIFO buffer
- STALL setting: A STALL response is returned to all generated transactions

Note: For setup transactions, an ACK response is always returned, regardless of the PID[1:0] bits setting, and the USB request is stored in the register.

Hardware response PID settings in host controller mode and Hardware response PID settings in device controller mode describe situations in which the USBHS writes to the PID[1:0] bits because of specific transaction results.

Hardware response PID settings in host controller mode

- NAK setting: The PID[1:0] bits are set to 00b (NAK response) in the following cases, and issuing of tokens is automatically stopped:
  - When a non-isochronous transfer is performed and an NRDY interrupt is generated. For details, see section 38.3.6.2. NRDY Interrupt
  - If a short packet is received when the PIPECFG.SHTNAK bit is set to 1 for bulk transfers
  - If transaction counting ends when the SHTNAK bit is set to 1 for bulk transfers
- BUF setting: The USBHS does not write this setting
- STALL setting: The PID[1:0] bits are set to STALL in the following cases, and issuing of tokens is automatically stopped:
  - When STALL is received in response to a transmitted token
  - When a received data packet exceeds the maximum packet size

Hardware response PID settings in device controller mode

- NAK setting: The PID[1:0] bits are set to 00b (NAK response) in the following cases, and a NAK response is returned to transactions:
  - When the setup token is received normally (DCP only)
  - If transaction counting ends or a short packet is received when the PIPECFG.SHTNAK bit is set to 1 for bulk transfers
- BUF setting: The USBHS does not write this setting
- STALL setting: The PID[1:0] bits are set to STALL in the following cases, and a STALL response is returned to transactions:
  - When a received data packet exceeds the maximum packet size
  - When a control transfer sequence error is detected

38.3.7.7 Data PID Sequence Bit

The USBHS automatically toggles the sequence bit in the data PID when data is transferred successfully in the control transfer data stage, bulk transfer, and interrupt transfer. The sequence bit of the next data PID to be transmitted can be confirmed with the SQMON bit in DCPCTR and PIPEnCTR. When data is transmitted, the sequence bit toggles on ACK handshake reception. When data is received, the sequence bit toggles on ACK handshake transmission. The SQCLR bit in DCPCTR and the SQSET bit in PIPEnCTR can be used to change the data PID sequence bit.

In device controller mode when control transfers are used, the USBHS automatically sets the sequence bit for stage transitions. DATA1 is returned when the setup stage ends. The sequence bit is not referenced and PID = DATA1 is returned in the status stage. Therefore, no software settings are required. However, in host controller mode when control transfers are used, the sequence bit must be set by software for the stage transitions.

For ClearFeature requests for transmission or reception, the data PID sequence bit must be set by software in both host and device controller modes.

38.3.7.8 Response PID = NAK Function

The USBHS provides a function for disabling pipe operation (PID[1:0] bits are set to 00b (NAK response)) when the final data packet of a transaction is received. The USBHS automatically distinguishes this based on reception of a short packet or the transaction counter. Enable this function by setting the PIPECFG.SHTNAK bit to 1.

When the double buffer mode is being used for the FIFO buffer, using this function enables reception of data packets in transfer units. If pipe operation is disabled, software must enable the pipe again (PID[1:0] bits are set to 01b (BUF response)).

The response PID = NAK function can be used only for bulk transfers.

38.3.7.9 Auto Response Mode

For bulk transfer pipes (1 to 5), when the PIPEnCTR.ATREPM bit is set to 1, a transition is made to auto response mode.

During an OUT transfer (PIPECFG.DIR = 0), OUT-NAK mode is invoked, and during an IN transfer (DIR = 1), null auto response mode is invoked.

38.3.7.10 OUT-NAK Mode

For bulk OUT transfer pipes, NAK is returned in response to an OUT token, and an NRDY interrupt is output when the PIPEnCTR.ATREPM bit is set to 1. To transition from normal mode to OUT-NAK mode, specify OUT-NAK mode while pipe operation is disabled (PID[1:0] = 00b for NAK response). Next, enable pipe operation (PID[1:0] = 01b for BUF response), on which OUT-NAK mode becomes valid. If an OUT token is received immediately before pipe operation is disabled, the token data is normally received, and an ACK is returned to the host.

To transition from OUT-NAK mode to normal mode, cancel OUT-NAK mode while pipe operation is disabled (NAK).

Next enable pipe operation (BUF). In normal mode, reception of OUT data is enabled.

38.3.7.11 Null Auto Response Mode

For bulk IN transfer pipes, zero-length packets are continuously transmitted when the PIPEnCTR.ATREPM bit is set to 1.

To transition from normal mode to null auto response mode, specify null auto response mode while pipe operation is disabled (PID[1:0] bits are set to 00b (NAK response)). Next, enable pipe operation (PID[1:0] bits are set to 01b (BUF response)), on which null auto response mode becomes valid. Before setting null auto response mode, check that PIPEnCTR.INBUFM = 0, because the mode can be set only when the buffer is empty. If the INBUFM bit is 1, empty the buffer using the PIPEnCTR.ACLRM bit. Do not write data from the FIFO port while a transition to null auto response mode is being made.

To transition from null auto response mode to normal mode, keep pipe operation disabled (PID[1:0] bits are set to 00b (NAK response)) for the period of the zero-length packet transmission (about 10 µs) before canceling the null auto response mode. In normal mode, data can be written from the FIFO port, so packet transmission to the host is enabled by enabling pipe operation (PID[1:0] bits are set to 01b (BUF response)).

38.3.8 FIFO Buffer

The USBHS provides a FIFO buffer for data transfers, and it manages the memory area used for each pipe. The FIFO buffer has two states depending on whether the access right is assigned to the system (CPU side) or the USBHS (SIE side).

38.3.8.1 Buffer Status

Table 38.23 and Table 38.24 show the buffer status in the USBHS. The FIFO buffer status can be confirmed using the DCPCTR.BSTS and PIPEnCTR.INBUFM bits. The transfer direction for the FIFO buffer can be specified in the PIPECFG.DIR or CFIFOSEL.ISEL bit (when DCP is selected).

The INBUFM bit is valid for pipes 1 to 5 in the transmitting direction.

When a transmitting pipe uses double buffering, the software can read the BSTS bit to monitor the FIFO buffer status on the CPU side and the INBUFM bit to monitor the FIFO buffer status on the SIE side. When write access to the FIFO port by the CPU or DMAC/DTC is slow and the buffer empty status cannot be determined using the BEMP interrupt, the software can use the INBUFM bit to confirm the end of transmission.

Table 38.23 Buffer status indicated in the BSTS flag

ISEL or DIR	BSTS	FIFO buffer status
0 (receiving direction)	0	There is no received data, or data is being received. Reading from the FIFO port is disabled.
0 (receiving direction)	1	There is received data, or a zero-length packet is received. Reading from the FIFO port is allowed. When a zero-length packet is received, reading is not possible and the buffer must be cleared.
1 (transmitting direction)	0	Transmission is not complete. Writing to the FIFO port is disabled.
1 (transmitting direction)	1	Transmission is complete. CPU write is allowed.

Table 38.24 Buffer status indicated in the INBUFM bit

DIR	INBUFM	FIFO buffer status
0 (receiving direction)	Invalid	Invalid
1 (transmitting direction)	0	Transmission is complete. There is no data waiting to be transmitted.
1 (transmitting direction)	1	The FIFO port has written data to the buffer. There is data to be transmitted.

38.3.8.2 FIFO Buffer Clearing

Table 38.25 shows the methods for clearing the FIFO buffer. The FIFO buffer can be cleared using the BCLR bit in the port control register, DnFIFOSEL.DCLRM, or the PIPEnCTR.ACLRM bit.

Single or double buffering can be selected for pipes 1 to 5 in the PIPECFG.DBLB bit.

Table 38.25 Buffer clearing methods

FIFO buffer clearing mode	Clearing the FIFO buffer on the CPU side	Mode for automatically clearing the FIFO buffer after reading the specified pipe data	Auto buffer clear mode for discarding all received packets
Register used	• CFIFOCTR • DnFIFOCTR	DnFIFOSEL	PIPEnCTR
Bit used	BCLR	DCLRM	ACLRM
Clearing condition	Cleared by writing 1	1: Mode valid 0: Mode invalid	1: Mode valid 0: Mode invalid

Auto buffer clear mode function

The USBHS discards all received data packets if the PIPEnCTR.ACLRM bit is set to 1. If a correct data packet is received, the ACK response is returned to the host controller. The auto buffer clear mode function can only be set in the FIFO buffer reading direction.

Setting the ACLRM bit to 1 and then to 0 clears the FIFO buffer of the selected pipe regardless of the access direction. An access cycle of at least 100 ns is required for the internal hardware sequence processing between ACLRM = 1 and ACLRM = 0.

38.3.8.3 FIFO Port Functions

Table 38.26 shows the settings for the FIFO port functions. In write access, writing data until the maximum packet size is reached automatically enables transmission of the data. To enable transmission before the maximum packet size is reached, set the BVAL flag in the port control register to end writing. To send a zero-length packet, use the BCLR bit to clear the buffer, and then set the BVAL flag to end writing.

In reading, reception of new packets is automatically enabled when all data is read. Data cannot be read when a zero-length packet is received (DTLN[11:0] = 0), so the buffer must be cleared with the BCLR bit. The length of the receive data can be confirmed in the DTLN[11:0] flags in the port control register.

Table 38.26 FIFO port function settings

Register name	Bit name	Description
CFIFOSEL, DnFIFOSEL (n = 0, 1)	RCNT	Selects DTLN[11:0] read mode
	REW	FIFO buffer rewind (re-read, rewrite)
	DCLRM	Automatically clears receive data for a specified pipe after the data is read (only for DnFIFO)
	DREQE	Enables DMAC/DTC transfers (only for DnFIFO)
	MBW[1:0]	FIFO port access bit width
	BIGEND	Selects FIFO port endian
	ISEL	FIFO port access direction (only for DCP)
	CURPIPE[3:0]	Selects the current pipe
CFIFOCTR, DnFIFOCTR (n = 0, 1)	BVAL	Ends writing to the FIFO buffer
	BCLR	Clears the FIFO buffer on the CPU side
	DTLN[11:0]	Checks the length of receive data

FIFO port selection

Table 38.27 shows the pipes that can be selected with the different FIFO ports. The pipe to be accessed must be selected in the CURPIPE[3:0] bits in the port selection register. After a pipe is selected, the software must check whether the written value can be correctly read from the CURPIPE[3:0] bits. (If the previous pipe number is read, it indicates that the USBHS is modifying the pipe.) Next, the software checks that the FRDY flag in the port control register is 1.

In addition, the software must specify the bus width to be accessed in the MBW[1:0] bits in the port selection register.

The FIFO buffer access direction conforms to the PIPECFG.DIR setting. For the DCP only, the ISEL bit in the port selection register determines the direction.

Table 38.27 FIFO port access by pipe

Pipe	Access Method	Port that can be used
DCP	CPU access	CFIFO port register
Pipes 1 to 9	CPU access	CFIFO port register D0FIFO/D1FIFO port register
	DMAC/DTC access	D0FIFO/D1FIFO port register

REW bit

It is possible to temporarily stop access to a pipe being accessed, access a different pipe, and then continue processing for the first pipe again. The REW bit in the port selection register is used for this processing.

If a pipe is selected in the CURPIPE[3:0] bits in the port selection register with the REW bit set to 1, the pointer used for reading from and writing to the FIFO buffer is reset, and reading or writing can be carried out from the first byte. If a pipe is selected with the REW bit set to 0, data can be read and written in continuation from the previous selection, without the pointer being reset.

To access the FIFO port, the software must check that the FRDY bit in the port control register is 1 after selecting a pipe.

38.3.8.4 DMAC/DTC Transfers (D0FIFO and D1FIFO Ports)

Overview of DMAC/DTC transfer

For pipes 1 to 9, the FIFO port can be accessed using the DMAC/DTC. When buffer access for the pipe targeted for DMAC/DTC transfer is enabled, a DMAC/DTC transfer request is issued.

Select the unit of transfer to the FIFO port in the DnFIFOSEL.MBW[1:0] bits, and select the pipe targeted for the DMAC/DTC transfer in the DnFIFOSEL.CURPIPE[3:0] bits. Do not change the selected pipe during the DMAC transfer.

DnFIFO auto clear mode (D0FIFO and D1FIFO port reading direction)

If 1 is set in the DnFIFOSEL.DCLRM bit, the USBHS automatically clears the FIFO buffer of the selected pipe when reading of data from the FIFO buffer is complete.

Table 38.28 shows the packet reception and FIFO buffer clearing processing by software for each of the different settings. As shown in the table, the buffer clearing conditions depend on the value set in the PIPECFG.BFRE bit. Using the DnFIFOSEL.DCLRM bit eliminates the need for the buffer to be cleared by software in any situation that requires buffer clearing. This enables DMAC/DTC transfers without involving software.

The DnFIFO auto clear mode can only be set in the FIFO buffer reading direction.

Table 38.28 Packet reception and FIFO buffer clearing processing by software

Buffer status when packet is received	Register setting			
	DCLRM = 0		DCLRM = 1	
	BFRE = 0	BFRE = 1	BFRE = 0	BFRE = 1
Buffer full	No clearing required	No clearing required	No clearing required	No clearing required
Zero-length packet reception	Clearing required	Clearing required	No clearing required	No clearing required
Normal short packet reception	No clearing required	Clearing required	No clearing required	No clearing required
Transaction count end	No clearing required	Clearing required	No clearing required	No clearing required

38.3.8.5 Allocating the FIFO Buffer

Figure 38.11 shows an example of a memory map of the FIFO buffer. The FIFO buffer is an area shared by the USBHS and the control CPU of the application. There are two situations for the FIFO buffer: (1) access rights are given to the application (CPU side), and (2) access rights are given to the USBHS (SIE side).

An independent area is set for the FIFO buffer for each pipe. A memory area is determined by the first block number and the number of blocks (specified in the BUFNMB[7:0] and BUFSIZE[4:0] bits in PIPEBUF), where 64 bytes is regarded as one block. When the continuous transfer mode is selected in the CNTMD bit in PIPECFG, set the BUFSIZE[4:0] bits to an integral multiple of the maximum packet size. When double buffering is selected in the DBLB bit in PIPECFG, twice the memory area specified in the BUFSIZE[4:0] bits in PIPEBUF is allocated to the same pipe.

Three FIFO ports are used to access (read data from and write data to) the FIFO buffer. Specify the number of the pipe to be allocated to the FIFO port in the CURPIPE[3:0] bits in C/DnFIFOSEL.

The FIFO buffer status of each pipe can be checked in the DCPCTR.BSTS, PIPEnCTR, and INBUFM bits. The FIFO port access rights can be checked in the FRDY flag in C/DnFIFOCTR.

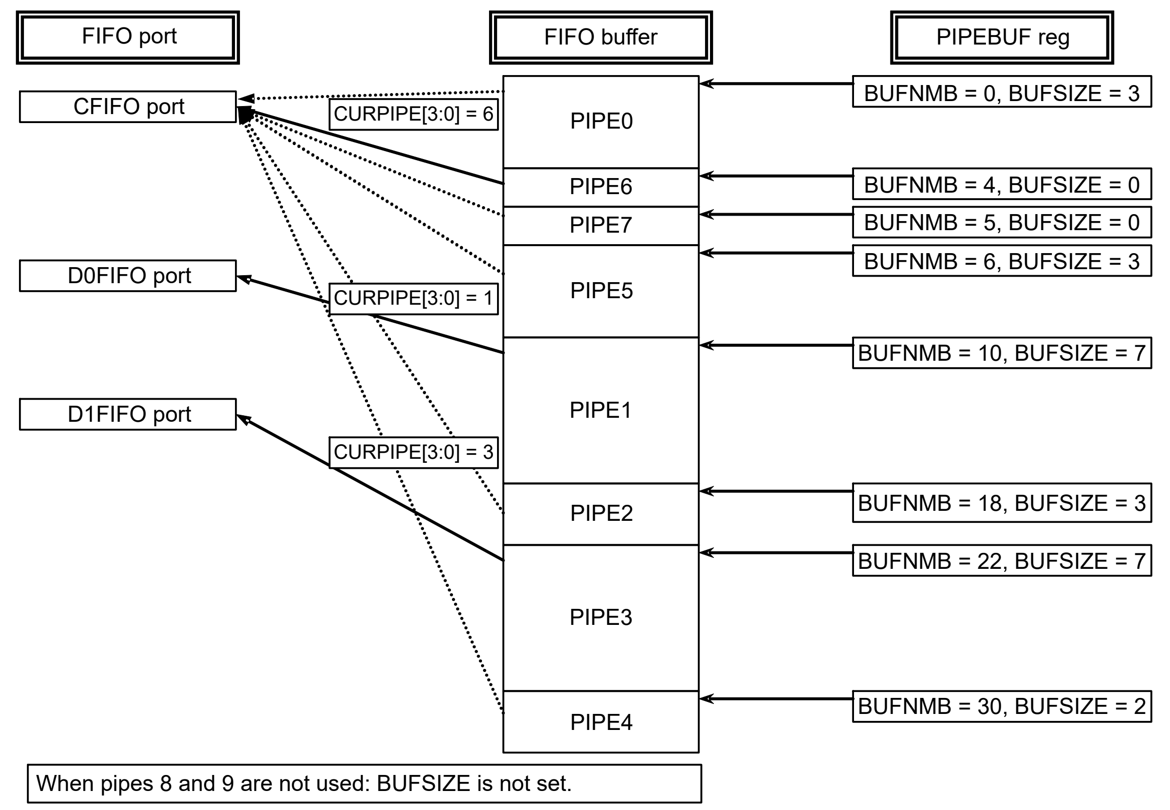


Figure 38.11 Example memory map of the FIFO buffer

38.3.9 Control Transfers Using the DCP

The Default Control Pipe (DCP) is used for data transfers in the control transfer data stage. The FIFO buffer of the DCP is a 64-byte single buffer with a fixed area for both control reads and control writes. The FIFO buffer can be accessed only through the CFIFO port.

38.3.9.1 Control Transfers in Host Controller Mode

Setup stage

The USBREQ, USBVAL, USBINDX, and USBLENG registers are used to transmit USB requests for setup transactions. Writing the setup packet data to the register and then writing 1 to the DCPCTR.SUREQ bit transmits the specified data for the setup transaction. On completion of the transaction, the SUREQ bit clears to 0. Do not change these USB request registers while SUREQ = 1.

When an attached function device is detected, the software must issue the first setup transaction for the device using this sequence with the DCPMAXP.DEVSEL[3:0] bits cleared to 0 and the DEVADD0.USBSPD[1:0] bits set appropriately.

When an attached function device is shifted to the Address state, the software must issue setup transactions using this sequence with the assigned USB Address set in the DEVSEL[3:0] bits and the bits in DEVADDn (n = 0 to A) corresponding to the specified USB Address set appropriately. For example, when PIPEMAXP.DEVSEL[3:0] = 0010b, make appropriate settings in DEVADD2. When PIPEMAXP.DEVSEL[3:0] = 0101b, make appropriate settings in DEVADD5.

When the setup transaction data is sent, an interrupt request is generated based on the response from the peripheral device (SIGN or SACK bit in INTSTS1). This interrupt request allows the software to check the setup transaction result.

The DATA0 data packet (USB request) for the setup transaction is always transmitted regardless of the status of the DCPCTR.SQMON flag.

Data stage

The data stage is used to transfer data using the DCP FIFO buffer.

Before accessing the DCP FIFO buffer, specify the access direction in the CFIFOSEL.ISEL bit. Specify the transfer direction in the DCPCFG.DIR bit.

For the first data packet of the data stage, the data PID must be transferred as DATA1. Set data PID to DATA1 in the DCPCTR.SQSET bit and set the PID[1:0] bits to 01b (BUF response). Completion of data transfer is detected using the BRDY or BEMP interrupt.

Data transfer of multiple packets is enabled in continuous transfer mode. However, when continuous transfer is specified in the receiving direction, the BRDY interrupt is not generated unless the buffer becomes full or a short packet is received (for 256 bytes or less, which is an integral multiple of the maximum packet size). If the transmit data size is an integral multiple of the maximum packet size, control the control write transfer through the software to transmit a zero-length packet last.

Status stage

The status stage is used for zero-length packet data transfers in the reverse direction of the data stage. As in the data stage, data is transferred using the DCP FIFO buffer. Transactions are executed using the same procedure as the data stage.

Data packets in the status stage must be transmitted and received with the data PID set to DATA1 using the DCPCTR.SQSET bit.

When a zero-length packet is received, check the receive-data length in the CFIFOCTR.DTLN[11:0] flags after a BRDY interrupt is generated, and then clear the FIFO buffer using the BCLR bit.

38.3.9.2 Control Transfers in Device Controller Mode

Setup stage

The USBHS returns an ACK response to a normal setup packet for the USBHS. The USBHS operates in the setup stage as follows:

On receiving a new setup packet, the USBHS sets the following bits:

- Sets the INTSTS0.VALID flag to 1
- Sets the DCPCTR.PID[1:0] bits to 00b (NAK response)
- Sets the DCPCTR.CCPL bit to 0

When the USBHS receives a data packet following a setup packet, it stores the USB request parameters in USBREQ, USBVAL, USBINDX, and USBLENG.

Before performing the response processing for a control transfer, set the VALID flag to 0. When the VALID flag = 1, the PID[1:0] bits cannot be set to 01b (BUF response), and the data stage cannot be terminated.

Using the VALID flag function, the USBHS can suspend a request being processed when it receives a new USB request during a control transfer and return a response to the latest request.

In addition, the USBHS automatically detects the direction bit (bmRequestType bit 8) and the request data length (wLength) in the received USB request. It distinguishes between control read transfers, control write transfers, and no data control transfers, and it controls stage transitions. For an incorrect sequence, a sequence error occurs in the control transfer stage transition interrupt, and the interrupt is reported to the software. For a diagram of the stage control by the USBHS, see Figure 38.9.

Data stage

The DCP must be used to execute data transfers for received USB requests. Before accessing the DCP FIFO buffer, specify the access direction in the CFIFOSEL.ISEL bit.

If the transfer data is larger than the size of the DCP FIFO buffer, execute the data transfer using the BRDY interrupt for control write transfers and the BEMP interrupt for control read transfers.

In high-speed control write transfers, a NYET handshake response is returned based on the FIFO buffer status.

Status stage

Control transfers are terminated by setting the DCPCTR.CCPL bit to 1 while the DCPCTR.PID[1:0] bits are set to 01b (BUF response). Control transfers are terminated by setting the DCPCTR.CCPL bit to 1 while the DCPCTR.PID[1:0] bits are set to 01b (BUF response).

After this setting is made, the USBHS automatically executes the status stage based on the data transfer direction determined at the setup stage. The status stage is executed as follows:

- For control read transfers:
  The USBHS receives a zero-length packet from the USB host and transmits an ACK response
- For control write transfer and no data control transfer:
  The USBHS transmits a zero-length packet and receives an ACK response from the USB host.

Control transfer auto response function

The USBHS automatically responds to a normal SET_ADDRESS request. If the SET_ADDRESS request contains any of the following errors, a response must be returned by software.

- When bmRequestType is not 0x00: except control write transfer
- When wIndex is not 0x00: request error
- When wLength is not 0x00: except no data control transfer
- When wValue is larger than 0x7F: request error
- When PL1CTRL.DVSQ[3:0] flags are 0011b (Configured): control transfer of device state error

A response by the corresponding software is required to all requests other than SET_ADDRESS.

38.3.10 Bulk Transfers (Pipes 1 to 5)

The FIFO buffer usage (setting of single buffer/double buffer or continuous/discontinuous transfer mode) is configurable for bulk transfers. The FIFO buffer size can be set up to 2 KB. The USBHS manages the FIFO buffer state and automatically responds to the PING packet and the NYET handshake.

38.3.10.1 PING Packet Control in Host Controller Mode

In the OUT direction, a PING packet is automatically transmitted by the USBHS. The USBHS starts communication in the transmitting direction beginning with the PING packet. When it receives an ACK handshake in response to the PING packet, the USBHS transmits an OUT packet. The USBHS returns to the PING transmission state on receiving a NAK or NYET response during an OUT transaction. The procedure is as follows:

Starting OUT data transmission

(1) Transmit PING packet
(2) Receive NAK handshake
(3) Transmit PING packet
(4) Receive ACK handshake
(5) Transmit OUT data packet
(6) Receive ACK handshake
(7) Transmit OUT data packet
⋮
(8) Return to (1) on receiving a NAK/NYET handshake

The USBHS returns to the PING packet transmission state when a hardware reset is issued, the NYET or NAK handshake is received, the sequence toggle bit is cleared (SQCLR), or the buffer clear bit (ACLRM) is set.

38.3.10.2 NYET Handshake Control in Device Controller Mode

Table 38.29 lists responses to received tokens during bulk and control transfers. The USBHS returns a NYET response when an available area for only one packet is left in the FIFO buffer when the USBHS has received an OUT token during a bulk or control transfer. When the USBHS receives a short packet, however, it returns an ACK response instead of NYET even when this condition occurs.

Table 38.29 Responses to received tokens

PID[1:0] bit setting	FIFO buffer state	Received token	Response	Note
NAK/STALL	—	SETUP	ACK	—
	—	IN/OUT/PING	NAK/STALL	—
BUF	—	SETUP	ACK	—
	RCV-BRDY	OUT/PING	ACK	When OUT token is received, data packet is received.*1
	RCV-BRDY	OUT	NYET	Data packet is received*2
	RCV-BRDY	OUT (Short)	ACK	Data packet is received*2
	RCV-BRDY	PING	ACK	*2
	RCV-NRDY	OUT/PING	NAK	—
	TRN-BRDY	IN	DATA0/1	Data packet is transmitted
	TRN-NRDY	IN	NAK	—

Note 1. RCV-BRDY: An available area for two packets is left in the FIFO buffer when an OUT token or a PING token is received.
Note 2. RCV-BRDY: An available area for only one packet is left in the FIFO buffer when an OUT token is received.
RCV-NRDY: No available area is left in the FIFO buffer when a PING token is received.
TRN-BRDY: The FIFO buffer contains transmit data when an IN token is received.
TRN-NRDY: The FIFO buffer contains no transmit data when an IN token is received.

38.3.11 Interrupt Transfers (Pipes 6 to 9)

In device controller mode, the USBHS performs interrupt transfers based on the timing dictated by the host controller. In the interrupt transfer, the USBHS ignores PING packets (no response) and does not transmit the NYET handshake, but returns an ACK, NAK, or STALL response.

In host controller mode, the software can set the timing for issuing tokens using the interval counter. The USBHS does not issue a PING token but issues an OUT token, including for transfers in the OUT direction.

The USBHS does not support high-bandwidth interrupt transfers.

38.3.11.1 Interval Counter for Interrupt Transfers in Host Controller Mode

Specify the transaction interval for interrupt transfers in the PIPEPERI.IITV[2:0] bits. The USBHS issues interrupt transfer tokens based on this interval.

(1) Initializing the counter

The USBHS initializes the interval counter under the following conditions:

- Power-on reset:
  This initializes the IITV[2:0] bits
- FIFO buffer initialization using the PIPEnCTR.ACLRM bit:
  This does not initialize the IITV[2:0] bits, but does initialize the count value. Setting the PIPEnCTR.ACLRM bit to 0 starts counting from the value set in IITV[2:0].

(2) Operation when tokens cannot be transmitted or received even on token generation

No token is generated in the following cases even at token generation time. In these cases, the USBHS tries to execute the transaction in the next interval.

- When the PID[1:0] bits are set to NAK or STALL
- When the FIFO buffer is full at token transmit time in the receiving (IN) direction
- When there is no data to be transmitted in the FIFO buffer at token transmit time in the transmitting (OUT) direction

38.3.12 Isochronous Transfers (Pipes 1 and 2)

The USBHS does not support high-bandwidth isochronous transfers but provides the following functions for isochronous transfers:

- Notification of isochronous transfer error

- Interval counter (specified in the PIPEPERI.IITV[2:0] bits)
- Isochronous IN transfer data setup control (IDLY function)
- Isochronous IN transfer buffer flush function (specified in the PIPEPERI.IFIS bit)
- SOF pulse output function

38.3.12.1 Error Detection in Isochronous Transfers

The USBHS provides a function for detecting the errors described in this section, so that when errors occur in isochronous transfers, they can be controlled by software. Table 38.30 and Table 38.31 show the priority order for errors detected by the USBHS and the associated interrupts.

PID errors

- The PID value of the received packet is invalid

CRC errors and bit stuffing errors

- A CRC error is found in a received packet or the bit stuffing is illegal

Maximum packet size exceeded

- The data size of the received packet exceeds the specified maximum packet size

Overrun and underrun errors

In host controller mode:

- The FIFO buffer is full at token transmit time in the IN (receiving) direction
- There is no data to be sent in the FIFO buffer at token transmit time in the OUT (transmitting) direction

In device controller mode:

- There is no data to be sent in the FIFO buffer at token receive time in the IN (transmitting) direction
- The FIFO buffer is full at token receive time in the OUT (receiving) direction

Interval error

In device controller mode, the following cases are treated as an interval error:

- Failure to receive an IN token in the interval frame during an isochronous IN transfer
- Failure to receive an OUT token in the interval frame during an isochronous OUT transfer

Table 38.30 Error detection for token transmission and reception

Detection priority	Error type	Interrupt generated at error detection and status
1	PID error	No interrupts are generated in either host or device controller mode. (Ignored as a corrupted packet.)
2	CRC or bit stuffing error	No interrupts are generated in either host or device controller mode. (Ignored as a corrupted packet.)
3	Overrun or underrun error	An NRDY interrupt is generated to set the OVRN flag to 1 in both host and device controller modes. In device controller mode, a zero-length packet is transmitted in response to an IN token. No data packets are received in response to the OUT token.
4	Interval error	An NRDY interrupt is generated in device controller mode. No interrupt is generated in host controller mode.

Table 38.31 Error detection for data packet reception (1 of 2)

Detection priority	Error type	Interrupt generated at error detection and status
1	PID error	No interrupt is generated. (Ignored as a corrupted packet.)
2	CRC or bit stuffing error	An NRDY interrupt is generated and the FRMNUM.CRCE bit sets to 1 in both host and device controller modes.

Table 38.31 Error detection for data packet reception (2 of 2)

Detection priority	Error type	Interrupt generated at error detection and status
3	Maximum packet size exceeded error	A BEMP interrupt is generated and the PID[1:0] bits set to STALL in both host and device controller modes.

38.3.12.2 DATA PID

The USBHS does not support high-bandwidth transfers. In device controller mode, the USBHS responds as follows to a received PID:

IN direction

- DATA0: Transmitted as data packet PID
- DATA1: Not transmitted
- DATA2: Not transmitted
- mDATA: Not transmitted

OUT direction (full-speed operation)

- DATA0: Received normally as data packet PID
- DATA1: Received normally as data packet PID
- DATA2: Packets ignored
- mDATA: Packets ignored

OUT direction (high-speed operation)

- DATA0: Received normally as data packet PID
- DATA1: Received normally as data packet PID
- DATA2: Received normally as data packet PID
- mDATA: Received normally as data packet PID

38.3.12.3 Interval Counter

The isochronous transfer interval can be set in the PIPEPERI.IITV[2:0] bits. In device controller mode, the interval counter enables functions as shown in Table 38.32. In host controller mode, the USBHS generates the token issuance timing, and the interval counter operation is the same as that for interrupt transfers.

Table 38.32 Interval counter functions in device controller mode

Transfer direction	Function	Conditions for detection
IN	Transmit buffer flush	Failure to receive an IN token successfully in the interval frame during an isochronous IN transfer
OUT	Notification of no reception of token	Failure to receive an OUT token successfully in the interval frame during an isochronous OUT transfer

The interval count is performed when an SOF is received or for complemented SOFs, so the isochronism can be maintained even if an SOF is corrupt. The frame interval can be set to 2IITV (μ) frames.

(1) Counter initialization in device controller mode

The USBHS initializes the interval counter under the following conditions:

- Power-on reset:
  This initializes the PIPEPERI.IITV[2:0] bits
- FIFO buffer initialization using the ACLRM bit:
  This does not initialize the IITV[2:0] bits, but does initialize the count value.

After the interval counter is initialized, the interval count starts under either of the following conditions when a packet is transferred successfully:

- An SOF is received after data is transmitted in response to an IN token, with the PID[1:0] bits set to 01b (BUF response)
- An SOF is received after data is received in response to an OUT token, with PID[1:0] bits set to 01b (BUF response)

The interval counter is not initialized under the following conditions:

- When the PID[1:0] bits are set to NAK or STALL
  This does not stop the interval timer. The USBHS attempts the transaction in the next interval.
- USB bus reset and USB suspension
  This does not initialize the IITV[2:0] bits. When an SOF is received, the interval counter starts counting from the value set before SOF was received.

(2) Interval counting and transfer control in host controller mode

The USBHS controls the interval between token issuance operations based on the PIPEPERI.IITV[2:0] bit settings. Specifically, the USBHS issues a token for a selected pipe once every 2IITV frames.

The USBHS starts counting the token issuance interval at the frame following the frame in which the PID[1:0] bits are set to 01b (BUF response) by software.

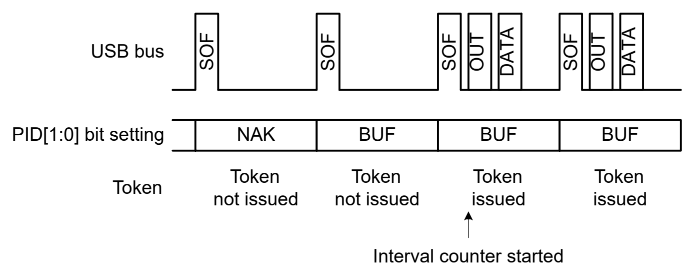


Figure 38.12 Token issuance when IITV[2:0] = 0

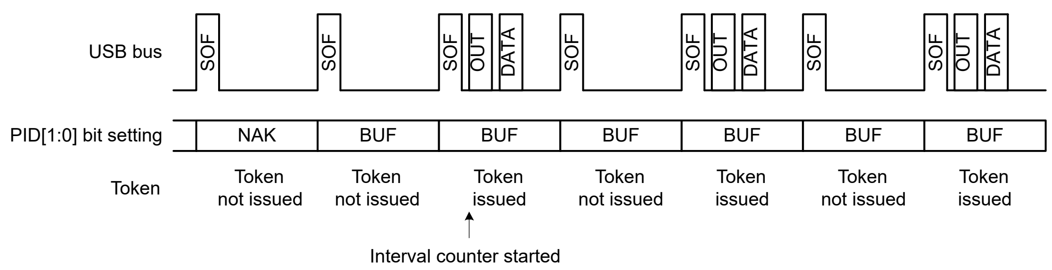


Figure 38.13 Token issuance when IITV[2:0] = 1

When the selected pipe is set for isochronous transfers, the USBHS carries out the following operation in addition to controlling the token issuance interval. The USBHS issues a token even when the NRDY interrupt generation condition is satisfied.

When the selected pipe is for isochronous IN transfers

The USBHS generates an NRDY interrupt when the USBHS issues an IN token but does not receive a packet successfully from a peripheral device (no response or packet error).

When the selected pipe is for isochronous OUT transfers

The USBHS sets the OVRN flag to 1, generating an NRDY interrupt and transmitting a zero-length packet, when the time to issue an OUT token comes while there is no data to be transmitted in the FIFO buffer, because the CPU or DMAC/DTC is too slow in writing data to the FIFO buffer.

The token issuance interval is reset on any of the following conditions:

- When the MCU is reset
  This initializes the IITV[2:0] bits
- When the PIPEnCTR.ACLRM bit is set to 1 by software

(3) Interval counting and transfer control in device controller mode

When the selected pipe is for isochronous OUT transfers

The USBHS generates an NRDY interrupt when it fails to receive a data packet within the interval set in the PIPEPERI.IITV[2:0] bits.

The USBHS also generates an NRDY interrupt when it fails to receive data because of a CRC error or other errors contained in the data packet or because of the FIFO buffer is full.

The NRDY interrupt is generated on SOF packet reception. Even if the SOF packet is corrupted, internal complementation allows the interrupt to be generated when the SOF packet is received. However, when the IITV[2:0] bits are set to a value other than 0, the USBHS generates an NRDY interrupt on receiving an SOF packet for every interval after interval counting starts.

When the PID[1:0] bits are set to 00b (NAK response) by software after starting the interval timer, the USBHS does not generate an NRDY interrupt on receiving an SOF packet.

The timing for starting interval counting depend on the IITV[2:0] setting as follows:

- When the IITV[2:0] bits = 0:
  The interval counting starts when the PID[1:0] bits of the selected pipe are changed to BUF
- When the IITV[2:0] bits ≠ 0:
  The interval counting starts on completion of successful reception of the first data packet after the PID[1:0] bits for the selected pipe are changed to 01b (BUF response)

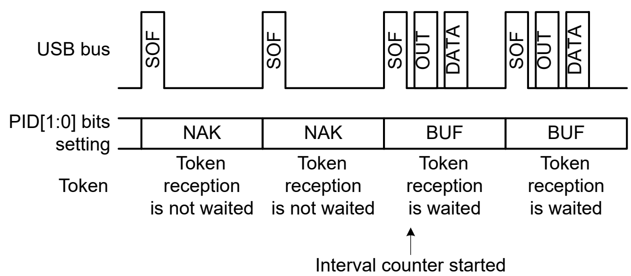


Figure 38.14 Relationship between frames and expected token reception when IITV[2:0] = 0

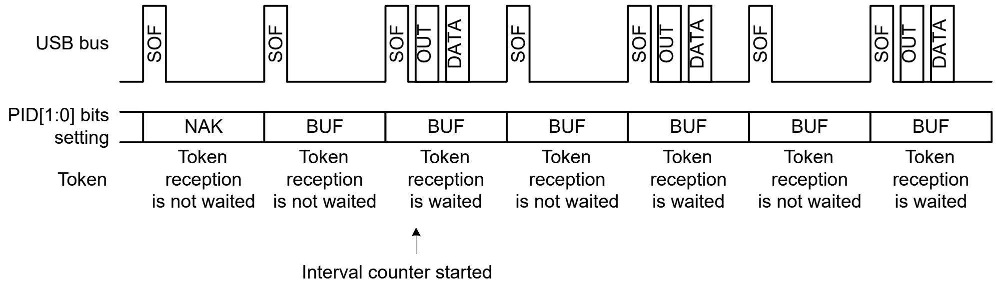


Figure 38.15 Relationship between frames and expected token reception when IITV[2:0] ≠ 0

When the selected pipe is for isochronous IN transfers

The PIPEPERI.IFIS bit must be 1 for this use case. When the IFIS bit is cleared to 0, the USBHS transmits a data packet in response to a received IN token, regardless of the PIPEPERI.IITV[2:0] setting.

When IFIS is 1 and there is data to be transmitted in the FIFO buffer, the USBHS clears the FIFO buffer when it fails to receive an IN token in the frame at the interval set in the IITV[2:0] bits.

The USBHS also clears the FIFO buffer when it fails to receive an IN token successfully because of a bus error, such as a CRC error, contained in the IN token.

The FIFO buffer is cleared on SOF packet reception. Even if the SOF packet is corrupted, the internal complementation allows the FIFO buffer to be cleared when the SOF packet is received.

The timing to start interval counting depends on the IITV[2:0] setting, as with OUT transfers.

The interval is counted on any of the following conditions in device controller mode:

- When a hardware reset is applied to the USBHS (which also sets the IITV[2:0] bits to 000b)
- When the PIPEnCTR.ACLRM bit is set to 1 by software
- When the USBHS detects a USB bus reset

(4) Transmit data setup for isochronous transfers in device controller mode

With isochronous data transmission using the USBHS in device controller mode, after data is written to the FIFO buffer, a data packet can be transmitted in the first frame after the SOF packet is detected. This isochronous transfer transmit data setup function can identify the frame that started transmission.

When the double buffering is used, transmission is only enabled for the buffer in which data writing was completed first, even after the data write to both buffers is complete. Accordingly, even if multiple IN tokens are received, only the one packet of FIFO buffer data is transmitted.

When the FIFO buffer is ready to transmit data when an IN token is received, the data is transferred and a normal response is returned. However, if the FIFO buffer cannot transmit data, a zero-length packet is transmitted and an underrun error occurs.

Figure 38.16 shows an example of transmission using the isochronous transfer transmit data setup function when the IITV[2:0] bits are set to 0 (every frame).

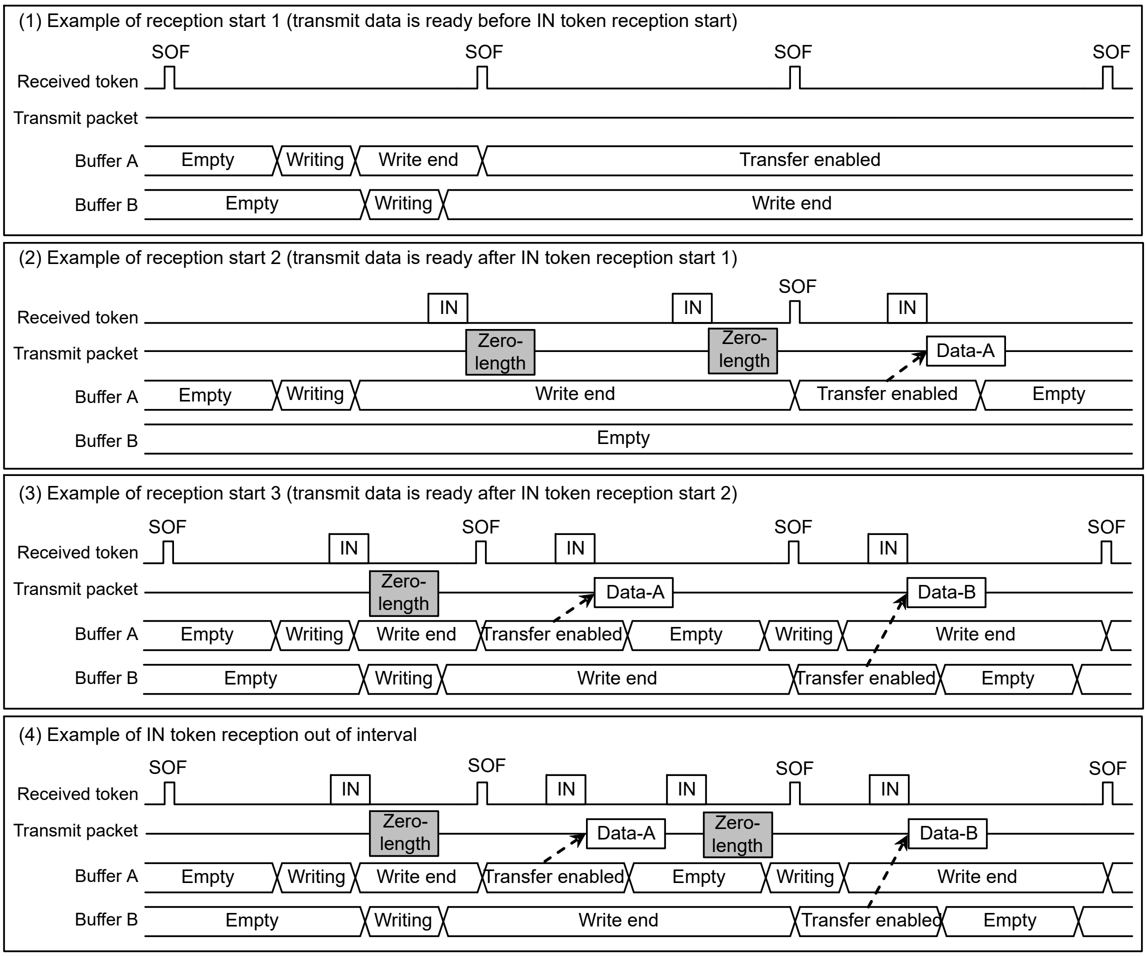


Figure 38.16 Example data setup operation

(5) Isochronous transfer transmit buffer flush in device controller mode

In device controller mode during isochronous data transmission, if the USBHS receives an SOF packet for the next frame without receiving an IN token in the interval frame, it operates as if the IN token is corrupt and clears the buffer that is enabled for transmission, putting that buffer in the writing enabled state.

When double buffering is used and writing to both buffers is complete, the cleared FIFO buffer is assumed to be the one where the data was transmitted in the interval frame, and transmission is enabled for the FIFO buffer that was not cleared on SOF packet reception.

The timing of the buffer flush function depends on the PIPEPERI.IITV[2:0] setting as follows:

- When IITV[2:0] = 0:
  The buffer flush operation starts from the first frame after the pipe is enabled
- When IITV[2:0] ≠ 0:
  The buffer flush operation starts after the first normal transaction

Figure 38.17 shows an example buffer flush. When an unanticipated token is received before the interval frame, the USBHS sends the write data or a zero-length packet as an underrun error, depending on the data setup status.

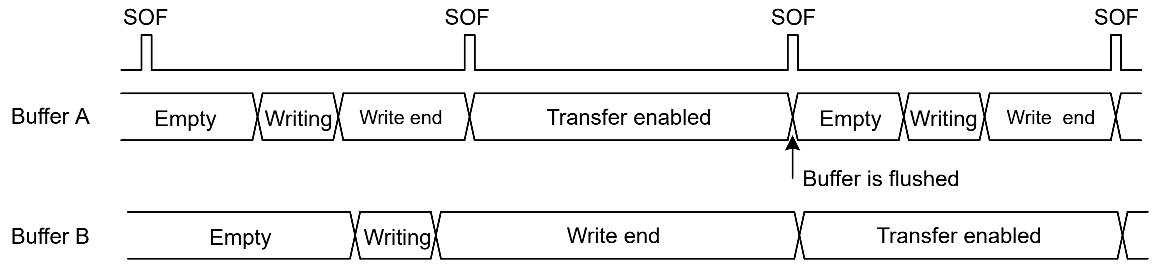


Figure 38.17 Example buffer flush operation

Figure 38.18 shows an example interval error occurrence. There are five types of interval errors, as shown in the figure. An interval error occurs at timing ①, and the buffer flush function is activated.

If an interval error occurs during an IN transfer, the buffer flush function is activated. If it occurs during an OUT transfer, an NRDY interrupt is generated. Use the FRMNUM.OVRN bit to distinguish between this and NRDY interrupts triggered by received packet errors and overrun errors.

For tokens that are shaded in the figure, responses are returned based on the FIFO buffer status.

- IN direction:
  - If the buffer is ready to transfer data, the data is transferred and a normal response is returned
  - If the buffer is not ready to transfer data, a zero-length packet is transmitted and an underrun error occurs
- OUT direction:
  - If the buffer is ready to receive data, the data is received and a normal response is returned
  - If the buffer is not ready to receive data, the received data is discarded and an overrun error occurs

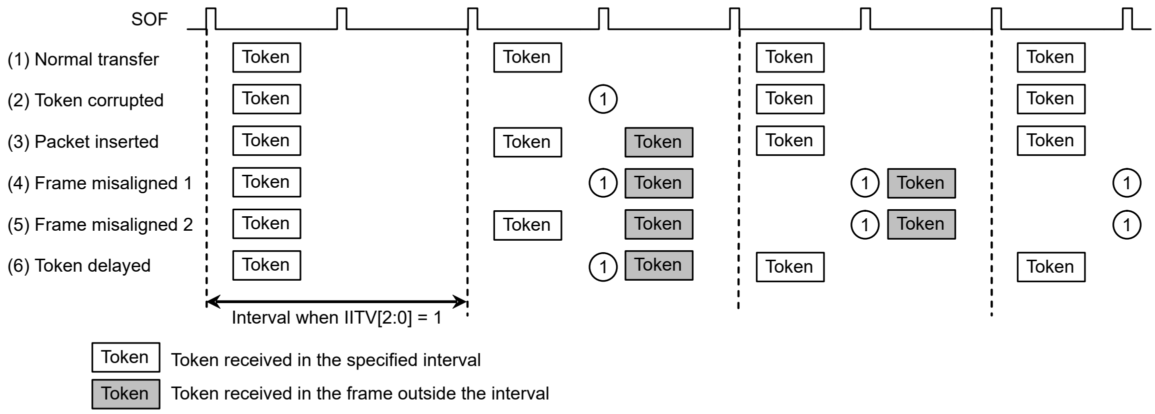


Figure 38.18 Example interval error occurrence when PIPEPERI.IITV[2:0] = 1

38.3.13 SOF Complementation Function

In device controller mode, if packet reception is disabled at intervals of 125 µs in high-speed mode or 1 ms in full-speed mode because the SOF packet is missing or corrupted, the USBHS complements the SOF. SOF complementation begins when the SYSCFG.USBE and LPSTS.SUSPENDM bits are set to 1 and an SOF packet is received. The complementation function is initialized under the following conditions:

- Power-on reset

- USB bus reset
- Suspend state detection

The SOF complementation function operates as follows:

- The frame interval (125 µs or 1 ms) is determined by the reset handshake protocol result
- The complementation function is not activated until an SOF packet is received
- When the first SOF packet is received, complementation is performed by counting 125 µs or 1 ms on the 48-MHz internal clock
- When the second or subsequent SOF packets are received, complementation is performed at the previous reception interval
- Complementation is not performed in the Suspend state or on reception of a USB bus reset. During high-speed operation, complementation continues for 3 ms from the last packet on transition to the Suspend state

The USBHS supports the following functions controlled by SOF packet reception. These functions operate normally with SOF complementation if the SOF packet is missing:

- Updating of the frame number and micro frame number
- SOFR interrupt and micro-SOF lock
- SOF pulse output
- Isochronous transfer interval count

If an SOF packet is missing during full-speed operation, the FFRMNUM.FRNM[10:0] flags are not updated. If a micro-SOF packet is missing during high-speed operation, the URMNUM.UFRNM[2:0] bits are updated.

However, if a micro-SOF packet is missing while the UFRNM[2:0] bits are set to 000b, the FRNM bits are not updated. In this case, even if a subsequent micro-SOF packet with a value other than UFRNM[2:0] bits = 000b is received successfully while UFRNM[2:0] bits are set to the value other than 000b, the FRNM bits are not updated.

38.3.14 Pipe Schedule

38.3.14.1 Conditions for Generating Transactions

In host controller mode and when the DVSTCTR0.UACT bit is set to 1, the USBHS generates transactions under the conditions as shown in Table 38.33.

Table 38.33 Conditions for generating transactions

Transaction	Conditions for generation				
	DIR	**PID[1:0]**	**IITV0**	**Buffer state**	**SUREQ**
Setup	—*1	—*1	—*1	—*1	1 setting
Control transfer data stage, status stage, bulk transfer	IN	BUF	—*1	Receive area exists	—*1
	OUT	BUF	—*1	Transmit data exists	—*1
Interrupt transfer	IN	BUF	Valid*2	Receive area exists	—*1
	OUT	BUF	Valid*2	Transmit data exists	—*1
Isochronous transfer	IN	BUF	Valid*2	*3	—*1
	OUT	BUF	Valid*2	*4	—*1

Note 1. An em dash (—) in the table indicates that the condition is unrelated to the generating of tokens.
Note 2. “Valid” indicates that, for interrupt transfers and isochronous transfers, a transaction is generated only in transfer frames that are based on the interval counter.
Note 3. This indicates that a transaction is generated regardless of whether there is a receive area. If there is no receive area, however, the received data is discarded.
Note 4. This indicates that a transaction is generated regardless of whether there is any data to be transmitted. If there is no data to be transmitted, however, a zero-length packet is transmitted.

38.3.14.2 Transfer Schedule

This section describes the transfer scheduling within a frame of the USBHS. After the USBHS sends an SOF, the transfer is carried out in the following sequence:

1. Execution of periodic transfers:
   A pipe is searched for in the order of pipe 1 → pipe 2 → pipe 6 → pipe 7 → pipe 8 → pipe 9, and then if there is a pipe for which an isochronous or interrupt transfer transaction can be generated, the transaction is generated.
2. Setup transactions for control transfers:
   The DCP is checked, and if a setup transaction is possible, it is sent.
3. Execution of bulk transfers, control transfer data stages, and control transfer status stages:
   A pipe is searched for in the order of DCP → pipe 1 → pipe 2 → pipe 3 → pipe 4 → pipe 5, and then if there is a pipe for which a transaction for a bulk transfer, a control transfer data stage, or a control transfer status stage can be generated, the transaction is generated.
   When a transaction is generated, processing moves to the next pipe transaction regardless of whether the response from the peripheral device is ACK or NAK. If there is time for transfer within the frame, step 3 is repeated.

38.3.14.3 Enabling USB Communication

Setting the DVSTCTR0.UACT bit to 1 initiates an SOF transmission, and transaction generation is enabled. Setting the UACT bit to 0 stops SOF transmission, and the Suspend state is invoked. If the UACT setting is changed from 1 to 0, processing stops after the next SOF is sent.

38.3.15 Battery Charging Detection Processing

The USBHS provides control over the data contact detection processing (D+ line contact checking), primary detection processing (charger detection processing), and secondary detection processing (charger determination processing) as defined in the Battery Charging Specification.

This section describes operations required in device and host controller modes.

38.3.15.1 Processing in Device Controller Mode

To operate a function device as a battery charging portable device:

1. Start primary detection processing after detecting contact with the D+ and D- lines. The Battery Charging Specification describes two processing methods for Data Contact Detection. The USBHS supports both methods as follows:
   - Software processing
     After a VBINT interrupt or polling of the VBSTS flag indicates a change in the state of the USBHS_VBUS input pin, software controls a wait from 300 to 900 ms. The BCCTRL.VDPSRCE and IDMSINKE bits are then both set to 1, enabling the VDP_SRC and IMP_SINK circuits, respectively, to start primary detection processing.
   - Hardware processing
     Apply 7 to 13 µA of current to the D+ line to hold the D+ line at the logical high level. This is done to detect the D+ and D- lines going to the logical low level because of pull-down resistors on the host device side when the D+ and D- lines come in contact with those of the host. Monitor the SYSSTS0.LNST[1:0] flags while the BCCTRL.IDPSRCE bit is set to 1, enabling the IDP_SRC circuit, to see when the level on the D+ line changes from high to low. After detecting a low level on the D+ line, clear the BCCTRL.IDPSRCE bit to 0, disabling the IDP_SRC circuit, and set both the BCCTRL.VDPSRCE and IDMSINKE bits to 1, enabling the VDP_SRC and IDM_SINK circuits, respectively, to start primary detection processing. The VDPSRCE and IDMSINKE bits must be set to 1 simultaneously.
2. After the start of primary detection processing followed by a software-controlled wait of 40 ms, check the BCCTRL.CHGDETSTS flag. A value of 1 indicates detection of a charger, and secondary detection processing starts.[*1]
3. To start secondary detection processing, clear both the BCCTRL.VDPSRCE and IDMSINKE bits to 0, disabling the VDP_SRC and IDM_SINK circuits, respectively. Next, set both the BCCTRL.VDMSRCE and IDPSINKE bits to 1, enabling the VDM_SRC and IDP_SINK circuits, respectively.
4. After the start of secondary detection processing followed by a software-controlled wait of 40 ms, check the BCCTRL.PDDETSTS flag. A value of 1 indicates that secondary detection processing is complete.

Note 1. In primary detection processing, detection of a voltage above the range from 0.25 to 0.4 V and below the range from 0.8 to 2.0 V on the D-Line indicates that the other device is a host device that supports battery charging (charging downstream port). The BCCTRL.CHGDETSTS flag in the PHY block only indicates whether the voltage on the D-line is higher than the range from 0.25 to 0.4 V, so add processing as required to read the SYSSTS0.LNST[1:0] flags and confirm that the voltage on the D- line is also below the range from 0.8 to 2.0 V.

Figure 38.19 illustrates this processing flow.

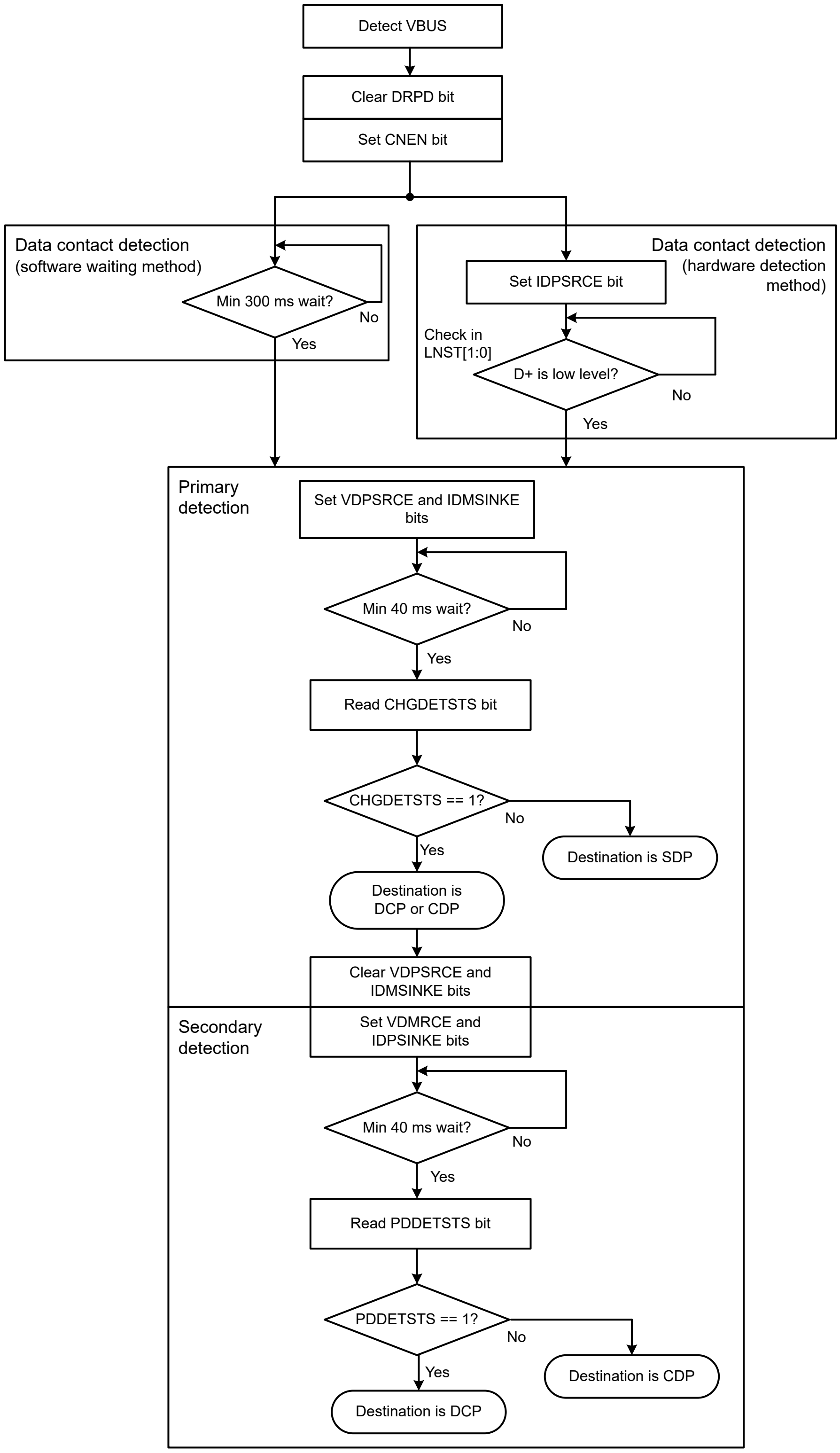


Figure 38.19 Processing flow as portable device

38.3.15.2 Processing in Host Controller Mode

In host controller mode, driving the D- line is required for a portable device to perform primary detection. The USBHS supports the following two primary detection methods:

- When the hardware has a portable device detection function
- When the hardware does not have the function or the function is present but not used

Figure 38.20 and Figure 38.21 show the processing flows for these methods.

(1) When the hardware has a portable device detection function

a. Start driving the USBHS_VBUS input pin.

b. Set the BCCTRL.IDMSINKE bit to 1 to enable the portable device detection circuit.

c. Monitor the portable device detection signal and start driving the D-line when the level of the portable device detection signal is high*1.

d. Stop driving the D-line when the portable device detection signal is at the low level*1.

Note 1. The PDDETINT interrupt indicates a change in the level of the portable device detection signal (EUH_CPDDET), and the current level can be obtained by reading the PDDETSTS flag.

(2) When the hardware does not have a portable device detection function or the function is not used

Software handles the timing of steps a. and b.

a. After a disconnect is detected, start driving the D-line within 200 ms.

b. After a connect is detected, stop driving the D-line within 10 ms.

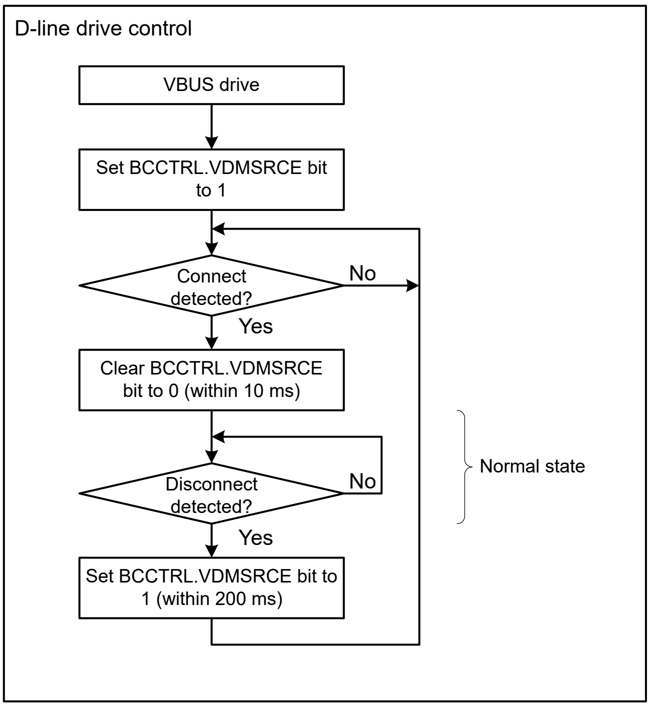


Figure 38.20 Processing flow as charging downstream port without hardware portable device detection function or when function is not used

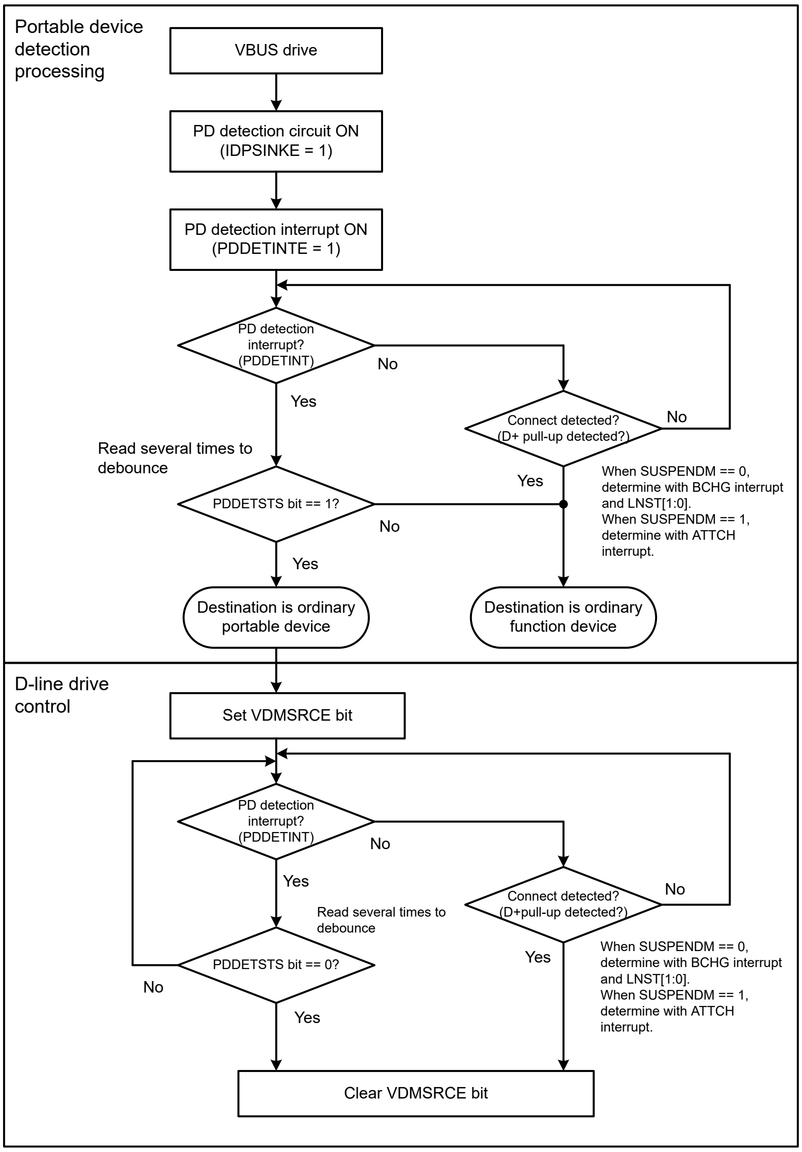


Figure 38.21 Processing flow as charging downstream port with hardware portable device detection function

38.3.16 Link Power Management Processing

The Link Power Management standard defines the existing Suspend state as the L2 state and also defines the L1 state as a state that allows transition and return with lower latency than the L2 state (Suspend). Table 38.34 provides a comparison between the L2 (Suspend) and L1 states.

Table 38.34 Comparison between L2 (Suspend) state and L1 state (1 of 2)

Parameter	L1 state	L2 (Suspend) state
Transition	LPM transaction	Idle for 3 ms

Table 38.34 Comparison between L2 (Suspend) state and L1 state (2 of 2)

Parameter	L1 state	L2 (Suspend) state
Return caused by host	Host: Minimum drive period (75 µs to 1.175 ms) can be specified by the host. Function: 10-µs K drive	Host: Minimum 20-ms K drive Function: 10-ms K drive
Return caused by function	Device: 50-µs K drive Function: 60- to 990-µs K drive Device: 10-µs K drive	Function: 1- to 15-ms K drive Host: Minimum 20-ms K drive Function: 10-ms K drive
Signaling	Full-speed and low-speed idle	Full-speed and low-speed idle

38.3.16.1 Processing in Device Controller Mode

(1) Descriptor contents

In device controller mode, the USBHS must return its descriptor on receiving the GetDescriptor command.

Change the content of the descriptor to be returned depending on whether the transition to and return from the L1 state corresponds to the processing for the LPM transaction. The following table shows the relationship between LPM correspondence and the descriptor.

Table 38.35 Relationship between LPM correspondence and descriptor

Correspondence with LPM	bcdUSB field	USB2.0 extension descriptor		Response to received LPM request	Notes
		Provided/ not provided	Value of LPM bit		
Does not correspond	0x0200	Not provided	—	No response	Normal operation when the LPM is not supported
	0x0201	Provided	0	STALL	Setting for clear non-correspondence to LPM. In this case, a STALL response must be returned.
Corresponds	0x0201	Provided	1	ACK or NYET	Normal operation when the LPM is supported

Declare whether to correspond to the transition to and return from L1 in the LPM bit in the USB 2.0 extension descriptor. To provide the USB2.0 extension descriptor, the bcdUSB field of the device descriptor must be set to a value of 0x0201 or larger.

When the LPM is not supported, the USB2.0 extension descriptor is not provided and the bcdUSB field value must be 0x0200. If an LPM token is received in this case, it must be ignored. It is also possible to set the bcdUSB field value to 0x0201 and the LPM bit in the USB2.0 extension descriptor to 0 (LPM tokens not supported). In this case, the LPM token cannot be ignored and a STALL response must be returned.

When the LPM token is supported, set the bcdUSB field value to 0x0201 and set the LPM bit in the USB 2.0 extension descriptor to 1 (LPM tokens supported). This allows acknowledgment when returning a NYET or ACK response to the LPM token.

(2) Processing during LPM token reception

Transition to and return from the L1 state in device controller mode is as follows:

a. When the USBHS receives an LPM token from the host, the L1RESPEN, L1RESPMD[1:0], and L1NEGOMD settings in PL1CTRL1 determine whether a response packet is sent or the token is ignored and, if a response is to be sent, whether it is an ACK, NYET, or STALL packet.

b. If an ACK response to the LPM token is sent and the host does not transmit another LPM token in 8 µs, the USBHS enters the L1 state. The USBHS handles detection of the newly transmitted packet and the transition to the L1 state. The DVST interrupt can be used to detect the transition.

c. Two types of processing can return the USBHS from the L1 state:

- When the host drives the D-line in the K-state:
  The function device detects the K-state and starts processing the return from the L1 state in response to an RESM interrupt request
- When the function device outputs a remote wakeup signal:
  If the software on the function device sets the DVSTCTR0.WKUP bit to 1, it sends a remote wakeup signal to the host

The software clears the DVSTCTR0.WKUP bit to 0 on returning from the L2 (Suspend) state, and the USBHS clears the DVSTCTR0.WKUP bit to 0 for return from the L1 state.

(3) HIRD field value negotiation function

The HIRD field value included in the LPM token indicates the host K-drive period on return from the L1 state. The HIRD field value can be adjusted according to the requirements of the target system. For example, a small HIRD field value is better for systems focusing on higher transfer efficiency, while a large HIRD field value is better for systems focusing on low power consumption.

Based on the L1NEGOMD and HIRDTHR[3:0] settings in PL1CTRL1, an ACK response is returned when the received HIRD field value is in the expected range, and otherwise a NYET response is returned, requesting the host to change the HIRD field value.

Note: This HIRD field value negotiation function at the host must also support negotiation processing.

38.3.16.2 Processing in Host Controller Mode

(1) Processing during LPM token transmission

Transition to and return from the L1 state in host controller mode is as follows:

a. When the HL1CTRL.L1REQ bit is set to 1, an LPM token is sent to the function device from the host device.

b. If an ACK response is received from the function device, a transition to the L1 state starts within 10 µs and is complete within 50 µs. If a transaction error is detected, another LPM token is transmitted within 8 µs. Retransmission can proceed up to two times. The USBHS handles all of this processing.

c. Two types of processing can return the USBHS from the L1 state:

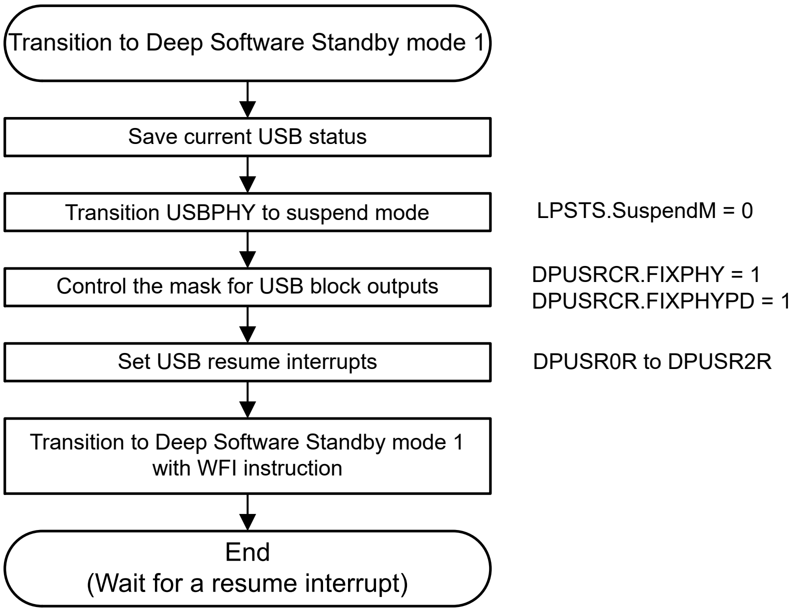
- When the host drives the D-line for the K state:
  When the DVSTCTR0.RESUME bit is set to 1, the host device starts driving the D-line for the K-state and starts processing the return
- When the function device generates a remote wakeup signal:
  When the host device detects a remote wakeup signal from the function device, it sets the DVSTCTR0.RESUME bit to 1 and starts driving the D-line for the K-state

Unlike when returning from the Suspend (L2) state, the USBHS clears the DVSTCTR0.RESUME bit to 0. After clearing the RESUME bit, it sets the DVSTCTR0.UACT bit to 1 and issues an L1RSMEND interrupt request.

38.3.17 Deep Software Standby Mode 1 Because of USB Suspend/Resume Interrupts

Deep Software Standby mode 1 can be canceled by a USB suspend/resume interrupt. USB suspend/resume interrupts are detected by the USB resume detecting unit, which controls and monitors the USB I/O pins to detect the interrupts.

Figure 38.22 shows the flow for setting the USBHS when entering Deep Software Standby mode 1 from either host or device controller mode. Figure 38.23 and Figure 38.24 show the flows for setting the USBHS when canceling Deep Software Standby mode 1 from host controller mode. Figure 38.25 shows the flow for setting the USBHS when canceling Deep Software Standby mode 1 from device controller mode.



Note: When connected to an external device, transition to Deep Software Standby mode 1 is enabled only on USB Suspend.

Note: Never invoke Deep Software Standby mode 1 in the following cases:

- In an L1 Suspend status with the USB-LPM protocol
- When remote wakeup is enabled in host controller mode.

Figure 38.22 USBHS setup flow for transition to Deep Software Standby mode 1 as a host or device controller

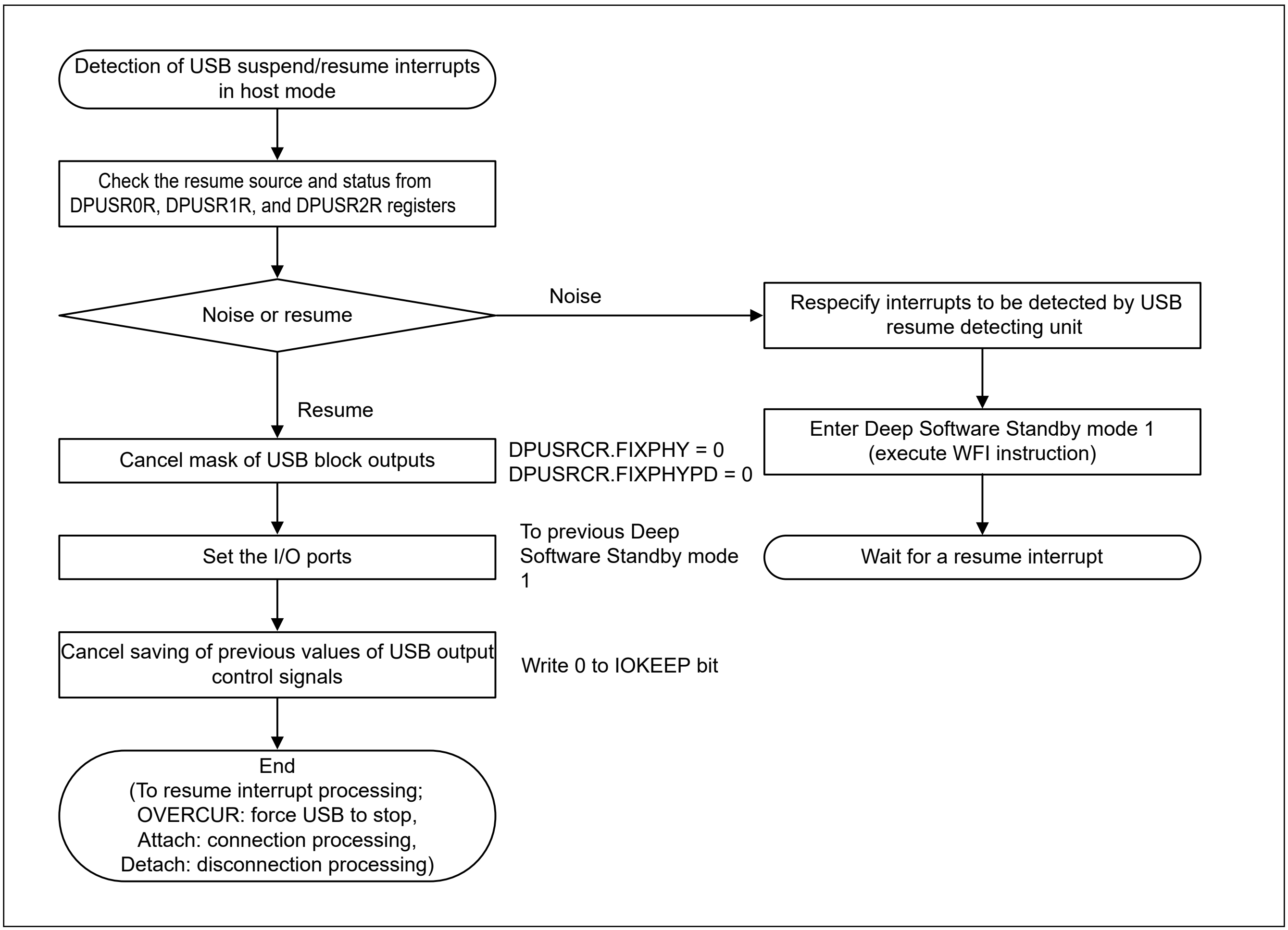


Figure 38.23 USBHS setup flow for canceling Deep Software Standby mode 1 as a host controller (1)

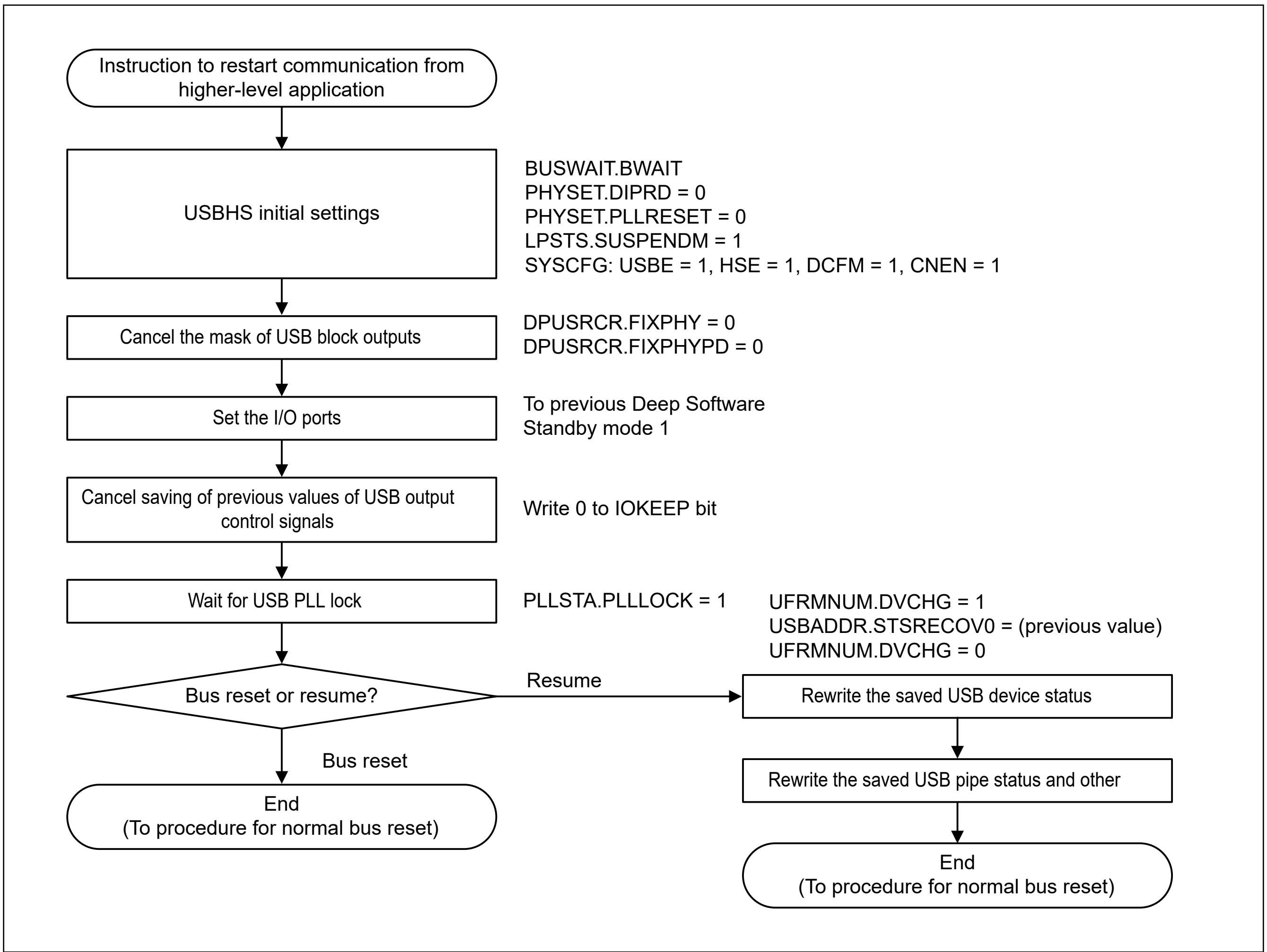
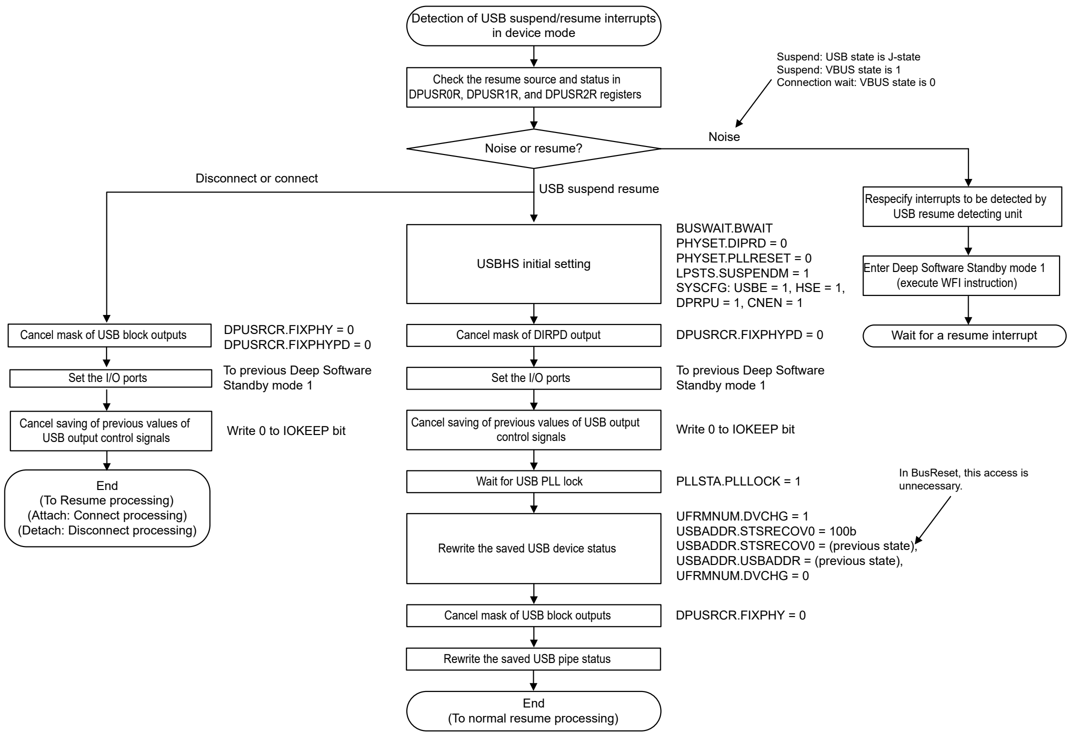


Figure 38.24 USBHS setup flow for canceling Deep Software Standby mode 1 as a host controller (2)



Note 1. In Bus Reset return, the write to the STSRECOV0 bit must be within 2.5 µs of USBHS-PHY PLL oscillation start.

Figure 38.25 USBHS setup flow for transition to Deep Software Standby mode 1 as a host or device controller

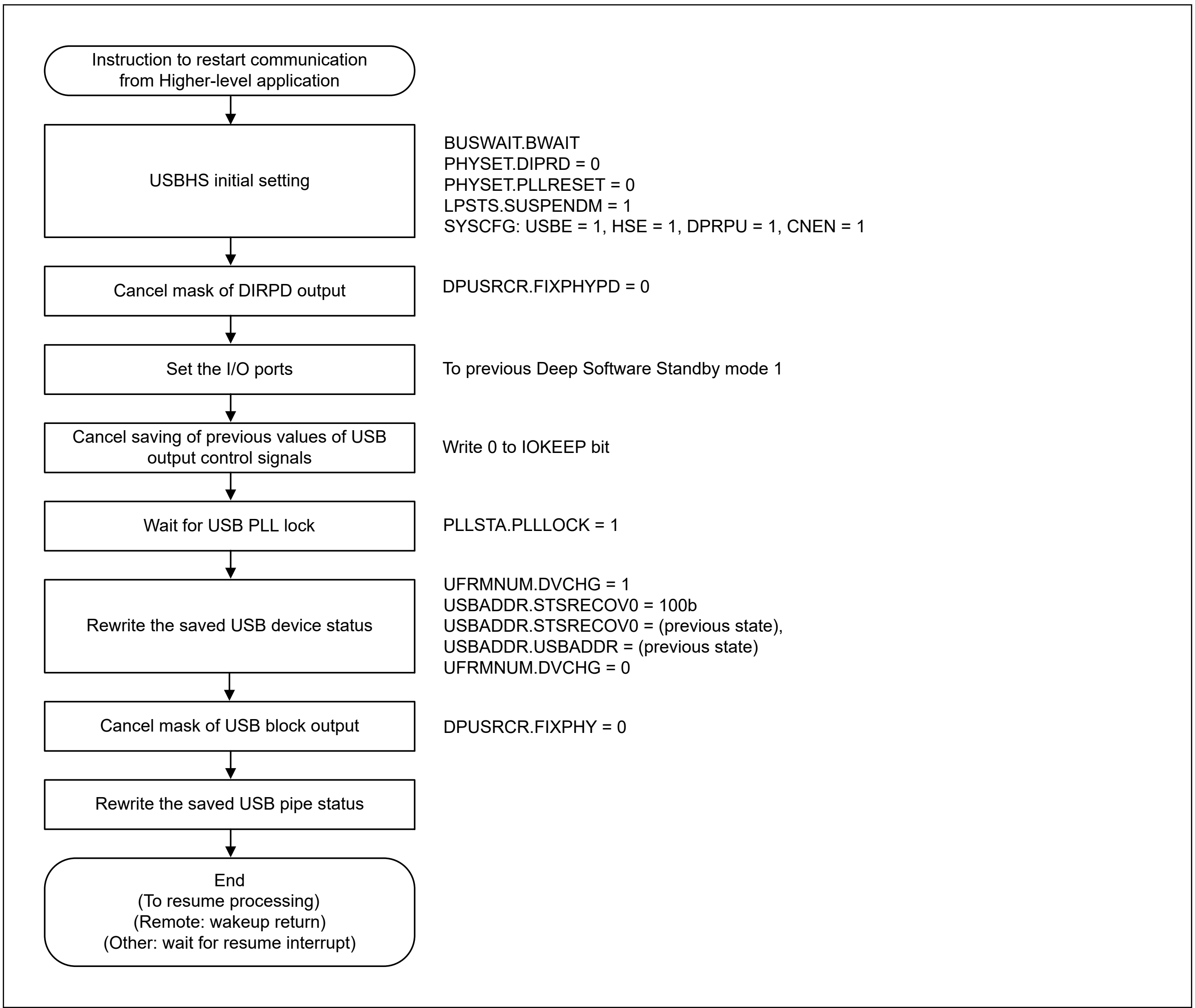


Figure 38.26 USBHS setup flow for canceling Deep Software Standby mode 1 as a device controller (2)

38.3.18 Example External Connection Circuits

Figure 38.27 shows an example OTG connection in a self-powered system. The USBHS controls the pull-up resistor of the D+ line and the pull-down resistor of D+ and D- lines. Select pull-up and pull-down for the lines in the SYSCFG.DPRPU and SYSCFG.DRPD bits. In device controller mode, the pull-up resistor of USB data line is disabled if SYSCFG.DPRPU bit is set to 0 while communicating with the USB host. The USBHS can use this to notify the USB host of a device disconnect.

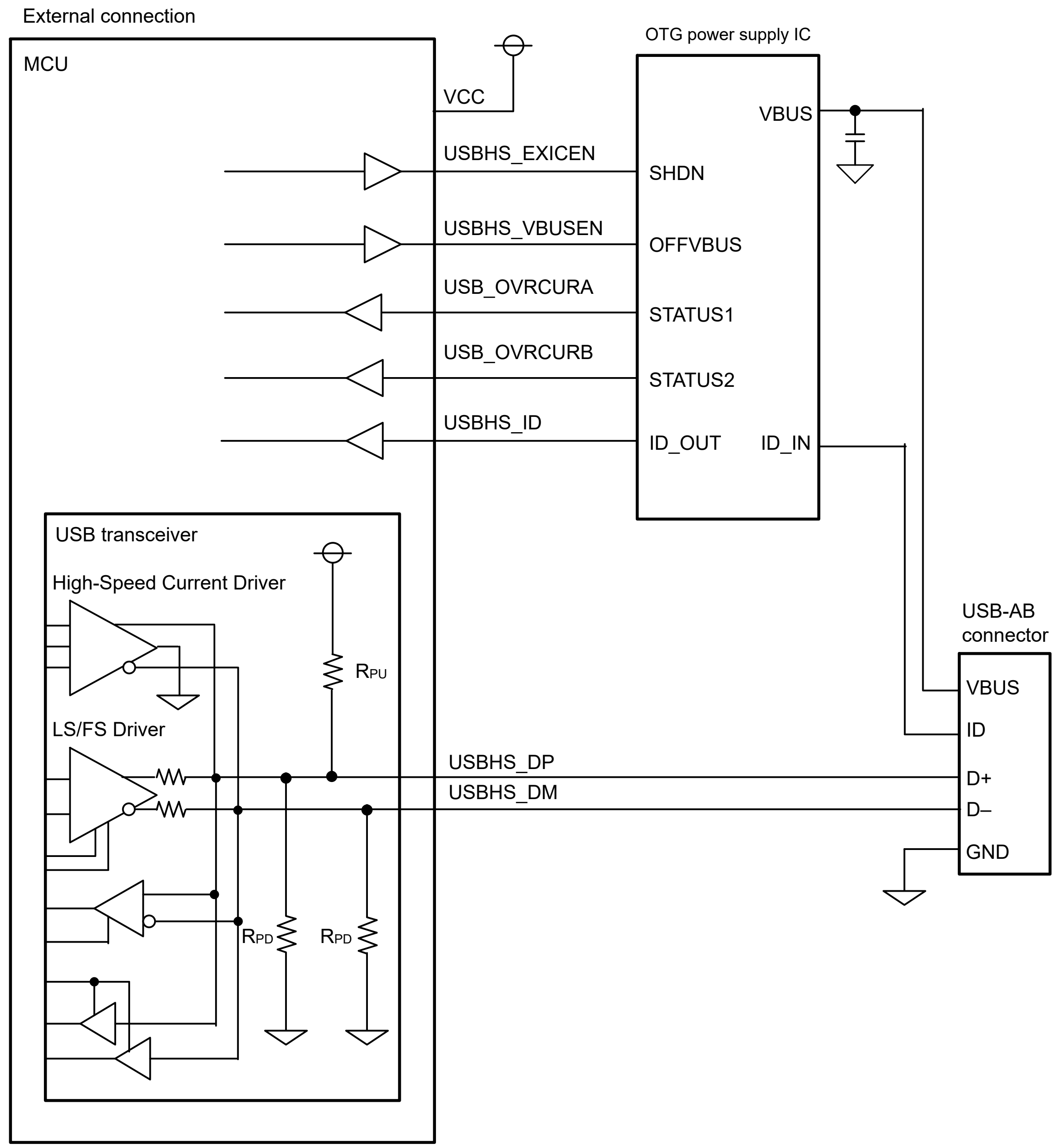
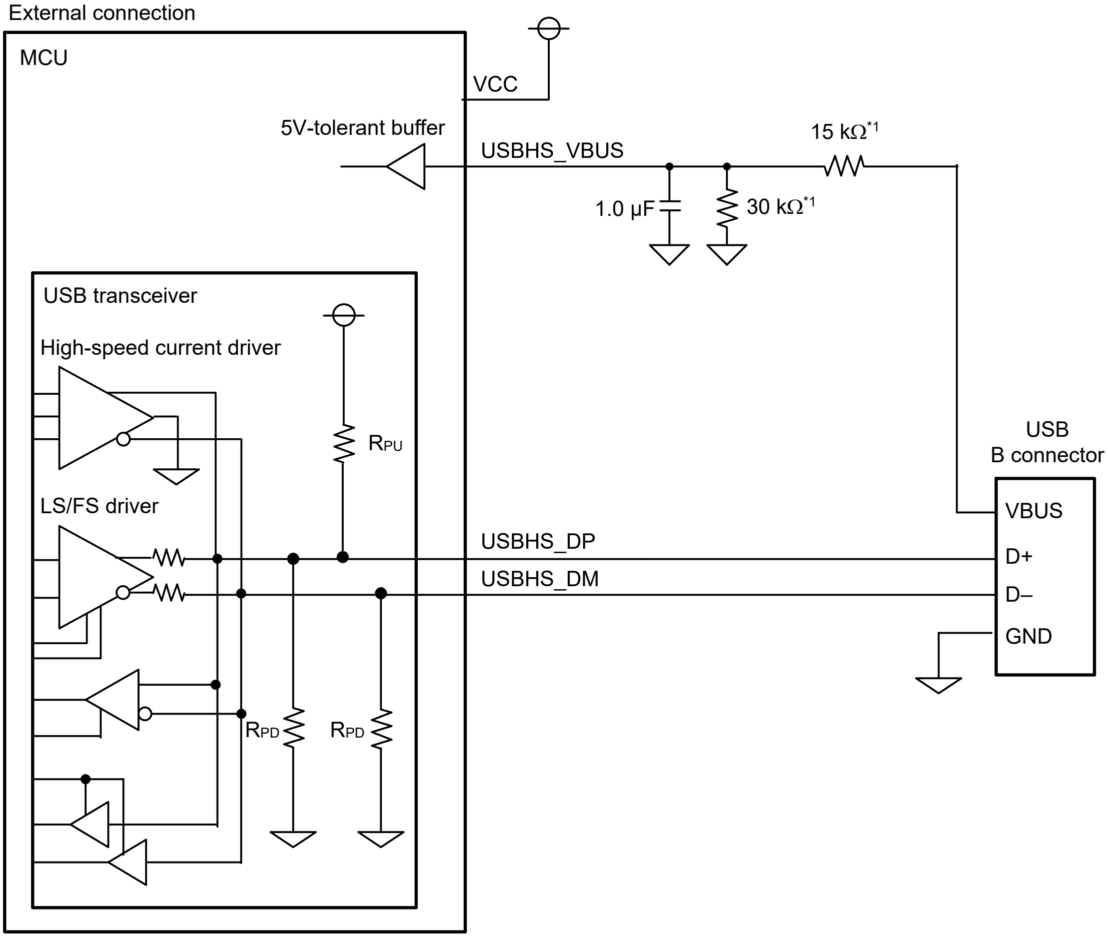


Figure 38.27 Example OTG connection in a self-powered system

Figure 38.28 shows an example USB device connection in a self-powered system.



Note 1. The VBUS (5 V) can be directly connected to the MCU if the VCC power supply of the MCU is not turned off when the USB is connected. If the VCC power supply of the MCU is turned off, the VBUS should be less than 3.6 V.

Figure 38.28 Example device connection in self-powered system

Figure 38.29 shows an example USB device connection in a bus-powered system.

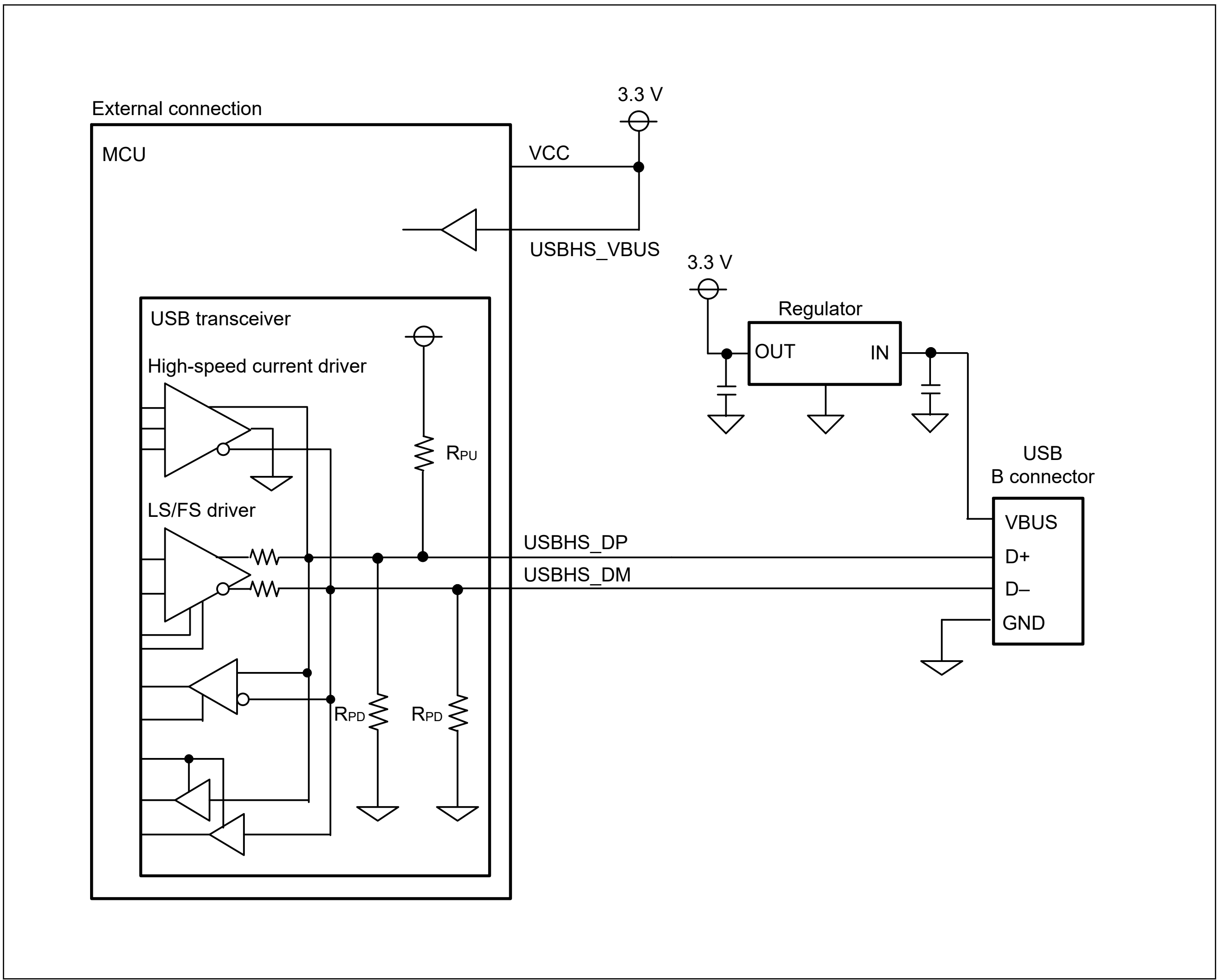


Figure 38.29 Example device connection in a bus-powered system

Figure 38.30 shows an example USB host connection.

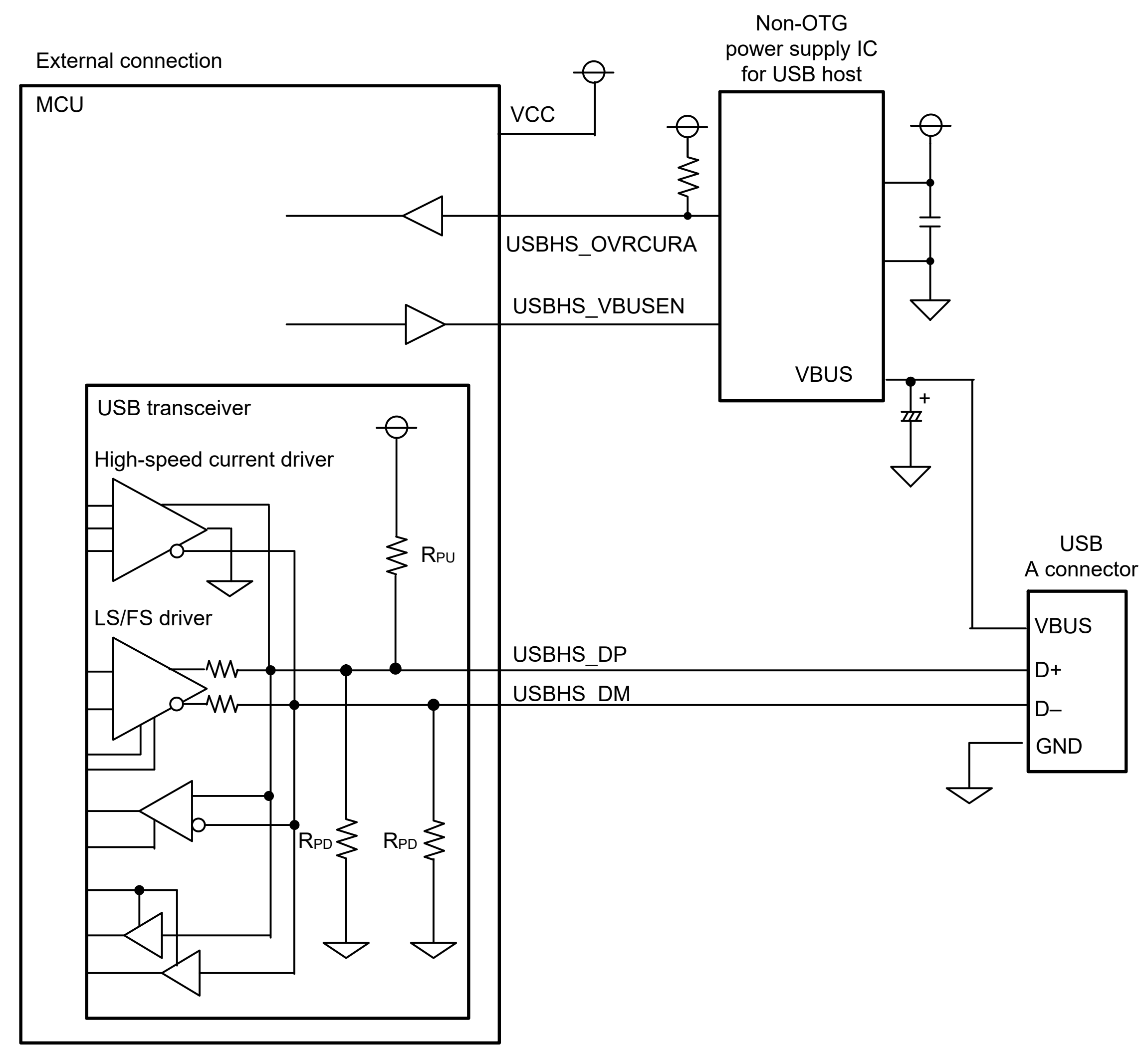


Figure 38.30 Example USB host connection

38.4 Usage Notes

38.4.1 Settings for the Module-Stop Function

USBHS operation can be disabled or enabled using Module Stop Control Register B (MSTPCRB). The USBHS is initially stopped after reset. Releasing the module-stop state enables access to the registers. After releasing module stop, make settings required to activate the PHY circuit, including the input clock frequency setting, and then clear the PHYSET.DIRPD bit to 0. For details, see section 11, Low Power Mode.

38.4.2 Setup for Transitioning to Deep Software Standby Mode 1

Before transitioning to Deep Software Standby mode 1, clear the DVSTCTR0.VBUSEN bit to 0.

38.4.3 Clearing the Interrupt Status Register on Exiting Software Standby Mode

Because the input buffer is always enabled in Software Standby mode, an unexpected interrupt might occur under the following conditions:

- When the interrupt is enabled in Normal mode

- When the interrupt is disabled in Software Standby mode
- When the input level of the pin that cancels Software Standby mode is changed in Software Standby mode

These conditions might cause the associated interrupt flag in the Interrupt Status Register to set unexpectedly. After the MCU exits Software Standby mode, the unexpected interrupt might be sent to the interrupt controller. To avoid this, always clear the INTSTS0 and INTSTS1 registers in the canceling sequence.

38.4.4 Clearing the Interrupt Status Register after Setting Up the Port Function

The input buffer is disabled before the PmnPFS.PSEL and PmnPFS.PMR ports are set up, so the internal signal is fixed high or low. The input buffer is enabled after the port is set so that the external pin state is propagated to the MCU. An unexpected interrupt might occur at this time, causing the VBINT and OVRCR bits in INTSTS0 and INTSTS1, or other interrupt status flags to set to 1. To avoid a malfunction, always clear the INTSTS0 and INTSTS1 registers after setting up the port.

38.4.5 Restriction of Register Access

When accessing the USBHS registers consecutively and writing to the Deep Software Standby-related registers (DPUSR0R, DPUSR1R, DPUSR2R, and DPUSRCR), there are constraints between the Deep Software Standby-related registers and the last accessed USBHS register. At least one of the following two constraints must be performed.

- Set an interval specified in the Table 38.36 between the last accessed USBHS register and write access to the Deep Software Standby-related registers.

Table 38.36 Required interval before write access to Deep Software Standby-related registers

Last accessed USBHS register	Write	Read
Other registers than following	BWAIT[3:0] + 3	BWAIT[3:0] + 3
DPUSR0R, DPUSR1R	130	BWAIT[3:0] + 3
DPUSR2R, DPUSRCR	130	7

Note: Unit: PCLKA cycles

[Examples]

1. Performing read access to other registers, then write access to Deep Software Standby-related registers.
   The required interval is BWAIT[3:0] + 3 cycles or longer.
2. Performing write access to the Deep Software Standby-related registers, then write access to the Deep Software Standby-related registers. The required interval is 130 cycles or longer.

- Before performing write access to the Deep Software Standby-related registers, when the read access to a USBHS register is performed, confirm the read value. When the write access to a USBHS register is performed, read it and confirm the written value.